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Algebra Toolkit

The tools every algebra topic uses: the words, collecting like terms, and substituting numbers — especially negatives — without a slip.

THE TOPIC ON ONE MAP

Algebra Toolkit

01 — Terms and like terms

$$4y + 2x - 3 + 3x + 8$$

$$3a - a \times 4a + 2a$$

identity: find a, b

02 — Substitution

$$f = 3g + 7h, g = -5$$

$$s = ut + \frac{1}{2}at^2$$

“all the figures on your calculator”

03 — Substitute, then solve

$$A = 4bc: \text{find } c$$

$$h = 3t^2: \text{find } t$$

“if x doubles ...”

THE FIVE WORDS Know which one you are looking at:

ZONE 1

Term

$$4x - 3y^2$$

ZONE 2

Expression

$$4x - 3y \text{ (no } = \text{)}$$

ZONE 3

Equation

$$4x - 3 = 9 \text{ (solve it)}$$

ZONE 4

Formula

$$A = \pi r^2 \text{ (links letters)}$$

ZONE 5

Identity

$\equiv$ true for every x

01 Terms and Like Terms

A term is a number, letters, or both — and the sign in front belongs to it. Only **like terms** (same letters, same powers) can be added.

WHEN YOU SEE → YOU DO

“Simplify”

→ collect like terms: same letters, same powers

$-4a$ **after** $3a$

→ the sign moves with the term: $3a - 4a = -a$

$a \times 4a$

→ multiply first (BIDMAS): $4a^2$ — not a like term of a

≡ **“find a and b in the identity”**

→ expand, then match the x terms and the numbers

Circle each term with its sign

Group like terms

Add the numbers in front

WORKED MODEL • SIMPLIFY

Simplify $3a - 5b - a + 2b$.

1 $(3a - a) + (-5b + 2b)$

group like terms

ANSWER $2a - 3b$

WORKED MODEL • IDENTITY

$5(7x + 8) + 3(2x + b) \equiv ax + 13$. Find a and b .

1 $35x + 40 + 6x + 3b = 41x + 40 + 3b$

expand

2 x terms: $a = 41$; numbers: $40 + 3b = 13$

match

ANSWER $a = 41, b = -9$

× **TRAP** $\frac{6c + 10}{2}$ is **not** $3c + 10$. Divide **both** terms: $3c + 5$.

IDEA 01

Practise: Terms and Like Terms

My questions first, then selected practice.

Q1 **DR ESLAM** Collect

CH 2 • TOPIC 1 2 marks

Simplify $5x - 2y + 3x + 7y$

SELECTED PRACTICE

Q2 **PRACTICE** Practice • Easy

CH 2 • TOPIC 1 2 marks

3 Simplify $4y + 2x - 3 + 3x + 8$

(2 marks)

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 1

2 marks

5 Simplify $ab - 5g + 5ab - 2g$

(2 marks)**Q4****PRACTICE**

Practice · Easy

CH 2 · TOPIC 1

1 mark

7 Sofia is trying to simplify $\frac{6c + 10}{2}$

Her method is

divide $6c$ by 2

then add 10

Evaluate her method.

(1 mark)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 1 2 marks

11 Simplify $5c - d - 3d - 2c$.

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1 3 marks

3 a and b are positive values.

Show that $\frac{7a + 2b - 3a}{8a + 6b + 2a - b}$ always simplifies to the same value.

(3 marks)

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1**1 mark**

5 Amy has x beads.

Billy has three more beads than Amy.

Carly has four times as many beads as Billy.

Circle the expression for the number of beads that Carly has.

$4x + 3$	$3x + 4$	$4(x + 3)$	$x + 12$
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(1 mark)

02 Substitution

Replace every letter by its number **in brackets**, then use the calculator. Brackets protect the negatives.

WHEN YOU SEE → YOU DO

a negative value: $c = -5$

→ write (-5) — always in brackets

c^2 **with** $c = -5$

→ $(-5)^2 = 25$, not -25

$2qx^2$

→ square first, then multiply: $2 \times 3 \times (-5)^2$

“write down all the figures on your calculator”

→ no rounding at all

Write the formula

Numbers in brackets

Calculator

Answer with units

WORKED MODEL • NEGATIVES

$G = c^2 - 4c$. Find G when $c = -5$.

1 $G = (-5)^2 - 4(-5)$

brackets

2 $G = 25 + 20$

BIDMAS

ANSWER $G = 45$

× **TRAP** Typing -5^2 in the calculator gives -25 . Type $(-5)^2$.

IDEA 02

Practise: Substitution

My questions first, then selected practice.

Q8 DR ESLAM Negative squared

CH 2 • TOPIC 1 2 marks

Work out the value of $2x^2 - 4x$ when $x = -3$.

Q9 DR ESLAM Calculator trap

CH 2 • TOPIC 1 1 mark

Ali types -3^2 and gets -9 . What should he type to square -3 ?

SELECTED PRACTICE

Q10**PRACTICE**

Practice • Easy

CH 2 • TOPIC 1 2 marks

1



You can work out the amount of medicine, c *ml*, to give to a child by using the formula

$$c = \frac{ma}{150}$$

m is the age of the child, in months.

a is an adult dose, in *ml*.

A child is 30 months old.

An adult's dose is 40 *ml*.

Work out the amount of medicine you can give to the child.

(2 marks)

Q11**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 4 marks

- 1 (a)** The body mass index, B , for a person of mass m kg and height h metres is given by the formula

$$B = \frac{m}{h^2}$$

Usman has a mass of 50 kg.

He has a height of 1.57 m.

Work out Usman's body mass index.

Give your answer correct to one decimal place.

(2 marks)

- (b)** Tom's height is 1.80 m.
He wants his body mass index to be 21

Work out the mass that will give Tom a body mass index of 21

(2 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 1 2 marks

$$7 \quad w = 5y^2 - y^3$$

Work out the value of w when $y = -2$

$$w = \dots\dots\dots$$

(2 marks)

Q13 **PRACTICE** Practice · Medium

CH 2 · TOPIC 1 2 marks

$$11 \quad y = mx + c$$

Find the value of y when $m = -3$, $x = -2$ and $c = -8$.

$$y = \dots\dots\dots$$

(2 marks)

Q14 PRACTICE Practice · Medium

CH 2 · TOPIC 1 2 marks

$$13 \quad s = ut + \frac{1}{2}at^2$$

Find the value of s when $u = 5.2$, $t = 7$ and $a = 1.6$.

$s =$

(2 marks)

Q15 PRACTICE Practice · Hard

CH 2 · TOPIC 1 4 marks

$$7 \quad s = ut + \frac{1}{2}at^2$$

Find s when $t = 26.5$, $u = 104.3$, $a = -2.2$.

Give your answer in standard form, correct to 4 significant figures.

$s =$

(4 marks)

03 Substitute, Then Solve

Sometimes the letter you want is **not** the subject. Substitute what you know, then solve the equation that is left.

WHEN YOU SEE → YOU DO

" $A = 100, b = 2$, **work out c** " in $A = 4bc$ " → substitute, then solve: $100 = 8c$

$h = 3t^2$, "**find t when $h = 108$** " → $t^2 = 36$, $t = 6$ (or -6)

"**if x doubles, what happens to y ?**" → try numbers: $x = 1$ then $x = 2$

"**One solution is $x = 7$** " → put $x = 7$ in, find the constant

WORKED MODEL • FIND A LETTER INSIDE

$C = \frac{5(F - 32)}{9}$. Is 77°F below the 20°C minimum?

1 $C = \frac{5(77 - 32)}{9} = \frac{5 \times 45}{9}$

substitute

2 $C = 25$

calculator

ANSWER $25^\circ\text{C} > 20^\circ\text{C}$: not below the minimum

WHERE YOU WILL USE IT Every formula question in the course: rearranging (Topic 7), graphs, mensuration, kinematics.

IDEA 03

Practise: Substitute, Then Solve

My questions first, then selected practice.

Q16 **DR ESLAM** Then solve

CH 2 • TOPIC 1 2 marks

 $v = 5t - 3$. Find t when $v = 22$.**SELECTED PRACTICE****Q17** **PRACTICE** Practice • Easy

CH 2 • TOPIC 1 1 mark

$$8 \quad y = \frac{10}{x}$$

If the value of x doubles, what happens to the value of y ? Circle your answer.

$\div 2$	$\times 2$	$\div 5$	$\times 5$
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(1 mark)

Q18**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 2 marks

$$A = 4bc$$

2

$$A = 100$$

$$b = 2$$

Work out the value of c .**(2 marks)**

Q19**PRACTICE**

Practice • Medium

CH 2 • TOPIC 1 3 marks

3 You can change temperatures from °F to °C by using the formula

$$C = \frac{5(F - 32)}{9}$$

F is the temperature in °F.

C is the temperature in °C.

The minimum temperature in an elderly person's home should be 20 °C.

Mrs Smith is an elderly person.

The temperature in Mrs Smith's home is 77 °F.

Decide whether or not the temperature in Mrs Smith's home is lower than the minimum temperature should be.

(3 marks)

Q20**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 3 marks

$$6 \quad h = 5t^2 + 2$$

(i) Work out the value of h when $t = -2$

[1]

(ii) Work out a value of t when $h = 47$

[2]

(3 marks)**Q21****PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 1 mark

14 Complete the statement.

When $5x = 15$, $12x = \dots\dots\dots$

(1 mark)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1 3 marks

8 (a) One solution of the equation $ax^2 + a = 150$ is $x = 7$.

Find the value of a .

 $a = \dots\dots\dots$ **(2 marks)**

(b) Find the other solution.

 $x = \dots\dots\dots [1]$ **(1 mark)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------|--|
| 01 | Like terms | Same letters and powers; the sign moves with the term. Identity: match coefficients. |
| 02 | Substitution | Every number in brackets; $(-5)^2 = 25$; BIDMAS; no early rounding. |
| 03 | Then solve | Substitute what you know, solve for the letter left. |

BOTTOM LINE Brackets round every substituted number → BIDMAS → the sign belongs to the term after it.

Answers

Algebra Toolkit • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TERMS AND LIKE TERMS

Q1	Dr Eslam • Collect	$8x + 5y$
Q2	Practice • Easy	$5x + 4y + 5$
Q3	Practice • Easy	$6ab - 7g$
Q4	Practice • Easy	wrong: 10 must be divided by 2 as well; $3c + 5$
Q5	Practice • Easy	$3c - 4d$
Q6	Practice • Hard	$\frac{4a + 2b}{10a + 5b} = \frac{2(2a + b)}{5(2a + b)} = \frac{2}{5}$
Q7	Practice • Hard	$4(x + 3)$

IDEA 02 • SUBSTITUTION

Q8	Dr Eslam • Negative squared	$2(-3)^2 - 4(-3) = 18 + 12 = 30$
Q9	Dr Eslam • Calculator trap	$(-3)^2 = 9$
Q10	Practice • Easy	$\frac{30 \times 40}{150} = 8 \text{ ml}$
Q11	Practice • Medium	(a) 20.3 (b) 68.04 kg
Q12	Practice • Medium	$w = 28$
Q13	Practice • Medium	$y = -2$
Q14	Practice • Medium	$s = 75.6$
Q15	Practice • Hard	1.991×10^3

IDEA 03 • SUBSTITUTE, THEN SOLVE

Q16 Dr Eslam • Then solve

$$22 = 5t - 3; t = 5$$

Q17 Practice • Easy

$$\div 2$$

Q18 Practice • Medium

$$100 = 8c; c = 12.5$$

Q19 Practice • Medium

$$C = 25^{\circ}\text{C} > 20^{\circ}\text{C}: \text{not lower}$$

Q20 Practice • Medium

$$(i) 22 (ii) t = 3 \text{ (or } -3)$$

Q21 Practice • Medium

$$12x = 36$$

Q22 Practice • Hard

$$(a) a = 3 (b) x = -7$$



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Algebraic Roots & Indices

Four laws, three special powers, one way to solve.
Every indices question is built from these.

THE TOPIC ON ONE MAP

Algebraic Roots & Indices

01 — The four laws

$$x^3 \times x^5$$

$$(3x^2)^3$$

$$\frac{12a^4h^6}{4ah^2}$$

02 — Zero and negative powers

$$g^0 = 1$$

$$x^{-2} = \frac{1}{x^2}$$

03 — Fractional powers

$$8^{\frac{2}{3}}$$

Flip → Root → Power

“exact value”

04 — Solving: unify the base

$$2^n = 8^3$$

“find n ”

“ x in terms of n ”

05 — Where you will use it

“in the form ax^n ”

differentiation

01 The Four Laws

Same base only. Numbers in front are worked out normally — the laws are for the powers.

WHEN YOU SEE → YOU DO

“simplify” with the same base

→ $\times$ add $\cdot \div$ subtract $\cdot ()^n$ multiply

“(3x²)³”

→ the number gets the power too

Multiply

$$a^m \times a^n = a^{m+n}$$

add the powers

Divide

$$a^m \div a^n = a^{m-n}$$

subtract the powers

Power of a power

$$(a^m)^n = a^{mn}$$

multiply the powers

Power of a product

$$(ab)^n = a^n b^n$$

everything inside gets the power

WORKED MODEL • NUMBERS AND LETTERS

$$\frac{36af^8}{12a^5f^2}$$

1 $36 \div 12 = 3$

numbers: normal

2 $a^{1-5} = a^{-4}$ $f^{8-2} = f^6$

letters: subtract

ANSWER $\frac{3f^6}{a^4}$

× **TRAP** $(3x^2)^3 = 3x^6$ is wrong — the 3 is inside the bracket too: $3^3x^6 = 27x^6$.

IDEA 01

Practise: The Four Laws

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM One law each

CH 2 · TOPIC 2 3 marks

Simplify (a) $x^5 \times x^3$ (b) $y^9 \div y^4$ (c) $(x^4)^3$

Q2 DR ESLAM Numbers inside the bracket

CH 2 · TOPIC 2 2 marks

Simplify $(2x^3y)^4$.

The 2 gets the power too: $2^4 = 16$.

Answer _____

Q3 **DR ESLAM** Numbers, then letters

CH 2 • TOPIC 2 2 marks

Simplify $\frac{12x^7y^2}{4x^3y^5}$.

Answer _____

SELECTED PRACTICE**Q4** **PRACTICE** Practice • Easy

CH 2 • TOPIC 2 3 marks

2 (a) Simplify $m^5 \div m^3$

(1 mark)

(b) Simplify $5x^4y^3 \times x^2y$

(2 marks)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark**13** Simplify.

$$a^6 \div a^2$$

(1 mark)**Q6****PRACTICE**

Practice · Medium

CH 2 · TOPIC 2 2 marks**9** Simplify $2a^3b \times 5a^2b^3$ **(2 marks)**

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 2 marks

13 $(ar^b)^4 = 16r^{20}$ where a and b are positive integers.

Work out a and b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q8**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q1

CH 2 · TOPIC 2 3 marks

You must write down all the stages in your working.

1 (a) Simplify $\frac{x^9}{x^2}$

(1)

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

(Total for Question 1 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q1

CH 2 · TOPIC 2 4 marks

- 1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

 $m = \dots\dots\dots$

- (b) Write down the value of n , given that $(5^3)^7 = 5^n$

 $n = \dots\dots\dots$ **(1)**

- (c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

 $p = \dots\dots\dots$ **(Total for Question 1 is 4 marks)**

Q10**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q1

CH 2 · TOPIC 2 6 marks**You must write down all the stages in your working.**

1 (a) Simplify $a^7 \times a^4$

.....

(b) Simplify $w^{15} \div w^3$

.....

(c) Simplify $(8x^5y^3)^2$

.....

(d) Make t the subject of $c = t^3 - 8v$

.....

(Total for Question 1 is 6 marks)

Q11**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q6

CH 2 · TOPIC 2 4 marks6 (a) Simplify $m^{10} \div m^3$

.....

$$k^n \times k^4 = k^{12}$$

(b) Write down the value of n $n =$ (c) Simplify $(3x^6y^8)^2$

.....

(Total for Question 6 is 4 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q4

CH 2 · TOPIC 2 8 marks

4 (a) Solve $\frac{2x+5}{6} = 2x-5$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(b) Simplify $h^{15} \div h^3$

$$\dots\dots\dots$$

(c) Simplify fully $(2g^3k^5)^4$

$$\dots\dots\dots$$

(2)

(d) Given that $\frac{y^5 \times y^n}{y^7} = y^{12}$

work out the value of n

$$n = \dots\dots\dots$$

(Total for Question 4 is 8 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q4

CH 2 · TOPIC 2 4 marks

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m = \dots\dots\dots$

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n = \dots\dots\dots$

(c) Simplify fully $(5a^4r^2)^3$

$\dots\dots\dots$

(Total for Question 4 is 4 marks)

02 Zero and Negative Powers

Zero means 1. Negative means **flip it**.

WHEN YOU SEE → YOU DO

“write down the value of g^0 ”

→ 1

a negative power

→ flip it

0

Power zero

Anything (except 0) to the power 0 is 1: $g^0 = 1$, $(3x^2y)^0 = 1$, but $5y^0 = 5 \times 1 = 5$.

-

Negative power = flip

$$a^{-n} = \frac{1}{a^n} \quad 2^{-3} = \frac{1}{8} \quad \left(\frac{x}{y}\right)^{-2} = \left(\frac{y}{x}\right)^2 = \frac{y^2}{x^2}$$

A negative power never makes the answer negative — it moves the term across the fraction line.

WORKED MODEL • FLIP, THEN THE LAWS

$$\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

$$1 \quad 3^{-2+5} = 3^3$$

top: add

$$2 \quad 3^{3-10} = 3^{-7}$$

divide: subtract

ANSWER $n = -7$

× **TRAP** $5y^0 \neq 1$. Only y has the power 0: $5y^0 = 5$. But $(5y)^0 = 1$.

IDEA 02

Practise: Zero and Negative Powers

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Zero and negative

CH 2 · TOPIC 2 3 marks

Work out (a) 5^0 (b) 2^{-3} (c) $\left(\frac{2}{3}\right)^{-2}$

Q15 DR ESLAM Negative = flip

CH 2 · TOPIC 2 1 mark

Write $\frac{1}{x^4}$ as a single power of x .

Answer _____

SELECTED PRACTICE**Q16** PRACTICE Practice · Easy

CH 2 · TOPIC 2 1 mark

1 Simplify $(m^{-2})^5$

(1 mark)

Q17**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark

9 Simplify $(3x^2y)^0$

(1 mark)

Q18**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark

11 Simplify $w^1 \times w^0$

Circle your answer.

1	0	w	w^2
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(1 mark)

Q19**PRACTICE**

Practice · Medium

CH 2 · TOPIC 2 7 marks

8 (a) Solve $3x^2 = 147$

(2 marks)

(b) Work out the value of 2^{-3}

(1 mark)

(c) Simplify $(3x^2)^3$

(2 marks)

(d) $w = 4p - 16$

Make p the subject of this formula.

(2 marks)

PAST PAPERS

Q20**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q4

CH 2 · TOPIC 2 4 marks

4 (a) Simplify $(3k^2)^4$

.....

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....

(Total for Question 4 is 4 marks)**Q21****PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1H · Q9

CH 2 · TOPIC 2 4 marks

9 (a) Simplify $8 \times (4t)^0$

.....

$x^6 \div x^{-5} = x^p$

(b) Find the value of p

$p =$

(c) Simplify fully $(2k^2m^4)^3$

.....

(Total for Question 9 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q7

CH 2 · TOPIC 2 2 marks7 (a) Simplify $(4^{-2})^0$

$$3^{-14} \times 3^8 = 3^m$$

(b) Find the value of m

$$m = \dots\dots\dots$$

(Total for Question 7 is 2 marks)**Q23****PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q10

CH 2 · TOPIC 2 4 marks10 (a) Simplify $(2p)^0$ where $p > 0$

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n

$$n = \dots\dots\dots$$

(c) Simplify fully $(5a^4c^2)^3$ **(Total for Question 10 is 4 marks)**

03 Fractional Powers

The **bottom** of the fraction is the root. The **top** is the power.

WHEN YOU SEE → YOU DO

a fractional power

→ bottom = root, top = power

“simplify fully $(\dots)^{-\frac{2}{3}}$ ”

→ Flip → Root → Power

“exact value”

→ no calculator decimals — keep fractions

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a} \right)^m$$

root first — the numbers stay small

F

Flip

If the power is negative, flip the fraction and make the power positive.

R

Root

Take the root of **every** part: numbers, and divide each letter's power.

P

Power

Raise to the top number.

WORKED MODEL · FLIP → ROOT → POWER

$$\left(\frac{64x^6}{25y^2}\right)^{-\frac{1}{2}}$$

$$1 \quad \left(\frac{25y^2}{64x^6}\right)^{\frac{1}{2}}$$

flip

$$2 \quad \sqrt{25} = 5 \quad \sqrt{64} = 8 \quad y^{2 \div 2} = y \quad x^{6 \div 2} = x^3$$

root

$$\text{ANSWER} \quad \frac{5y}{8x^3}$$

IDEA 03

Practise: Fractional Powers

My questions first, then selected practice, then real past papers.

Q24 **DR ESLAM** Bottom is the root

CH 2 • TOPIC 2 3 marks

Work out (a) $8^{\frac{2}{3}}$ (b) $27^{-\frac{1}{3}}$ (c) $\left(\frac{4}{9}\right)^{-\frac{3}{2}}$

Q25 **DR ESLAM** Flip → Root → Power

CH 2 • TOPIC 2 3 marks

Simplify fully $\left(\frac{x^6}{64y^3}\right)^{-\frac{2}{3}}$.

Flip first. Then cube-root every part. Then square.

Answer _____

SELECTED PRACTICE

Q26**PRACTICE**

Practice · Medium

CH 2 · TOPIC 2 2 marks

2 Simplify $(9x^8y^3)^{\frac{1}{2}}$

(2 marks)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 3 marks

3 (a) Simplify fully $(3e)^0$

(1 mark)

(b) Simplify fully $\left(\frac{64x^6}{25y^2}\right)^{-\frac{1}{2}}$

(2 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 3 marks

9 Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 4 marks

16 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

(2 marks)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

(2 marks)

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q17

CH 2 • TOPIC 2 6 marks

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

(Total for Question 17 is 6 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q12

CH 2 · TOPIC 2 9 marks

12 (a) Simplify $(16e^{10}f^6)^{\frac{1}{2}}$

.....
(2)

(b) Write $\frac{2x+1}{4} + \frac{x-2}{3}$ as a single fraction in its simplest form.

Given that $4^{k+3} = 16 \times 2^k$

(c) find the value of k .
Show your working clearly.

$k =$

(Total for Question 12 is 9 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q11

CH 2 • TOPIC 2 3 marks

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

.....

(Total for Question 11 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q14

CH 2 • TOPIC 2 3 marks

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

.....

(Total for Question 14 is 3 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q11

CH 2 • TOPIC 2 2 marks

- 11 Express $\left(\frac{m^6 k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\frac{m^a k^b}{c}$ where a , b and c are integers to be found.
-

(Total for Question 11 is 2 marks)

Q35**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q15

CH 2 · TOPIC 2 7 marks

15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

.....
(2)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

.....
(2)

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

$y =$

(Total for Question 15 is 7 marks)

Q36**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q12

CH 2 · TOPIC 2 4 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$$x = \dots\dots\dots$$

(2)

(b) Simplify fully $(125p^6y^{24})^{\frac{2}{3}}$

.....

(Total for Question 12 is 4 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q17

CH 2 · TOPIC 2 4 marks

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$n = \dots\dots\dots$

(1)

(b) Express $\frac{7}{2 - \sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

 $\dots\dots\dots$ **(Total for Question 17 is 4 marks)****Q38****PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q15

CH 2 · TOPIC 2 3 marks

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

 $\dots\dots\dots$ **(Total for Question 15 is 3 marks)**

Q39**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q13

CH 2 · TOPIC 2 4 marks

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$

(Total for Question 13 is 4 marks)

04 Solving: Unify the Base

Make both sides a single power of the **same base**. Then the powers are equal.

WHEN YOU SEE → YOU DO

“find the value of n / k ”, $2^{\dots} = \dots$

→ unify the base, equate the powers

“express x in terms of n ”

→ same method, answer contains n

1

Unify the base

Write every number as a power of one prime: $4 = 2^2$, $8 = 2^3$, $16 = 2^4$, $\sqrt{2} = 2^{\frac{1}{2}}$.

2

One power each side

Use the four laws until each side is $2^{\text{something}}$.

3

Equate the powers

If $a^x = a^y$ then $x = y$. Solve — usually a linear equation.

WORKED MODEL • JAN 2021 P1HR

$$4^{k+3} = 16 \times 2^k$$

$$1 \quad (2^2)^{k+3} = 2^4 \times 2^k$$

unify: base 2

$$2 \quad 2^{2k+6} = 2^{k+4}$$

one power each side

$$3 \quad 2k + 6 = k + 4$$

equate powers

$$\text{ANSWER} \quad k = -2$$

CALCULATOR CHECK Check with the calculator: $4^{(-2)+3} = 4$ and $16 \times 2^{-2} = 4$. Same → correct.

IDEA 04

Practise: Solving: Unify the Base

My questions first, then selected practice, then real past papers.

Q40 DR ESLAM Unify the base

CH 2 · TOPIC 2 2 marks

Solve $3^{x+1} = 81$.

$81 = 3^?$

$x =$ _____

Q41 DR ESLAM Both sides need changing

CH 2 · TOPIC 2 3 marks

Solve (a) $2^{3x} = 4^{x+1}$ (b) $9^x \times 3 = 27^2$

SELECTED PRACTICE

Q42 PRACTICE Practice • Hard

CH 2 • TOPIC 2 3 marks

$$1 \quad 16^{\frac{1}{5}} \times 2^x = 8^{\frac{3}{4}}$$

Work out the exact value of x .

(3 marks)

Q43 PRACTICE Practice • Hard

CH 2 • TOPIC 2 1 mark

$$5 \quad \text{Given that } \left(\sqrt[3]{\frac{1}{x}} \right)^4 = x^m$$

find the value of m $m = \dots\dots\dots$

(1 mark)

Q44**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 4 marks

8 Given that $4^{k+3} = 16 \times 2^k$

find the value of k .

Show your working clearly.

$$k = \dots\dots\dots$$

(4 marks)**Q45****PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 3 marks

15 $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n . Show your working clearly.

(3 marks)

PAST PAPERS

Q46**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q21

CH 2 • TOPIC 2 7 marks

21 (a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

.....
(3)

(b) Find an expression for n in terms of y .

Show clear algebraic working and simplify your expression.

.....

(Total for Question 21 is 7 marks)

Q47**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q21

CH 2 · TOPIC 2 5 marks

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

.....
(Total for Question 21 is 5 marks)

Q48**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q19

CH 2 · TOPIC 2 7 marks**19 (a)** Simplify $8^2 \times \sqrt[3]{4^6}$ Give your answer in the form 2^a where a is an integer.

Show each stage of your working clearly.

.....
(3)Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$ (b) find the value of n . $n =$ **(Total for Question 19 is 7 marks)**

Q49**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q11

CH 2 · TOPIC 2 4 marks

11 $\sqrt{2} \times 16 = 2^x$

- (a) Find the value of x .
Show your working clearly.

$$x = \dots\dots\dots (2)$$

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

- (b) Find the value of n .
Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 11 is 4 marks)

Q50**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q6

CH 2 · TOPIC 2 4 marks

- 6 (a) Write down the value of x^0

.....

Given that $2^{-3} \times 2^9 = 2^n$

- (b) find the value of n

$$n = \text{.....}$$

(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

- (c) find the value of m

$$m = \text{.....}$$

(Total for Question 6 is 4 marks)**Q51****PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q7

CH 2 · TOPIC 2 2 marks

7 $\frac{2^k}{4^n} = 2^x$

Find an expression for x in terms of k and n

$$x = \text{.....}$$

(Total for Question 7 is 2 marks)

Q52**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2H · Q24

CH 2 · TOPIC 2 3 marks**24**

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ **(Total for Question 24 is 3 marks)**

Q53**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q24

CH 2 · TOPIC 2**5 marks****24** Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n State any restrictions on n

(Total for Question 24 is 5 marks)**Q54****PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q7

CH 2 · TOPIC 2**2 marks**

$$7 \quad \frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

Find the value of n

$$n = \dots\dots\dots$$

(Total for Question 7 is 2 marks)

Q55 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2H • Q23**CH 2 • TOPIC 2** **3 marks**

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

Q56 **PAST PAPER** Edexcel 4MA1 • November 2024 • Paper 2H • Q14**CH 2 • TOPIC 2** **3 marks**

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

$n =$

(Total for Question 14 is 3 marks)

Q57**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q6

CH 2 · TOPIC 2 5 marks

- 6 (a) Write down the value of
- 5^0

.....
(1)

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^k$$

- (b) Find the value of
- k

 $k =$

- (c) Simplify fully
- $(2d^4e^5)^3$

.....

(Total for Question 6 is 5 marks)**Q58****PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q8

CH 2 · TOPIC 2 2 marks

$$8 \quad \frac{8^{-2} \times 8^9}{8^{10}} = 8^n$$

Work out the value of n $n =$ **(Total for Question 8 is 2 marks)**

05 Where You Will Use It

Indices are the language of later topics. Learn to write everything as x^n .

WHEN YOU SEE → YOU DO

“in the form ax^n ”

$$\rightarrow \sqrt{x} = x^{\frac{1}{2}}, \frac{1}{x^2} = x^{-2}$$

differentiate $\frac{3}{x^2}$ or $\sqrt{x}$

→ rewrite as powers first

Differentiation

Rewrite first: $\frac{3}{x^2} = 3x^{-2}$,

$$\sqrt{x} = x^{\frac{1}{2}}$$

Surds

$\sqrt{a} = a^{\frac{1}{2}}$ — the same laws apply.

Indices

Standard form

$$10^m \times 10^n = 10^{m+n}$$

Growth & decay

Compound interest: $P \times 1.05^n$

WORKED MODEL • WRITE IN THE FORM ax^n

$$\frac{5}{\sqrt[3]{x^2}}$$

$$1 \quad \sqrt[3]{x^2} = x^{\frac{2}{3}}$$

bottom is the root

$$2 \quad \frac{5}{x^{\frac{2}{3}}} = 5x^{-\frac{2}{3}}$$

flip = negative

ANSWER $5x^{-\frac{2}{3}}$

IDEA 05

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q59 DR ESLAM Differentiation: rewrite first

CH 2 • TOPIC 2 3 marks

Write each in the form ax^n : (a) $\frac{1}{x^3}$ (b) $\sqrt{x}$ (c) $\frac{5}{\sqrt[3]{x^2}}$

Q60 DR ESLAM Surds are powers

CH 2 • TOPIC 2 2 marks

Write $\sqrt{x} \times x^2$ as a single power of x .

Answer _____

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|---|
| 01 | Four laws | $\times$ add $\cdot \div$ subtract $\cdot ()^n$ multiply $\cdot$ everything inside gets the power. |
| 02 | Zero & negative | $a^0 = 1$. $a^{-n} = \frac{1}{a^n}$ — negative means flip. |
| 03 | Fractional | Bottom = root, top = power. Flip $\rightarrow$ Root $\rightarrow$ Power. |
| 04 | Solving | Unify the base (primes) $\rightarrow$ one power each side $\rightarrow$ equate powers. |
| 05 | Use it | Differentiation, surds, standard form, growth. |

BOTTOM LINE Same base $\rightarrow$ use the laws $\rightarrow$ negative = flip, fraction = root $\rightarrow$ to solve, unify the base and equate the powers.

Answers

Algebraic Roots & Indices • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE FOUR LAWS

- | | | |
|------------|--|--|
| Q1 | Dr Eslam • One law each | (a) x^8 (b) y^5 (c) x^{12} |
| Q2 | Dr Eslam • Numbers inside the bracket | $16x^{12}y^4$ |
| Q3 | Dr Eslam • Numbers, then letters | $\frac{3x^4}{y^3}$ |
| Q4 | Practice • Easy | (a) m^2 (b) $5x^6y^4$ |
| Q5 | Practice • Easy | a^4 |
| Q6 | Practice • Medium | $10a^5b^4$ |
| Q7 | Practice • Hard | $a = 2, b = 5$ |
| Q8 | Edexcel 4MA1 • January 2020 • Paper 2H • Q1 | (a) x^7 (b) 7^9 |
| Q9 | Edexcel 4MA1 • November 2021 • Paper 2H • Q1 | (a) $m = 9$ (b) $n = 21$ (c) $p = 4$ |
| Q10 | Edexcel 4MA1 • January 2022 • Paper 1H • Q1 | (a) a^{11} (b) w^{12} (c) $64x^{10}y^6$ (d) $t = \sqrt[3]{c + 8v}$ |
| Q11 | Edexcel 4MA1 • May 2023 • Paper 1HR • Q6 | (a) m^7 (b) $n = 8$ (c) $9x^{12}y^{16}$ |
| Q12 | Edexcel 4MA1 • November 2023 • Paper 2H • Q4 | (a) $x = 3.5$ (b) h^{12} (c) $16g^{12}k^{20}$ (d) $n = 14$ |
| Q13 | Edexcel 4MA1 • June 2025 • Paper 1H • Q4 | (a) $m = 12$ (b) $n = 5$ (c) $125a^{12}r^6$ |

IDEA 02 • ZERO AND NEGATIVE POWERS

- Q14** Dr Eslam • Zero and negative (a) 1 (b) $\frac{1}{8}$ (c) $\frac{9}{4}$
- Q15** Dr Eslam • Negative = flip x^{-4}
- Q16** Practice • Easy m^{-10}
- Q17** Practice • Easy 1
- Q18** Practice • Easy w
- Q19** Practice • Medium (a) $x = \pm 7$ (b) $\frac{1}{8}$ (c) $27x^6$ (d) $p = \frac{w + 16}{4}$
- Q20** Edexcel 4MA1 • January 2021 • Paper 2HR • Q4 (a) $81k^8$ (b) $7m^4n^6$
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q9 (a) 8 (b) $p = 11$ (c) $8k^6m^{12}$
- Q22** Edexcel 4MA1 • June 2023 • Paper 2HR • Q7 (a) 1 (b) $m = -6$
- Q23** Edexcel 4MA1 • May 2024 • Paper 1H • Q10 (a) 1 (b) $n = 6$ (c) $125a^{12}c^6$

IDEA 03 • FRACTIONAL POWERS

- Q24** Dr Eslam • Bottom is the root (a) 4 (b) $\frac{1}{3}$ (c) $\frac{27}{8}$
- Q25** Dr Eslam • Flip → Root → Power $\frac{16y^2}{x^4}$
- Q26** Practice • Medium $3x^4y^{\frac{3}{2}}$
- Q27** Practice • Hard (a) 1 (b) $\frac{5y}{8x^3}$
- Q28** Practice • Hard $\frac{y^{15}}{8w^6}$
- Q29** Practice • Hard (a) $8a^9$ (b) $1000x^3$
- Q30** Edexcel 4MA1 • January 2020 • Paper 1H • Q17 (a) $36 + 24\sqrt{12} + 48 = 84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$
(b) $\frac{t^{10}}{9a^8}$
- Q31** Edexcel 4MA1 • January 2021 • Paper 1HR • Q12 (a) $4e^5f^3$ (b) $\frac{10x-5}{12}$ (c) $k = -2$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2H • Q11 $4t^4w^2$
- Q33** Edexcel 4MA1 • May 2021 • Paper 2H • Q14 $\frac{4y^5}{3x^2}$
- Q34** Edexcel 4MA1 • June 2023 • Paper 2HR • Q11 $\frac{m^9k^{15}}{125}$
- Q35** Edexcel 4MA1 • June 2023 • Paper 2H • Q15 (a) $8a^9$ (b) $1000x^3$ (c) $y = 2.9$
- Q36** Edexcel 4MA1 • November 2023 • Paper 2H • Q12 (a) $x = 8.5$ (b) $25p^4y^{16}$
- Q37** Edexcel 4MA1 • May 2024 • Paper 1HR • Q17 (a) $n = 15$ (b) $\sqrt{147} + 14$
- Q38** Edexcel 4MA1 • June 2025 • Paper 2H • Q15 $125a^{12}c^6$
- Q39** Edexcel 4MA1 • November 2025 • Paper 2H • Q13 (a) $\frac{25x^4}{y^{10}}$ (b) $n = m - 3p$

IDEA 04 · SOLVING: UNIFY THE BASE

- Q40** Dr Eslam • Unify the base $x = 3$
- Q41** Dr Eslam • Both sides need changing (a) $x = 2$ (b) $x = \frac{5}{2}$
- Q42** Practice • Hard $x = \frac{29}{20}$
- Q43** Practice • Hard $m = -\frac{4}{3}$
- Q44** Practice • Hard $k = -2$
- Q45** Practice • Hard $n = -\frac{5}{2}$
- Q46** Edexcel 4MA1 • January 2020 • Paper 2HR • Q21 (a) $\frac{5x+4}{2x-3}$ (b) $n = 5y - 1$
- Q47** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q21 $n = 5$ or $n = -\frac{1}{3}$
- Q48** Edexcel 4MA1 • January 2021 • Paper 2HR • Q19 (a) 2^{10} (b) $n = 32$
- Q49** Edexcel 4MA1 • May 2021 • Paper 1H • Q11 (a) $x = 4.5$ (b) $n = -34$
- Q50** Edexcel 4MA1 • January 2022 • Paper 2HR • Q6 (a) 1 (b) $n = 6$ (c) $m = 5$
- Q51** Edexcel 4MA1 • June 2022 • Paper 2HR • Q7 $x = k - 2n$
- Q52** Edexcel 4MA1 • June 2022 • Paper 2H • Q24 $x = 2n - 6$
- Q53** Edexcel 4MA1 • May 2023 • Paper 1HR • Q24 $x = \sqrt{n+1} - 2, n > 3$
- Q54** Edexcel 4MA1 • June 2024 • Paper 2HR • Q7 $n = -7$
- Q55** Edexcel 4MA1 • June 2024 • Paper 2H • Q23 5^{2x+10}
- Q56** Edexcel 4MA1 • November 2024 • Paper 2H • Q14 $n = \frac{5}{4}$

IDEA 05 · WHERE YOU WILL USE IT

Q59 Dr Eslam · Differentiation:
rewrite first

$$(a) x^{-3} \quad (b) x^{\frac{1}{2}} \quad (c) 5x^{-\frac{2}{3}}$$

Q60 Dr Eslam · Surds are powers

$$x^{\frac{5}{2}}$$

DR ESLAM AHMED
eliteigcse.com

Expanding Brackets

Multiply every term by every term. Then collect. That is the whole topic — and you will use it in every algebra topic after this one.

THE TOPIC ON ONE MAP

Expanding Brackets

01 — One bracket

$3(2x - 5)$

$-2(x - 4)$

expand and simplify

02 — Two brackets

$(m - 8)(m + 5)$

the grid

03 — The special two

$(x + 3)^2$

$(a + b)(a - b)$

04 — Three brackets

$(x + 1)(x - 2)(x + 3)$

$ax^3 + bx^2 + cx + d$

05 — Where you will use it

show that ... = 0

single fraction

proof

Five ideas. After each idea you practise it straight away: my questions first, then selected practice, then real past papers.

01 One Bracket

The term outside multiplies **every** term inside — and the sign in front belongs to the term.

WHEN YOU SEE → YOU DO

“expand” / “multiply out”

→ outside term $\times$ every inside term

“expand and simplify”

→ expand, then collect like terms

“-(...)” or “-2(...)”

→ the minus changes every sign inside

$$-3 (2x - 5) = -6x + 15$$

1

Multiply

Outside term $\times$ each inside term. Take the sign with it.

2

Collect like terms

x terms together, numbers together. That is “simplify”.

WORKED MODEL • EXPAND AND SIMPLIFY

$$5(y - 2) - 2(y - 3)$$

$$\textcircled{1} \quad 5y - 10$$

first bracket

$$\textcircled{2} \quad -2y + 6$$

 $-2 \times -3 = +6$

$$\textcircled{3} \quad 5y - 2y - 10 + 6$$

line up like terms

$$\text{ANSWER} \quad 3y - 4$$

× **TRAP** A minus in front of a bracket changes **every** sign inside: $-(5c - 1) = -5c + 1$, not $-5c - 1$.

IDEA 01

Practise: One Bracket

My questions first, then selected practice.

Q1 **DR ESLAM** The sign goes with the term

CH 2 • TOPIC 3 2 marks

Expand (a) $-3(2x - 5)$ (b) $-x(4 - x)$

Q2 **DR ESLAM** Minus in front of a bracket

CH 2 • TOPIC 3 2 marks

Expand and simplify $4(x + 3) - 2(x - 5)$.

The -2 multiplies *both* terms: $-2 \times -5 = +10$.

Answer _____

Q3 **DR ESLAM** Letters outside

CH 2 · TOPIC 3 2 marks

Expand and simplify $2x(3x - 4) - x(x + 2)$.

Answer _____

SELECTED PRACTICE**Q4** **PRACTICE** Practice · Easy

CH 2 · TOPIC 3 1 mark

1 Expand $3(2y - 5)$

(1 mark)

Q5 **PRACTICE** Practice · Easy

CH 2 · TOPIC 3 1 mark

4 Expand $4(x + 2)$

(1 mark)

Q6**PRACTICE**

Practice · Easy

CH 2 · TOPIC 3

3 marks**7 (a)** Expand $x(x + 2)$ **(1 mark)****(b)** Expand and simplify $3(y + 2) + 4(x - 1)$ **(2 marks)****Q7****PRACTICE**

Practice · Easy

CH 2 · TOPIC 3

2 marks**11** Expand and simplify $3(x + 4) + 2(5x - 1)$ **(2 marks)**

Q8**PRACTICE**

Practice · Medium

CH 2 · TOPIC 3**2 marks**

1 Expand and simplify $3(m + 4) - 2(4m + 1)$

(2 marks)**Q9****PRACTICE**

Practice · Medium

CH 2 · TOPIC 3**2 marks**

16 Expand and simplify fully $4(2c + 3) - (5c - 1)$

(2 marks)

02 Two Brackets

Every term in the first bracket $\times$ every term in the second. Four products, then collect the middle two.

WHEN YOU SEE $\rightarrow$ YOU DO

$(\)(\)$

$\rightarrow$ grid: four products, collect the middle two

“circle the term which is part of the answer”

$\rightarrow$ only the middle terms are needed

$\times$	x	-4
$2x$	$2x^2$	$-8x$
$+3$	$+3x$	-12

$(2x + 3)(x - 4)$ — the gold cells are the like terms.

1

Fill the grid

Row $\times$ column in every cell.

2

Collect the gold cells

$$-8x + 3x = -5x$$

ANSWER $2x^2 - 5x - 12$

CALCULATOR CHECK Put $x = 2$ into the question and into your answer. $(7)(-2) = -14$ and $8 - 10 - 12 = -14$. Same number $\rightarrow$ correct. (On the calculator: type both with ALPHA X, press CALC, enter 2.)

$\times$ TRAP $(y + 4)(2 - y)$: the x^2 term is $-y^2$. Keep the order the question gives and watch the minus.

IDEA 02

Practise: Two Brackets

My questions first, then selected practice, then real past papers.

Q10 **DR ESLAM** Grid first

CH 2 · TOPIC 3 2 marks

Expand and simplify $(x + 5)(x - 2)$ using the grid.

×	x	+5
x		
-2		

Answer _____

Q11 **DR ESLAM** Numbers in front of x

CH 2 · TOPIC 3 2 marks

Expand and simplify $(3x - 2)(2x + 5)$.

Answer _____

SELECTED PRACTICE

Q12**PRACTICE**

Practice · Easy

CH 2 · TOPIC 3 1 mark

15 $(2x - 4)(3x + 5)$ is expanded and simplified.

Circle the term which is part of the answer.

$2x$	$-2x$	$22x$	$-22x$
------	-------	-------	--------

(1 mark)

Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 3 2 marks

6 Expand and simplify $(x - 5)(x + 3)$

(2 marks)

Q14 **PRACTICE** Practice · Medium

CH 2 · TOPIC 3 2 marks

9 Expand and simplify $(m + 7)(m + 3)$

(2 marks)

Q15 **PRACTICE** Practice · Medium

CH 2 · TOPIC 3 2 marks

17 Expand and simplify $(m + 9)(m - 5)$

(2 marks)

PAST PAPERS

Q16**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q1

CH 2 · TOPIC 3 8 marks**You must write down all the stages in your working.**

1 $w = 5y^2 - y^3$

(a) Work out the value of w when $y = -2$ $w =$ (b) Factorise fully $8p^2 - 2p$

.....

(c) Expand $4t(3t - 2)$

.....

(d) Expand and simplify $(5x - 2)(x + 4)$

.....

(Total for Question 1 is 8 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q1

CH 2 • TOPIC 3**3 marks**

You must write down all the stages in your working.

1 (a) Simplify $e^8 \div e^2$

(b) Expand and simplify $(x - 3)(x + 1)$

(Total for Question 1 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q1

CH 2 • TOPIC 3**4 marks**

You must write down all the stages in your working.

1 (a) Expand and simplify $(y + 4)(2 - y)$

(b) Factorise fully $15b^5c - 35b^3c^9$

(Total for Question 1 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q1

CH 2 · TOPIC 3 4 marks**You must write down all the stages in your working.**

1 $P = m^2 - 4c$

(a) Work out the value of P when $m = -5$ and $c = 3$

$P = \dots\dots\dots$

(b) Expand and simplify $(x + 5)(x - 7)$

$\dots\dots\dots$

(Total for Question 1 is 4 marks)

03 The Special Two

Two patterns come back everywhere. Learn them so well you write the answer in one line.

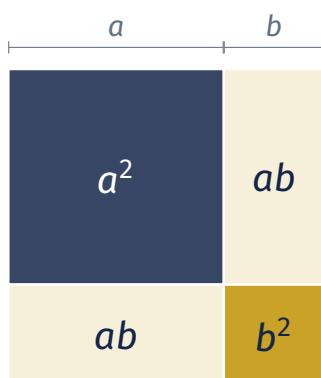
WHEN YOU SEE → YOU DO

"()²"

→ square · double · square

"(a + b)(a - b)", same numbers, opposite signs

→ the middle cancels: $a^2 - b^2$



$(a + b)^2$ is a square: a^2 , two ab strips, b^2 .

A

Perfect square

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Square the first · double the product · square the last.

B

Difference of two squares

$$(a + b)(a - b) = a^2 - b^2$$

Same numbers, opposite signs → the middle terms cancel.

WORKED MODEL • PERFECT SQUARE IN ONE LINE

1 $(2x - 3)^2$

square · double · square

2 $(2x)^2 - 2(2x)(3) + 3^2$

ANSWER $4x^2 - 12x + 9$

× **TRAP** $(x + 3)^2 \neq x^2 + 9$. Where is the middle term? $(x + 3)^2 = x^2 + 6x + 9$.

IDEA 03

Practise: The Special Two

My questions first, then selected practice.

Q20 DR ESLAM Perfect-square drill

CH 2 • TOPIC 3 5 marks

Square the first • double the product • square the last.

(a) $(x + 4)^2$ (b) $(x - 6)^2$ (c) $(2x + 3)^2$ (d) $(3x - 1)^2$ (e) $\left(x + \frac{1}{2}\right)^2$

Q21 DR ESLAM Spot the mistake

CH 2 • TOPIC 3 2 marks

Sara writes $(x + 3)^2 = x^2 + 9$.

Explain her mistake and write the correct expansion.

Correct _____

Q22 DR ESLAM Difference of two squares

CH 2 · TOPIC 3 2 marks

Expand (a) $(x + 7)(x - 7)$ (b) $(2x - 5)(2x + 5)$

Same numbers, opposite signs: the middle terms cancel.

SELECTED PRACTICE**Q23** PRACTICE Practice · Easy

CH 2 · TOPIC 3 2 marks

16 Multiply out and simplify $(x - 8)^2$

(2 marks)

Q24 PRACTICE Practice · Medium

CH 2 · TOPIC 3 2 marks

13 Expand and simplify $(w - 5)^2$

(2 marks)

04 Three Brackets

The most common exam question on this topic: 13 of the 17 past-paper questions here have three brackets. Two first, then the third.

WHEN YOU SEE → YOU DO

three brackets

→ two first, then the third — single term last

“in the form $ax^3 + bx^2 + cx + d$ ”

→ expand fully, highest power first

“≡” / “work out the values of a, b, c ”

→ compare coefficients

1

Expand two brackets

Pick the two easiest. If one bracket is a single term like $5x$, leave it for last.

2

Multiply by the third

Every term $\times$ every term again — usually 6 products.

3

Collect, highest power first

Write it as $ax^3 + bx^2 + cx + d$.

WORKED MODEL • JUNE 2025 P2H STYLE

$$3x(2x + 5)(7x - 4)$$

$$1 \quad (2x + 5)(7x - 4) = 14x^2 - 8x + 35x - 20$$

two brackets

$$2 \quad = 14x^2 + 27x - 20$$

collect

$$3 \quad 3x(14x^2 + 27x - 20)$$

single term last

ANSWER $42x^3 + 81x^2 - 60x$

“SHOW THAT” / IDENTITIES Expand the left side fully, then **compare coefficients**: x^3 with x^3 , x^2 with x^2 , and so on.

IDEA 04

Practise: Three Brackets

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Single term last

CH 2 · TOPIC 3 3 marks

Expand and simplify $2x(x + 3)(x - 1)$.

Brackets first, then multiply everything by $2x$.

Answer _____

Q26 DR ESLAM Two first, then the third

CH 2 · TOPIC 3 3 marks

Expand and simplify $(x + 1)(x + 2)(x - 3)$.

Answer _____

SELECTED PRACTICE

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 1 Show that $(x + 1)(x + 2)(x + 3)$ can be written in the form $ax^3 + bx^2 + cx + d$ where a, b, c and d are positive integers.

(3 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 1 mark

- 11 Show that $x(x - 1)(x + 1) = x^3 - x$

(1 mark)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 4 marks

15 Expand and simplify $(3x^2 + 2)(2x + 5) - 6x(x^2 - 3)$

(4 marks)**Q30****PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

18 Expand and simplify $(x + 6)(3x - 2)(x + 6)$

(3 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q14

CH 2 · TOPIC 3 3 marks

14 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

(Total for Question 14 is 3 marks)

Q32 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1HR · Q16

CH 2 · TOPIC 3 5 marks

16 (a) Expand and simplify $(x + 4)(x - 2)(x + 1)$ (b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

(Total for Question 16 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q12

CH 2 • TOPIC 3 6 marks

12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

(b) Expand and simplify $2x(x-5)(x-3)$

(Total for Question 12 is 6 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q13

CH 2 • TOPIC 3 3 marks

13 Expand and simplify $4x(3x+1)(2x-3)$
Show your working clearly.

(Total for Question 13 is 3 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q12

CH 2 • TOPIC 3 7 marks

12 (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$

.....

(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

.....

(c) Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly.

.....

(Total for Question 12 is 7 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q13

CH 2 • TOPIC 3 6 marks

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$

.....
(3)

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

.....
(Total for Question 13 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q16

CH 2 • TOPIC 3 6 marks

16 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$

.....
(3)

(b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$

.....
(Total for Question 16 is 6 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q16

CH 2 • TOPIC 3 3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....
(Total for Question 16 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q14

CH 2 • TOPIC 3 7 marks

- 14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(b) Solve $\frac{2x + 3}{5} + \frac{6x - 5}{4} = \frac{163}{100}$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 14 is 7 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q14

CH 2 • TOPIC 3 6 marks

14 (a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$

.....
(3)

(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

.....
(Total for Question 14 is 6 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q15

CH 2 • TOPIC 3 3 marks

15 Write $3x(2x - 1)(5x + 4)$ in the form $ax^3 + bx^2 + cx$ where a , b and c are integers.

.....
(Total for Question 15 is 3 marks)

Q42 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2HR • Q13**CH 2 • TOPIC 3** **5 marks**13 (a) Factorise $4x^2 - 25y^2$ (b) Show that $4x(x + 3)(2x - 5)$ can be written in the form $ax^3 + bx^2 + cx$ where a , b and c are integers to be found.**(Total for Question 13 is 5 marks)****Q43** **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2H • Q12**CH 2 • TOPIC 3** **3 marks**12 Expand and simplify $3x(2x + 5)(7x - 4)$ **(Total for Question 12 is 3 marks)**

Q44**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q14

CH 2 • TOPIC 3 6 marks

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$

(Total for Question 14 is 6 marks)

05 Where You Will Use It

This topic is a tool. Every one of these later topics starts by expanding brackets.

WHEN YOU SEE → YOU DO

“show that ... = 0”

→ expand both sides, rearrange to = 0

“as a single fraction”

→ expand the numerator

“prove”

→ expand, then factorise to prove

Solving quadratics

ERCS: Expand first.

$$x(x - 5) = 14$$

Algebraic fractions

Adding: expand the top.

$$3(x + 1) - 2(x - 4)$$

Expanding brackets

Completing the square

Check by expanding back.

$$(x + 3)^2 + 2 = x^2 + 6x + 11$$

Proof & functions

$$(n + 3)^2 - (n - 3)^2 = 12n$$

fg(x), area “show that”

WORKED MODEL • SOLVE AFTER EXPANDING

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$1 \quad 4x^2 + 20x + 25$$

perfect square

$$2 \quad 4x^2 + 4x - 3$$

two brackets

$$3 \quad 20x + 25 = 4x - 3$$

the $4x^2$ cancel

ANSWER $x = -\frac{7}{4}$

IDEA 05

Practise: Where You Will Use It

My questions first, then selected practice.

Q45 **DR ESLAM** Quadratics: the E of ERCS

CH 2 • TOPIC 3 2 marks

Show that $(x + 2)^2 = 3x + 10$ can be written as $x^2 + x - 6 = 0$.

Q46 **DR ESLAM** Algebraic fractions: the numerator

CH 2 • TOPIC 3 3 marks

Write $\frac{3}{x-4} - \frac{2}{x+1}$ as a single fraction.

The top becomes $3(x + 1) - 2(x - 4)$ — expand and simplify it.

Answer _____

Q47 DR ESLAM Proof

CH 2 • TOPIC 3 3 marks

Show that $(n + 3)^2 - (n - 3)^2 = 12n$ for all values of n .

SELECTED PRACTICE**Q48** PRACTICE Practice • Hard

CH 2 • TOPIC 3 3 marks

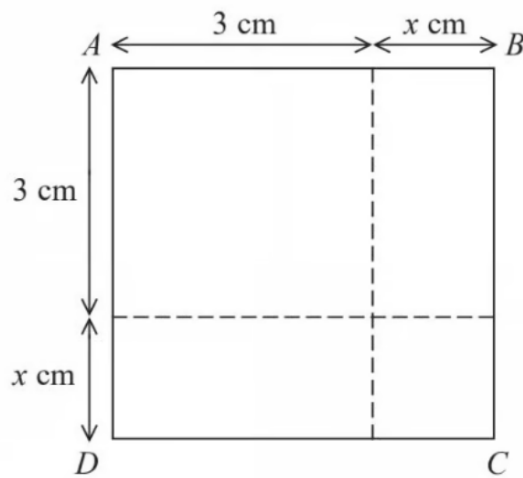
3 Simplify fully $(\sqrt{a} + \sqrt{4b})(\sqrt{a} - 2\sqrt{b})$

(3 marks)

Q49**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3

3 marks**5**

The area of square $ABCD$ is 10 cm^2 .

Show that $x^2 + 6x = 1$

(3 marks)

Q50**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 7 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer,

find the value of x and the value of y .

Show each stage of your working clearly.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(3 marks)

Q51**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 8 Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x = \dots\dots\dots$

(3 marks)

Q52**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3

3 marks

9 The functions f and g are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

Solve the equation $fg(x) = f(x)$

Show clear algebraic working.

(3 marks)**Q53****PRACTICE**

Practice · Hard

CH 2 · TOPIC 3

3 marks

13 Here is an identity.

$$a(3x - 10) \equiv 21x + 2b$$

Work out the values of a and b .

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 One bracket** Outside $\times$ every term inside. The sign goes with the term.
- 02 Two brackets** Four products (grid), collect the middle two. Check with $x = 2$.
- 03 Special two** $(a \pm b)^2 = a^2 \pm 2ab + b^2$ $(a + b)(a - b) = a^2 - b^2$
- 04 Three brackets** Two first, then the third. Single term last. $ax^3 + bx^2 + cx + d$.
- 05 Use it** Quadratics (ERCS), fractions, completing the square, proof.

BOTTOM LINE Every term $\times$ every term $\rightarrow$ collect like terms $\rightarrow$ check with $x = 2$.

Answers

Expanding Brackets • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ONE BRACKET

- | | | |
|-----------|--|----------------------------------|
| Q1 | Dr Eslam • The sign goes with the term | (a) $-6x + 15$ (b) $-4x + x^2$ |
| Q2 | Dr Eslam • Minus in front of a bracket | $2x + 22$ |
| Q3 | Dr Eslam • Letters outside | $5x^2 - 10x$ |
| Q4 | Practice • Easy | $6y - 15$ |
| Q5 | Practice • Easy | $4x + 8$ |
| Q6 | Practice • Easy | (a) $x^2 + 2x$ (b) $3y + 4x + 2$ |
| Q7 | Practice • Easy | $13x + 10$ |
| Q8 | Practice • Medium | $10 - 5m$ |
| Q9 | Practice • Medium | $3c + 13$ |

IDEA 02 • TWO BRACKETS

- Q10** Dr Eslam • Grid first $x^2 + 3x - 10$
- Q11** Dr Eslam • Numbers in front of x $6x^2 + 11x - 10$
- Q12** Practice • Easy $-2x$ (full: $6x^2 - 2x - 20$)
- Q13** Practice • Medium $x^2 - 2x - 15$
- Q14** Practice • Medium $m^2 + 10m + 21$
- Q15** Practice • Medium $m^2 + 4m - 45$
- Q16** Edexcel 4MA1 • January 2021 • Paper 2HR • Q1 (a) 28 (b) $2p(4p-1)$ (c) $12t^2-8t$ (d) $5x^2+18x-8$
- Q17** Edexcel 4MA1 • November 2021 • Paper 1H • Q1 (a) e^6 (b) $x^2 - 2x - 3$
- Q18** Edexcel 4MA1 • January 2022 • Paper 2H • Q1 (a) $8 - 2y - y^2$ (b) $5b^3c(3b^2 - 7c^8)$
- Q19** Edexcel 4MA1 • June 2023 • Paper 2HR • Q1 (a) 13 (b) $x^2 - 2x - 35$

IDEA 03 • THE SPECIAL TWO

- Q20** Dr Eslam • Perfect-square drill (a) $x^2+8x+16$ (b) $x^2-12x+36$ (c) $4x^2+12x+9$
(d) $9x^2-6x+1$ (e) $x^2+x+\frac{1}{4}$
- Q21** Dr Eslam • Spot the mistake She missed the middle term $2 \times x \times 3$: x^2+6x+9
- Q22** Dr Eslam • Difference of two squares (a) $x^2 - 49$ (b) $4x^2 - 25$
- Q23** Practice • Easy $x^2 - 16x + 64$
- Q24** Practice • Medium $w^2 - 10w + 25$

IDEA 04 • THREE BRACKETS

- Q25** Dr Eslam • Single term last $2x^3 + 4x^2 - 6x$
- Q26** Dr Eslam • Two first, then the third $x^3 - 7x - 6$
- Q27** Practice • Hard $x^3 + 6x^2 + 11x + 6$
- Q28** Practice • Hard $x(x^2 - 1) = x^3 - x$
- Q29** Practice • Hard $15x^2 + 22x + 10$
- Q30** Practice • Hard $3x^3 + 34x^2 + 84x - 72$
- Q31** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q14 $20x^3 - 31x^2 - 81x - 18$
- Q32** Edexcel 4MA1 • January 2021 • Paper 1HR • Q16 (a) $x^3 + 3x^2 - 6x - 8$ (b) $(x - 5)^2 + 15$
- Q33** Edexcel 4MA1 • January 2021 • Paper 1H • Q12 (a) $\frac{x + 10}{(x - 2)(x + 1)}$ (b) $2x^3 - 16x^2 + 30x$
- Q34** Edexcel 4MA1 • May 2021 • Paper 1H • Q13 $24x^3 - 28x^2 - 12x$
- Q35** Edexcel 4MA1 • November 2021 • Paper 2H • Q12 (a) $16p^6q^8$ (b) $\frac{17}{30x}$ (c) $8x^3 - 28x^2 - 60x$
- Q36** Edexcel 4MA1 • January 2022 • Paper 1H • Q13 (a) $15x^3 + 10x^2 - 40x$ (b) $\frac{y^{15}}{8w^6}$
- Q37** Edexcel 4MA1 • June 2022 • Paper 1HR • Q16 (a) $9x^3 + 18x^2 - x - 2$ (b) $\frac{16y^4}{x^8}$
- Q38** Edexcel 4MA1 • November 2023 • Paper 2H • Q16 $2x^3 + x^2 - 43x - 60$
- Q39** Edexcel 4MA1 • June 2024 • Paper 2H • Q14 (a) $6y^3 + 57y^2 + 105y$ (b) $x = 1.2$
- Q40** Edexcel 4MA1 • May 2024 • Paper 1HR • Q14 (a) $-3x^3 - 7x^2 + 22x + 8$ (b) a^3b^2
- Q41** Edexcel 4MA1 • November 2024 $30x^3 + 9x^2 - 12x$

IDEA 05 • WHERE YOU WILL USE IT

Q45 Dr Eslam • Quadratics: the E of ERCS

$$x^2 + 4x + 4 = 3x + 10 \Rightarrow x^2 + x - 6 = 0$$

Q46 Dr Eslam • Algebraic fractions: the numerator

$$\frac{x + 11}{(x - 4)(x + 1)}$$

Q47 Dr Eslam • Proof

$$n^2 + 6n + 9 - (n^2 - 6n + 9) = 12n$$

Q48 Practice • Hard

$$a - 4b$$

Q49 Practice • Hard

$$(x + 3)^2 = 10 \Rightarrow x^2 + 6x = 1$$

Q50 Practice • Hard

$$40 + 3\sqrt{x} - x: x = 19, y = 3$$

Q51 Practice • Hard

$$x = -\frac{7}{4}$$

Q52 Practice • Hard

$$(x - 10)^2 + 6 = x^2 + 6 \Rightarrow x = 5$$

Q53 Practice • Hard

$$a = 7, b = -35$$

DR ESLAM AHMED
eliteigcse.com

Factorisation

Factorising is expanding backwards. Take out the HCF, then count the terms — the number of terms tells you the method.

THE TOPIC ON ONE MAP

Factorisation

01 — Count the terms

 2 terms 3 terms 4 terms

02 — Two terms: the HCF

 $12pq - 18p$

"factorise fully"

03 — Two terms: DOTS

 $x^2 - 49$ $2x^2 - 18$

04 — Three terms: the CC Rule

 $x^2 + 2x - 24$ $2x^2 - x - 15$

hence solve

05 — Four terms: grouping

 $ax + ay + bx + by$

00 Count the Terms

Before anything: take out the **HCF**. Then count what is left — the count chooses the method.

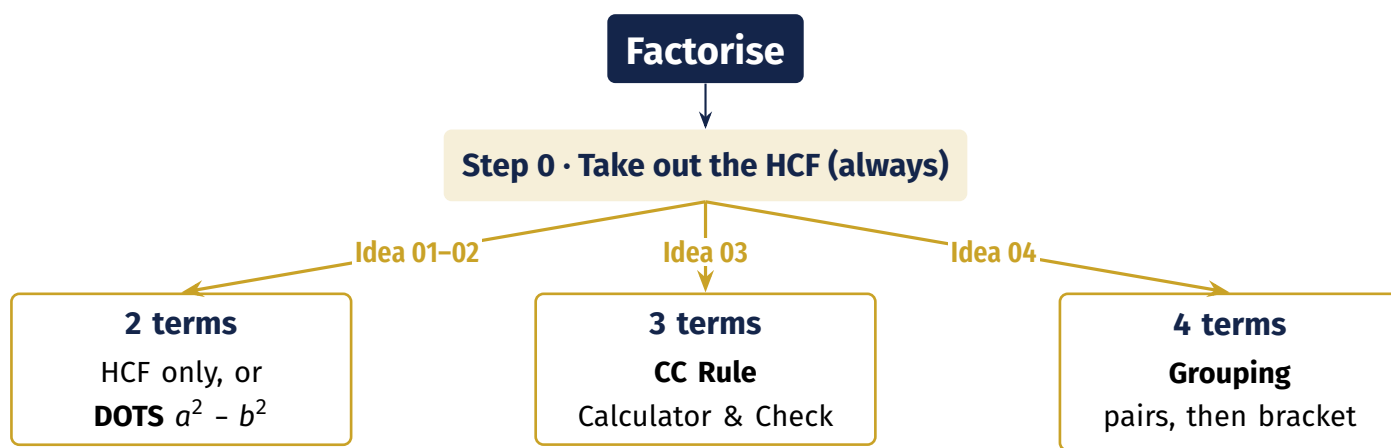
WHEN YOU SEE → YOU DO

“factorise fully” / “completely”

→ take out the biggest HCF first

count the terms

→ 2 · 3 · 4 terms → the method



BOTTOM LINE Factorising is the **first step** of algebraic fractions and of solving quadratics — and it comes on its own as a 1–3 mark question in almost every paper.

01 Two Terms: the HCF

Find what **every** term shares — the biggest number and the lowest power of each letter — and put it outside.

WHEN YOU SEE → YOU DO

“factorise fully” + two terms

→ biggest number, lowest power of each letter

“explain why ... is not correct”

→ something common is still inside the bracket

1

Numbers

HCF of the numbers: 16 and 20 → 4.

2

Letters

Each letter in **every** term, at its **lowest** power: a^2b^3 and $a^3b \rightarrow a^2b$.

3

Divide to fill the bracket

Each term $\div$ HCF. Then expand back in your head to check.

WORKED MODEL • FACTORISE FULLY

$$16a^2b^3 + 20a^3b$$

1 HCF = $4a^2b$

number • letters

2 $\frac{16a^2b^3}{4a^2b} = 4b^2$ $\frac{20a^3b}{4a^2b} = 5a$

divide each term

ANSWER $4a^2b(4b^2 + 5a)$

× **TRAP** “Fully” means the **biggest** HCF. $2x^2 + 6x = x(2x + 6)$ is not finished — 2 is still common: $2x(x + 3)$.

IDEA 01

Practise: Two Terms: the HCF

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Numbers and letters

CH 2 · TOPIC 4 2 marks

Factorise fully $15a^2b + 10ab^2$.

Answer _____

Q2 DR ESLAM Lowest power of each letter

CH 2 · TOPIC 4 2 marks

Factorise fully $12x^3y - 18x^2y^2$.

Lowest power of x: x^2 . Lowest power of y: y.

Answer _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 4 1 mark

1 Factorise $y^2 - 2y$

(1 mark)

Q4**PRACTICE**

Practice · Easy

CH 2 · TOPIC 4 1 mark

8 Factorise $y^2 + 27y$

(1 mark)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 2 marks

13 Factorise completely $6y^2 - 9xy$

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks

7 Factorise fully $25a^4c^7d + 45a^9c^3h$

(2 marks)

PAST PAPERS**Q7****PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q9

CH 2 · TOPIC 4 5 marks

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

 $x =$

(Total for Question 9 is 5 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q10

CH 2 · TOPIC 4 7 marks10 (a) Simplify $(6m)^0$ (b) Solve $5y + 20 < 7y + 1$ (c) Factorise fully $15w^2x^5 + 25w^3x^2$ (d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$ **(Total for Question 10 is 7 marks)**

02 Two Terms: Difference of Two Squares

A minus sign between two perfect squares. It factorises in one line — and the HCF comes out first.

WHEN YOU SEE → YOU DO

two squares with a minus

→ $(a + b)(a - b)$

"...² - ..." after an HCF

→ HCF first, then DOTS

?

Is it DOTS?

1. A minus sign in the middle
2. First term is a square
3. Last term is a square

$$a^2 - b^2 = (a + b)(a - b)$$

$\sqrt{\text{first}}$ and $\sqrt{\text{last}}$, + and -

WORKED MODEL • HCF, THEN DOTS

$$50g^2 - 18$$

$$1 \quad 2(25g^2 - 9)$$

HCF first

$$2 \quad \sqrt{25g^2} = 5g \quad \sqrt{9} = 3$$

roots

ANSWER $2(5g + 3)(5g - 3)$

× **TRAP** $x^2 + 9$ does **not** factorise — DOTS needs a **minus**.

CALCULATOR CHECK Put $g = 2$ into both: $50(4) - 18 = 182$ and $2(13)(7) = 182$. Same number → correct.

IDEA 02

Practise: Two Terms: DOTS

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM HCF first, then DOTS

CH 2 · TOPIC 4 3 marks

Factorise fully (a) $5x^2 - 20$ (b) $12x^2 - 27$ (c) $2x^3 - 8x$

Q10 DR ESLAM Is it DOTS?

CH 2 · TOPIC 4 1 mark

Which of these can be factorised as a difference of two squares? Explain.

(a) $x^2 - 36$ (b) $x^2 + 36$ (c) $9y^2 - 1$

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 1 mark

6 Factorise $x^2 - 49$

(1 mark)

Q12**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks

1 Factorise fully $20x^2 - 5$

(2 marks)

Q13**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 4 marks

4 (a) Factorise $a^2 - b^2$

(1 mark)

(b) Hence, or otherwise, simplify fully $(x^2 + 4)^2 - (x^2 - 2)^2$

(3 marks)

Q14**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 1 mark

9 Factorise $4c^2 - 9d^2$

(1 mark)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q11

CH 2 · TOPIC 4 5 marks

11 (a) Factorise $9x^2 - 4y^2$

(2)

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

(Total for Question 11 is 5 marks)

Q16 PAST PAPER Edexcel 4MA1 · May 2023 · Paper 1H · Q14

CH 2 · TOPIC 4 3 marks

14 Factorise fully $50g^2 - 18$

(Total for Question 14 is 3 marks)

03 Three Terms: the CC Rule

Calculator & Check. The calculator gives you the roots; you turn the roots into brackets.

WHEN YOU SEE → YOU DO

“factorise $x^2 + bx + c$ ”

→ CC Rule: calculator roots → brackets

“(i) factorise (ii) hence solve”

→ each bracket = 0 — the calculator roots

a root like $\frac{3}{2}$

→ the bracket $(2x - 3)$

C1

Calculator

MODE → 5 → 3 (or MENU → Equation → Polynomial → 2). Enter a , b , c .
Read the two roots.

Root is a whole number $x = 5$ → $(x - 5)$ flip the sign

Root is a fraction $x = \frac{3}{2}$ → $(2x - 3)$ bottom goes with x

C2

Check

Multiply the first terms of your brackets. Do they make the x^2 term?
If not, put the missing number in front.

WORKED MODEL · THE CHECK STEP

$$4x^2 - 10x + 6$$

1 roots $x = \frac{3}{2}, x = 1$

calculator

2 $(2x - 3)(x - 1)$

roots → brackets

3 $2x \cdot x = 2x^2$, but we need $4x^2$

check

ANSWER $2(2x - 3)(x - 1)$

“HENCE SOLVE” Each bracket = 0 — the answers are the roots the calculator already gave you.

IDEA 03

Practise: Three Terms: the CC Rule

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Whole-number roots

CH 2 · TOPIC 4 2 marks

Factorise $x^2 + x - 12$.

Calculator roots: → flip the signs.

Answer _____

Q18 DR ESLAM Fraction roots

CH 2 · TOPIC 4 2 marks

Factorise (a) $2x^2 + 7x + 3$ (b) $6x^2 - x - 2$

A root of $\frac{2}{3}$ gives the bracket $(3x - 2)$.

Q19 **DR ESLAM** The Check step

CH 2 · TOPIC 4 2 marks

Factorise fully $4x^2 - 10x + 6$.

After the brackets, multiply the first terms. Is it $4x^2$?

Answer _____

SELECTED PRACTICE**Q20** **PRACTICE** Practice · Medium

CH 2 · TOPIC 4 2 marks

1 Factorise $x^2 + 3x - 10$

(2 marks)

Q21 **PRACTICE** Practice · Medium

CH 2 · TOPIC 4 2 marks

5 Factorise $x^2 + 3x - 4$

(2 marks)

Q22**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 2 marks

16 Factorise $x^2 - 11x + 24$

(2 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks

12 Factorise $3x^2 + 14x + 8$
(2 marks)

PAST PAPERS

Q24**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q7

CH 2 • TOPIC 4 • 8 marks

- 7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

 $x = \dots\dots\dots$

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

 $\dots\dots\dots$

- (c) (i) Factorise $y^2 - 2y - 48$

 $\dots\dots\dots$

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

 $\dots\dots\dots$ **(Total for Question 7 is 8 marks)**

Q25**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q9

CH 2 • TOPIC 4 5 marks

9 (a) Simplify $(2x^3y^5)^4$

.....

(b) (i) Factorise $x^2 + 5x - 36$

.....

(ii) Hence, solve $x^2 + 5x - 36 = 0$

.....

(Total for Question 9 is 5 marks)**Q26****PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q9

CH 2 • TOPIC 4 3 marks

9 (i) Factorise $x^2 + 2x - 24$

.....

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....

(Total for Question 9 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q8

CH 2 · TOPIC 4 4 marks8 (a) Simplify $(3x^2y)^0$

.....

(b) (i) Factorise $x^2 - 5x - 36$ (ii) Hence solve $x^2 - 5x - 36 = 0$

.....

(Total for Question 8 is 4 marks)**Q28****PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q9

CH 2 · TOPIC 4 5 marks9 (a) Simplify $x^4 \times x^5$

.....

(b) Simplify $(4y^2)^3$

.....

(c) Factorise $n^2 - 7n + 12$

.....

(Total for Question 9 is 5 marks)

Q29**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q6

CH 2 · TOPIC 4 5 marks

6 (a) Solve the inequality $5x - 7 \leq 2$

.....

(b) (i) Factorise $y^2 - 2y - 35$

.....

(ii) Hence, solve $y^2 - 2y - 35 = 0$

.....

(Total for Question 6 is 5 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q5

CH 2 · TOPIC 4 5 marks

5 (a) Expand $3c^3(c + 4)$

.....

(b) (i) Factorise $x^2 + 8x - 9$

.....

(ii) Hence, solve $x^2 + 8x - 9 = 0$

.....

(Total for Question 5 is 5 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2H · Q11

CH 2 · TOPIC 4 3 marks

11 (i) Factorise $x^2 + 5x - 24$

.....

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....

(Total for Question 11 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q7

CH 2 • TOPIC 4 3 marks7 (a) Expand $4x(x - 5)$ (b) Factorise $y^2 - 9y + 20$ **(Total for Question 7 is 3 marks)**

Q33**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1H · Q6

CH 2 · TOPIC 4 · 8 marks6 (a) Simplify $(2c^4d^7)^3$

.....

(b) Find the value of $5y^0$ where $y > 0$

.....

(c) Factorise fully $16a^2b^3 + 20a^3b$

.....

(d) (i) Factorise $x^2 + 9x - 22$

.....

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....

(Total for Question 6 is 8 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2H • Q11**CH 2 • TOPIC 4** **3 marks**

11 (i) Factorise $x^2 + 9x - 22$

(ii) Hence, solve $x^2 + 9x - 22 = 0$

(Total for Question 11 is 3 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q7

CH 2 • TOPIC 4 9 marks7 (a) Simplify $g^9 \div g^2$

.....

(b) Expand $5k^2(k^3 + 4)$

.....

(c) (i) Factorise $x^2 - 2x - 63$

.....

(ii) Hence, solve $x^2 - 2x - 63 = 0$

.....

(d) Solve the inequality $7 - 2y < 3y - 12$

.....

(Total for Question 7 is 9 marks)

Q36**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q8

CH 2 · TOPIC 4 6 marks

8 (a) (i) Factorise $x^2 + 5x - 24$

.....

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

.....

(Total for Question 8 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q8

CH 2 · TOPIC 4 5 marks8 (a) Simplify $a^6 \times a^{10}$

.....

(b) Simplify $c^{30} \div c^{12}$

.....

(c) (i) Factorise $y^2 - 10y + 21$

.....

(ii) Hence, solve $y^2 - 10y + 21 = 0$

.....

(Total for Question 8 is 5 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q6

CH 2 · TOPIC 4 6 marks

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(b) (i) Factorise $y^2 - 11y + 30$

.....

(ii) Hence solve $y^2 - 11y + 30 = 0$

.....

(Total for Question 6 is 6 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q7

CH 2 · TOPIC 4 3 marks

7 (i) Factorise $x^2 + 2x - 48$

.....

(ii) Hence, solve $x^2 + 2x - 48 = 0$

.....

(Total for Question 7 is 3 marks)

04 Four Terms: Grouping

Two pairs. Factorise each pair — the same bracket appears twice. Take it out.

WHEN YOU SEE → YOU DO

four terms

→ group in pairs → the same bracket twice

1

Pair up

First two terms, last two terms.

2

HCF of each pair

The two brackets must come out **identical**. If not, swap the middle terms.

3

Take the bracket out

The repeated bracket is the common factor.

WORKED MODEL • GROUPING

$$xy + 3x + 2y + 6$$

1 $x(y + 3) + 2(y + 3)$

HCF of each pair

ANSWER $(y + 3)(x + 2)$

IDEA 04

Practise: Four Terms: Grouping

Questions written to practise exactly this idea.

Q40 DR ESLAM Grouping

CH 2 · TOPIC 4 2 marks

Factorise $xy + 3x + 2y + 6$.

Answer _____

Q41 DR ESLAM Grouping with minus signs

CH 2 · TOPIC 4 2 marks

Factorise $ax - ay + bx - by$.

Answer _____

05 Where You Will Use It

Factorising is the tool behind four later topics. Each one starts with “factorise”.

WHEN YOU SEE → YOU DO

“simplify fully” (a fraction)

→ factorise top and bottom, then cancel

“solve” a quadratic with clean roots

→ factorise

“prove ... multiple of”

→ factorise to prove

Algebraic fractions

Golden rule: factorise everything, then cancel.

Solving quadratics

Clean answers on the calculator → factorise.

Factorising

Algebraic proof

Factorise to prove: $12n = 2(6n)$ is even.

Rearranging formulas

Subject twice → take it out as a factor.

WORKED MODEL • ALGEBRAIC FRACTIONS

$$\frac{x^2 - 9}{x^2 + 5x + 6}$$

$$\textcircled{1} \quad \frac{(x+3)(x-3)}{(x+2)(x+3)}$$

factorise top and bottom

$$\textcircled{2} \quad \text{cancel } (x+3)$$

same bracket

ANSWER $\frac{x-3}{x+2}$

IDEA 05

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q42 DR ESLAM Algebraic fractions

CH 2 · TOPIC 4 3 marks

Simplify fully $\frac{x^2 + 4x}{x^2 - 16}$.

Factorise the top (HCF) and the bottom (DOTS), then cancel.

Answer _____

Q43 DR ESLAM Solving quadratics

CH 2 · TOPIC 4 2 marks

Solve $x^2 + x - 12 = 0$.

You factorised it in Idea 03: $(x + 4)(x - 3)$. Each bracket = 0.

$x =$ _____

Q44 DR ESLAM Proof: factorise to prove

CH 2 · TOPIC 4 3 marks

Show that $(n + 2)^2 - (n - 2)^2$ is a multiple of 8 for all integers n .

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 00 | Always first | Take out the HCF. Then count the terms. |
| 01 | 2 terms • HCF | Biggest number, lowest power of each letter.
“Fully” = biggest HCF. |
| 02 | 2 terms • DOTS | minus + two squares: $a^2 - b^2 = (a + b)(a - b)$ |
| 03 | 3 terms • CC Rule | Calculator → roots → brackets (flip sign; bottom goes with x) → Check the x^2 . |
| 04 | 4 terms | Group in pairs → same bracket → take it out. |
| 05 | Use it | Fractions (factorise first), quadratics, proof, rearranging. |

BOTTOM LINE HCF first → count the terms → 2: DOTS • 3: CC Rule • 4: grouping → expand back to check.

Answers

Factorisation • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TWO TERMS: THE HCF

- | | | |
|-----------|---|--|
| Q1 | Dr Eslam • Numbers and letters | $5ab(3a + 2b)$ |
| Q2 | Dr Eslam • Lowest power of each letter | $6x^2y(2x - 3y)$ |
| Q3 | Practice • Easy | $y(y - 2)$ |
| Q4 | Practice • Easy | $y(y + 27)$ |
| Q5 | Practice • Medium | $3y(2y - 3x)$ |
| Q6 | Practice • Hard | $5a^4c^3(5c^4d + 9a^5h)$ |
| Q7 | Edexcel 4MA1 • May 2021 • Paper 1H • Q9 | (a) $5a^4c^3(5c^4d + 9a^5h)$ (b) $x = -\frac{7}{4}$ |
| Q8 | Edexcel 4MA1 • November 2025 • Paper 1H • Q10 | (a) 1 (b) $y > 9.5$ (c) $5w^2x^2(3x^3 + 5w)$ (d) $3a^3c^4$ |

IDEA 02 • TWO TERMS: DOTS

- Q9** Dr Eslam • HCF first, then DOTS (a) $5(x + 2)(x - 2)$ (b) $3(2x + 3)(2x - 3)$ (c) $2x(x + 2)(x - 2)$
- Q10** Dr Eslam • Is it DOTS? (a) yes $(x + 6)(x - 6)$ (b) no — no minus (c) yes $(3y + 1)(3y - 1)$
 $(x + 7)(x - 7)$
- Q11** Practice • Medium $5(2x + 1)(2x - 1)$
- Q12** Practice • Hard (a) $(a + b)(a - b)$ (b) $(2x^2 + 2)(6) = 12(x^2 + 1)$
- Q13** Practice • Hard $(2c + 3d)(2c - 3d)$
- Q14** Practice • Hard (a) $(3x + 2y)(3x - 2y)$ (b) $\frac{5x - 6}{8x}$
- Q15** Edexcel 4MA1 • June 2022 • Paper 1H • Q11 $2(5g + 3)(5g - 3)$
- Q16** Edexcel 4MA1 • May 2023 • Paper 1H • Q14

IDEA 03 • THREE TERMS: THE CC RULE

- Q17** Dr Eslam • Whole-number roots $(x + 4)(x - 3)$
- Q18** Dr Eslam • Fraction roots (a) $(2x + 1)(x + 3)$ (b) $(3x - 2)(2x + 1)$
- Q19** Dr Eslam • The Check step $2(2x - 3)(x - 1)$
- Q20** Practice • Medium $(x + 5)(x - 2)$
- Q21** Practice • Medium $(x + 4)(x - 1)$
- Q22** Practice • Medium $(x - 3)(x - 8)$
- Q23** Practice • Hard $(3x + 2)(x + 4)$
- Q24** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q7 (a) $x = 6.5$ (b) $8m^2g^3(2m + 3g^2)$ (c) $(y - 8)(y + 6); y = 8, -6$
- Q25** Edexcel 4MA1 • January 2021 • Paper 1H • Q9 (a) $16x^{12}y^{20}$ (b) $(x + 9)(x - 4); x = -9, 4$
- Q26** Edexcel 4MA1 • May 2021 • Paper 2H • Q9 $(x + 6)(x - 4); x = -6, 4$
- Q27** Edexcel 4MA1 • November 2021 • Paper 2H • Q8 (a) 1 (b) $(x - 9)(x + 4); x = 9, -4$
- Q28** Edexcel 4MA1 • January 2022 • Paper 1HR • Q9 (a) x^9 (b) $64y^6$ (c) $(n - 3)(n - 4)$
- Q29** Edexcel 4MA1 • January 2022 • Paper 1H • Q6 (a) $x \leq \frac{9}{5}$ (b) $(y - 7)(y + 5); y = 7, -5$
- Q30** Edexcel 4MA1 • June 2022 • Paper 1HR • Q5 (a) $3c^4 + 12c^3$ (b) $(x + 9)(x - 1); x = -9, 1$
- Q31** Edexcel 4MA1 • June 2022 • Paper 2H • Q11 $(x + 8)(x - 3); x = -8, 3$
- Q32** Edexcel 4MA1 • May 2023 • Paper 1HR • Q7 (a) $4x^2 - 20x$ (b) $(y - 4)(y - 5)$
- Q33** Edexcel 4MA1 • May 2023 • Paper 1H • Q6 (a) $8c^{12}d^{21}$ (b) 5 (c) $4a^2b(4b^2 + 5a)$ (d) $(x + 11)(x - 2); x = -11, 2$
- Q34** Edexcel 4MA1 • June 2024 • Paper 1H • Q6 (a) $(x + 11)(x - 2); x = -11, 2$

IDEA 04 • FOUR TERMS: GROUPING**Q40** Dr Eslam • Grouping

$$(x + 2)(y + 3)$$

Q41 Dr Eslam • Grouping with minus signs

$$(a + b)(x - y)$$

IDEA 05 • WHERE YOU WILL USE IT**Q42** Dr Eslam • Algebraic fractions

$$\frac{x}{x - 4}$$

Q43 Dr Eslam • Solving quadratics

$$x = -4 \text{ or } x = 3$$

Q44 Dr Eslam • Proof: factorise to prove

$$(n^2 + 4n + 4) - (n^2 - 4n + 4) = 8n, \text{ and } 8n = 8 \times n$$

Completing the Square

One structure. Fill the blanks. Read the turning point straight off it.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Completing the Square

01 – The two-step method

$$(x + p)^2 + q$$

$$a(x + b)^2 + c$$

02 – Negative a

$$28 + 24x - 6x^2$$

$$a - b(x - c)^2$$

03 – Read the turning point

minimum / maximum

line of symmetry

sketch

04 – The “hence” questions

hence solve

surd form

transformations

Four ideas. After each idea you practise it straight away: my questions first, then selected practice, then real past papers.

01 The Two-Step Method

Any quadratic $ax^2 + bx + c$ — list four numbers, then drop them into one structure.

WHEN YOU SEE → YOU DO

“in the form $(x + p)^2 + q$ ”

→ complete the square

“express ... in the form $a(x + b)^2 + c$ ”

→ list $a, b, c, \frac{b}{2a}$ → structure

“where a, b and c are integers”

→ match the letters to your answer, not to the question

1

List your elements

$$a = ?$$

$$b = ?$$

$$c = ?$$

$$\frac{b}{2a} = ?$$

Always write $ax^2 + bx + c$ in this order first, even when the question writes it backwards.

2

Use the structure

Write the same structure every time — the **same** $\frac{b}{2a}$ goes in **both** brackets.

$$\underbrace{a\left(x + \frac{b}{2a}\right)^2}_{\text{keep it as it is}} \quad - \underbrace{a\left(\frac{b}{2a}\right)^2 + c}_{\text{one number: use the calculator}}$$

WORKED MODEL • $a = 1$ Write $x^2 - 6x + 11$ in the form $(x - p)^2 + q$.

$$1 \quad a = 1 \quad b = -6 \quad c = 11 \quad \frac{b}{2a} = \frac{-6}{2} = -3$$

list

$$2 \quad (x - 3)^2 - 1(-3)^2 + 11$$

structure

$$3 \quad (x - 3)^2 - 9 + 11$$

calculator: right part only

$$\text{ANSWER} \quad (x - 3)^2 + 2$$

WORKED MODEL • $a \neq 1$ Write $2x^2 + 8x + 5$ in the form $a(x + b)^2 + c$.

$$1 \quad a = 2 \quad b = 8 \quad c = 5 \quad \frac{b}{2a} = \frac{8}{4} = 2$$

list

$$2 \quad 2(x + 2)^2 - 2(2)^2 + 5$$

structure

$$3 \quad 2(x + 2)^2 - 8 + 5$$

calculator: right part only

$$\text{ANSWER} \quad 2(x + 2)^2 - 3$$

× **TRAP** Writing $-(2)^2$ instead of $-a(2)^2$. The a multiplies the second bracket too.

× **TRAP** The question's letters are not your a, b, c . In " $a(x + b)^2 + c$ ", the answer's b is $\frac{b}{2a}$ — read the form, then match.

IDEA 01

Practise: The Two-Step Method

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Step 1 first · $a = 1$

CH 2 · TOPIC 5 2 marks

Write $x^2 + 6x + 11$ in the form $(x + p)^2 + q$.

List first: $a =$ $b =$ $c =$ $\frac{b}{2a} =$

Answer _____

Q2 DR ESLAM $a \neq 1$ · the trap

CH 2 · TOPIC 5 3 marks

Write $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$.

The second bracket is multiplied by a too.

Answer _____

SELECTED PRACTICE

Q3

PRACTICE

Practice · Medium

CH 2 · TOPIC 5 2 marks

1 Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where a and b are integers.

(2 marks)

Q4

PRACTICE

Practice · Medium

CH 2 · TOPIC 5 3 marks

6 Write $x^2 - 6x + 11$ in the form $(x - a)^2 + b$

(3 marks)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 5 3 marks

10

$$x^2 + 4x - 9 = (x + a)^2 + b$$

Find the value of a and the value of b .

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ **(3 marks)****Q6****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 3 marks

7 $2x^2 - 6x + 5$ can be written in the form $a(x - b)^2 + c$

where a , b and c are positive numbers. Work out the values of a , b and c .

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$ **(3 marks)**

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q19

CH 2 · TOPIC 5 3 marks

19 Given that $\left(\sqrt[3]{\frac{1}{x}}\right)^4 = x^m$

(a) find the value of m

$m = \dots\dots\dots$

Given that a , b and c are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

$\dots\dots\dots$

(Total for Question 19 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q19

CH 2 · TOPIC 5 3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

$\dots\dots\dots$

(Total for Question 19 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q21

CH 2 · TOPIC 5 3 marks

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

.....

(Total for Question 21 is 3 marks)

02 Negative a ? Same structure.

No new method. Rewrite in the order $ax^2 + bx + c$ and carry the sign of a everywhere.

WHEN YOU SEE → YOU DO

“in the form $a - b(x - c)^2$ ”

→ negative a : same structure, write the number first

“ $28 + 24x - 6x^2$ ”

→ reorder to $-6x^2 + 24x + 28$ first

1

Reorder

$28 + 24x - 6x^2 \rightarrow -6x^2 + 24x + 28$, so $a = -6$, $b = 24$, $c = 28$.

2

Same structure, sign of a included

$-a(\dots)^2$ with $a = -6$ becomes $+6(\dots)^2$ — the minus-minus turns into plus.

EXAM FORM $a - b(x - c)^2$ • JUNE 2025 P2H

1 $\frac{b}{2a} = \frac{24}{-12} = -2$

list

2 $-6(x - 2)^2 - (-6)(-2)^2 + 28$

structure

3 $-6(x - 2)^2 + 24 + 28$

calculator

ANSWER $52 - 6(x - 2)^2$ so $a = 52$, $b = 6$, $c = 2$

× **TRAP** Answering in the wrong order. The form asks for “number first”: write $52 - 6(x - 2)^2$, not $-6(x - 2)^2 + 52$.

IDEA 02

Practise: Negative a

My questions first, then selected practice, then real past papers.

Q10 DR ESLAM Negative $a = -1$

CH 2 · TOPIC 5 3 marks

Write $10 - 6x - x^2$ in the form $p - (x + q)^2$.

Reorder first: $-x^2 - 6x + 10$, so $a = -1$.

Answer _____

Q11 DR ESLAM Negative $a \neq -1$

CH 2 · TOPIC 5 3 marks

Write $5 + 8x - 2x^2$ in the form $a - b(x - c)^2$.

Answer _____

SELECTED PRACTICE

Q12 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 5 2 marks

4 The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leq 3$

Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

(2 marks)

Q13 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 5 4 marks

5 Write $5 + 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(4 marks)

Q14**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 2 marks

6 Express $7 - 4x - x^2$ in the form $p - (x + q)^2$ where p and q are constants.

(2 marks)

PAST PAPERS**Q15****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q23

CH 2 · TOPIC 5 3 marks

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(Total for Question 23 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q20

CH 2 · TOPIC 5 4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

C is the curve with equation $y = 7 + 12x - 3x^2$

The point A is the maximum point on C

(b) Use your answer to part (a) to write down the coordinates of A

(.....,)

(Total for Question 20 is 4 marks)

Q17 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q24

CH 2 · TOPIC 5 4 marks

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Q18 PAST PAPER Edexcel 4MA1 · May 2024 · Paper 1H · Q25

CH 2 · TOPIC 5 4 marks

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$$f(x) = \dots\dots\dots$$

(Total for Question 25 is 4 marks)

03 Read the Turning Point

From $a(x + p)^2 + q$ the turning point is $(-p, q)$ — **flip** the inside, **keep** the outside.

WHEN YOU SEE → YOU DO

“turning point”, “minimum point”, “vertex”

→ flip the inside, keep the outside

“maximum point”

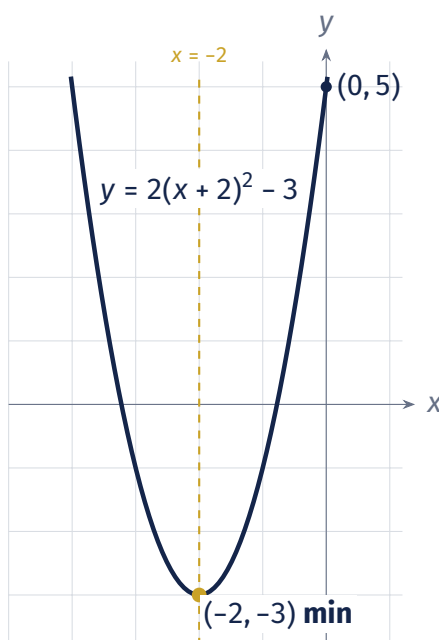
→ $a < 0$ — cap shape

“line of symmetry”

→ x = the flipped inside number

“sketch ... show the coordinates”

→ turning point + y-cut + shape



Inside $+2 \rightarrow \text{flip} \rightarrow x = -2$

Outside $-3 \rightarrow \text{keep} \rightarrow y = -3$

Shape $a = 2 > 0 \rightarrow \text{minimum}$

Symmetry the line $x = -2$

y-cut $x = 0$ in the original: 5

$a > 0$: cup, minimum. $a < 0$: cap, maximum.



Sketch questions

Mark three things and nothing else: the turning point, the y-cut, and the right shape (cup or cap).

CALCULATOR CHECK On a ClassWiz (fx-991EX / CW): MENU → Equation → Polynomial → 2, enter a, b, c and keep pressing = past the roots — it shows the *x-Value Minimum / Maximum* and the *y-Value*. Get the answer first, then write the working.

IDEA 03

Practise: Read the Turning Point

My questions first, then selected practice, then real past papers.

Q19 **DR ESLAM** Read it straight off

CH 2 · TOPIC 5 3 marks

The curve C has equation $y = 3(x - 1)^2 + 2$.

Write down (a) the turning point, (b) whether it is a maximum or a minimum, (c) the line of symmetry, (d) where C cuts the y -axis.

Q20 **DR ESLAM** Complete, then read

CH 2 · TOPIC 5 3 marks

Write $x^2 - 10x + 4$ in the form $(x - p)^2 - q$.

Hence write down the turning point of $y = x^2 - 10x + 4$.

Turning point _____

SELECTED PRACTICE

Q21**PRACTICE**

Practice · Medium

CH 2 · TOPIC 5 1 mark

4 The equation of a curve is $y = (x + 3)^2 + 5$

Circle the coordinates of the turning point.

(5, 3)	(5, -3)	(3, 5)	(-3, 5)
--------	---------	--------	---------

(1 mark)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5**3 marks**

2 Given that $x^2 - 6x + 1 = (x - a)^2 - b$ for all values of x .

(i) Find the value of a and the value of b .

[2]

(ii) Hence write down the coordinates of the turning point of the graph of $y = x^2 - 6x + 1$.

[1]

(3 marks)

Q23**PRACTICE**

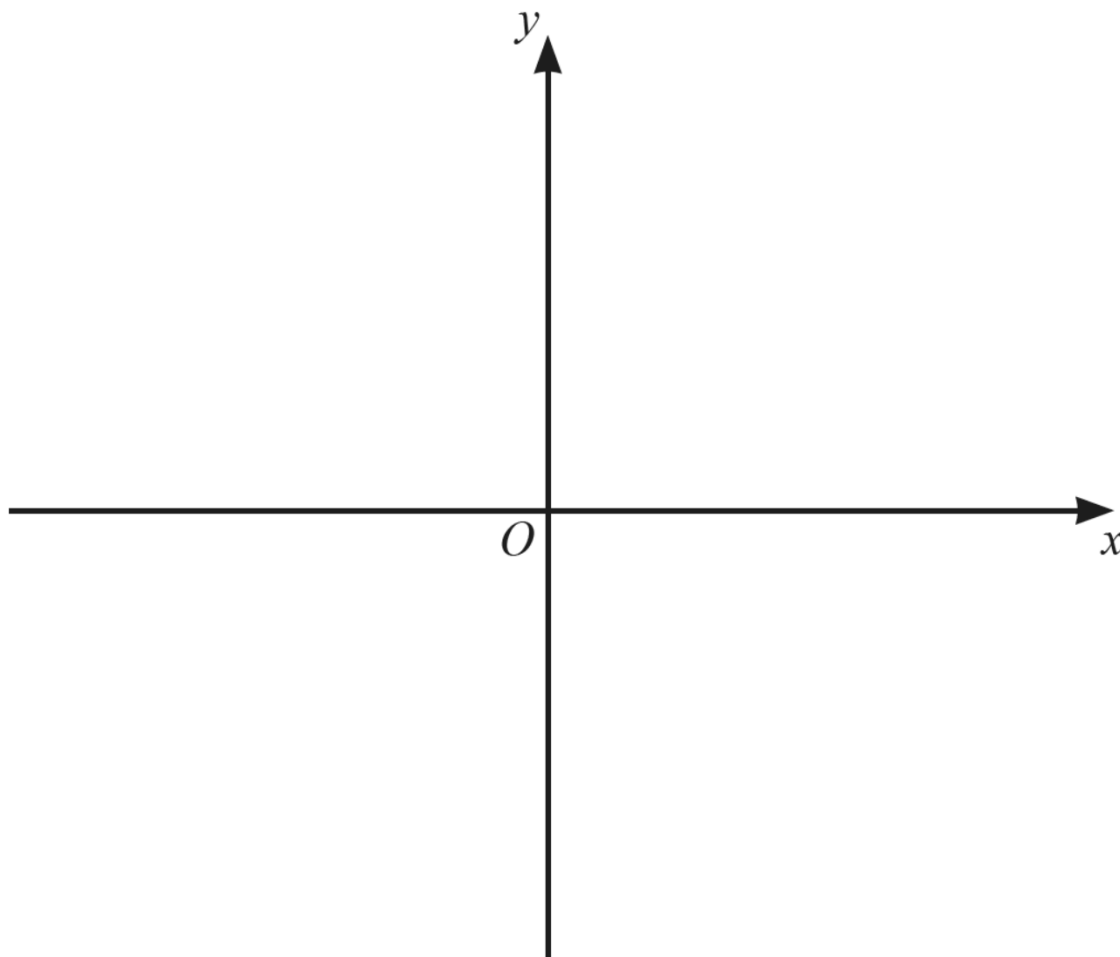
Practice · Hard

CH 2 · TOPIC 5 5 marks

12 (a) Write $x^2 + 10x + 14$ in the form $(x + a)^2 + b$.

(2 marks)

(b) Sketch the graph of $y = x^2 + 10x + 14$, indicating the coordinates of the turning point.

**(3 marks)**

Q24**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 4 marks

8 Find the turning point of $y = x^2 + 4x - 3$ by completing the square.

(..... ,)

(4 marks)

PAST PAPERS

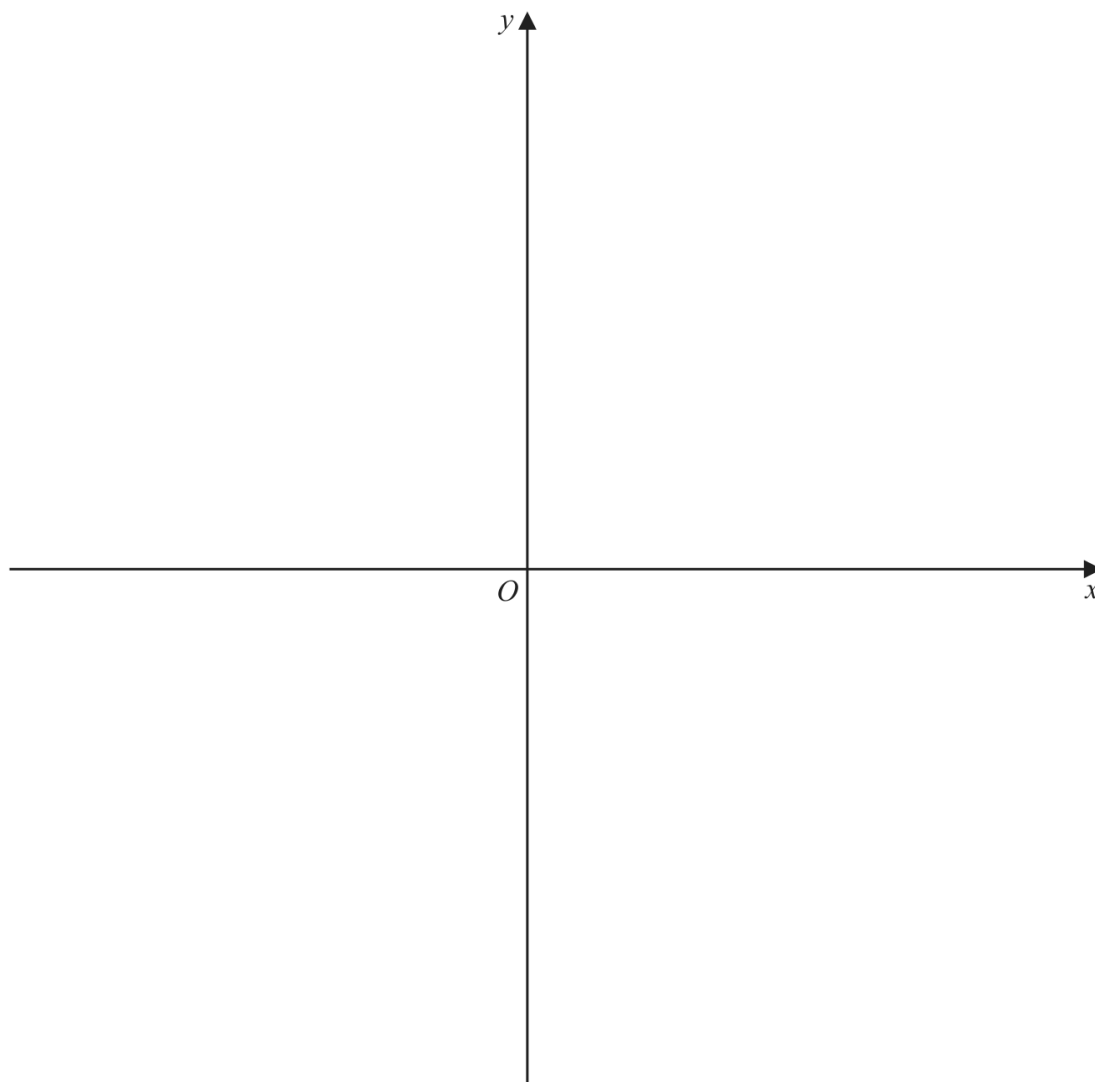
Q25 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 2H · Q25

CH 2 · TOPIC 5 6 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(Total for Question 25 is 6 marks)

04 The “Hence” Questions

Part (a) is always the completed square. Part (b) uses it — there are only three things it can ask.

WHEN YOU SEE → YOU DO

“hence solve”

→ Move → Root ($\pm$) → Solve

“surd form” / “exact”

→ keep the $\sqrt{}$, do not round

“ $y = f(x + 4)$ ”

→ find the turning point of f , then move it

“series of transformations ... maps S onto T”

→ translate · stretch · translate

A

Hence solve = Move → Root → Solve

$$x^2 + 4x - 16 = 0 \Rightarrow (x + 2)^2 - 20 = 0$$

$$\text{Move: } (x + 2)^2 = 20 \quad \text{Root: } x + 2 = \pm\sqrt{20} \quad \text{Solve: } x = -2 \pm 2\sqrt{5}$$

Never forget the $\pm$ — two answers.

B

The curve is moved: $y = f(x + 4)$

Find the turning point of f first, then move it.

$f(x) = 2(x - 3)^2 - 15$ has minimum $(3, -15)$. $f(x + 4)$ moves it **4 left** → $(-1, -15)$.

Inside the bracket moves x the opposite way; outside moves y the same way.

C

Transformations from $y = x^2$

$a(x + p)^2 + q$: translate $\begin{pmatrix} -p \\ 0 \end{pmatrix}$ → stretch parallel to the y -axis, s.f. a →

translate $\begin{pmatrix} 0 \\ q \end{pmatrix}$. Order matters.

$2(x - 3)^2 - 5$: translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, stretch s.f. 2, then translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.

IDEA 04

Practise: The “Hence” Questions

My questions first, then selected practice, then real past papers.

Q26 DR ESLAM Move → Root → Solve

CH 2 · TOPIC 5 4 marks

(a) Write $x^2 - 4x - 5$ in the form $(x - p)^2 - q$.

(b) Hence solve $x^2 - 4x - 5 = 0$.

$x =$ _____

Q27 DR ESLAM The curve is moved

CH 2 · TOPIC 5 2 marks

$$f(x) = (x - 2)^2 - 9.$$

Write down the coordinates of the turning point of the curve $y = f(x + 3)$.

Turning point _____

SELECTED PRACTICE

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 7 marks

8 (i) Write $x^2 + 4x - 16$ in the form $(x + a)^2 - b$.

[3]

(ii) Solve the equation $x^2 + 4x - 16 = 0$.

Give your answers in surd form as simply as possible.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

(7 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 4 marks

10 (i) Write $x^2 + 8x - 9$ in the form $(x + k)^2 + h$.

[2]

(ii) Use your answer to **part (i)** to solve the equation $x^2 + 8x - 9 = 0$.

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

(4 marks)

Q30**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 4 marks

11 (a) Write $x^2 - 18x - 27$ in the form $(x + k)^2 + h$.

(2 marks)

(b) Use your answer to **part (a)** to solve the equation $x^2 - 18x - 27 = 0$.

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 2H · Q22

CH 2 · TOPIC 5 4 marks

- 22 The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
describe fully a series of transformations that map the curve **S** onto the curve **T**.

.....

.....

.....

(Total for Question 22 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q23

CH 2 · TOPIC 5 5 marks

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.....,)

(Total for Question 23 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q20

CH 2 · TOPIC 5 5 marks

- 20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(..... ,)

(Total for Question 20 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

$$\text{List } a, b, c, \frac{b}{2a} \rightarrow a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$$

01 Two-step method

02 Negative a

Reorder to $ax^2 + bx + c$; carry the minus. Write the number first.

03 Turning point

$a(x + p)^2 + q \rightarrow (-p, q)$. $a > 0$ min, $a < 0$ max.
Symmetry $x = -p$.

04 “Hence”

Solve: Move $\rightarrow$ Root ($\pm$) $\rightarrow$ Solve. $f(x+k)$: move k left. Transform: translate, stretch, translate.

BOTTOM LINE List $a, b, c, \frac{b}{2a} \rightarrow$ structure $\rightarrow$ calculator on the right-hand part. Turning point: flip inside, keep outside. “Hence” = solve, move, or transform.

Answers

Completing the Square • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE TWO-STEP METHOD

- | | | |
|-----------|---|---|
| Q1 | Dr Eslam • Step 1 first • $a = 1$ | $(x + 3)^2 + 2$ |
| Q2 | Dr Eslam • $a \neq 1$ • the trap | $2(x + 3)^2 - 11$ |
| Q3 | Practice • Medium | $(x + 3)^2 - 16$ |
| Q4 | Practice • Medium | $(x - 3)^2 + 2$ |
| Q5 | Practice • Medium | $a = 2, b = -13$ |
| Q6 | Practice • Very Hard | $a = 2, b = \frac{3}{2}, c = \frac{1}{2}$ |
| Q7 | Edexcel 4MA1 • January 2022 • Paper 1HR • Q19 | (a) $m = -\frac{4}{3}$ (b) $3(x + 2)^2 + 7$ |
| Q8 | Edexcel 4MA1 • June 2023 • Paper 2H • Q19 | $3(x - 1)^2 + 2$ |
| Q9 | Edexcel 4MA1 • November 2025 • Paper 1H • Q21 | $5(x - 2)^2 + 3$ |

IDEA 02 • NEGATIVE a

- Q10** Dr Eslam • Negative $a = -1$ $19 - (x + 3)^2$, so $p = 19$, $q = 3$
- Q11** Dr Eslam • Negative $a \neq -1$ $13 - 2(x - 2)^2$, so $a = 13$, $b = 2$, $c = 2$
- Q12** Practice • Very Hard $14 - (x - 3)^2$: $p = 14$, $q = 3$
- Q13** Practice • Very Hard $23 - 2(x - 3)^2$: $a = 23$, $b = -2$, $c = -3$
- Q14** Practice • Very Hard $11 - (x + 2)^2$: $p = 11$, $q = 2$
- Q15** Edexcel 4MA1 • January 2020 • Paper 2HR • Q23 $25 - 2(x + 3)^2$
- Q16** Edexcel 4MA1 • January 2022 • Paper 1H • Q20 (a) $19 - 3(x - 2)^2$ (b) $A(2, 19)$
- Q17** Edexcel 4MA1 • June 2022 • Paper 1H • Q24 $a = q + \frac{36}{q}$, $b = q$, $c = \frac{6}{q}$
- Q18** Edexcel 4MA1 • May 2024 • Paper 1H • Q25 $29 - 3(x - 2)^2$

IDEA 03 • READ THE TURNING POINT

- Q19** Dr Eslam • Read it straight off (a) $(1, 2)$ (b) minimum (c) $x = 1$ (d) $(0, 5)$
- Q20** Dr Eslam • Complete, then read $(x - 5)^2 - 21$; turning point $(5, -21)$
- Q21** Practice • Medium $(-3, 5)$
- Q22** Practice • Hard (i) $a = 3$, $b = 8$ (ii) $(3, -8)$
- Q23** Practice • Hard (a) $(x + 5)^2 - 11$ (b) U-shape, min $(-5, -11)$, y-cut $(0, 14)$
- Q24** Practice • Very Hard $(-2, -7)$
- Q25** Edexcel 4MA1 • June 2025 • Paper 2H • Q25 (a) $52 - 6(x - 2)^2$ (b) cap shape, max $(2, 52)$, y-cut $(0, 28)$

IDEA 04 • THE “HENCE” QUESTIONS

- Q26** Dr Eslam · Move → Root → Solve (a) $(x - 2)^2 - 9$ (b) $x = 5$ or $x = -1$
- Q27** Dr Eslam · The curve is moved $(-1, -9)$
- Q28** Practice · Hard (i) $(x + 2)^2 - 20$ (ii) $x = -2 \pm 2\sqrt{5}$
- Q29** Practice · Hard (i) $(x + 4)^2 - 25$ (ii) $x = 1$ or $x = -9$
- Q30** Practice · Hard (a) $(x - 9)^2 - 108$ (b) $x = 9 \pm 6\sqrt{3}$
- Q31** Edexcel 4MA1 · May 2021 · Paper 2H · Q22 $g(x) = 2(x - 3)^2 - 5$: translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, stretch || y-axis s.f. 2, translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$
- Q32** Edexcel 4MA1 · June 2022 · Paper 1HR · Q23 (a) $2(x - 3)^2 - 15$ (b) $M(-1, -15)$
- Q33** Edexcel 4MA1 · May 2024 · Paper 1HR · Q20 (a) $2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$ (b) $P\left(\frac{23}{4}, -\frac{49}{8}\right)$



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Algebraic Fractions

Factorise everything first. Then it is just fractions: cancel, flip, or find the common denominator.

THE TOPIC ON ONE MAP

Algebraic Fractions

01 Simplify: cancel brackets

$$\frac{x^2 - 9}{x^2 + x - 6} \quad (3 - x) \text{ vs } (x - 3)$$

02 Multiply and divide: KFC

× cancel first ÷ Keep-Flip-Change

03 Add and subtract: the LCM

"as a single fraction"

04 Mixed operations

BODMAS fractions inside fractions

05 Solving with fractions

× every term by the LCM reject negative lengths

THE GOLDEN RULE

The first step of **any** algebraic fraction question:
factorise every top and every bottom.

01 Simplifying: Cancel Brackets

After factorising, cancel a bracket that is on the top **and** the bottom. Whole brackets only.

WHEN YOU SEE → YOU DO

“simplify fully” (fraction)

→ golden rule: factorise everything

$(3 - x)$ against $(x - 3)$

→ take out -1 , then cancel

WORKED MODEL • FACTORISE, THEN CANCEL

$$\frac{x^2 - 16}{2x^2 - 5x - 12}$$

$$\textcircled{1} \quad \frac{(x + 4)(x - 4)}{(2x + 3)(x - 4)}$$

DOTS on top • CC Rule below

$$\textcircled{2} \quad \frac{(x + 4)(\cancel{x - 4})}{(2x + 3)(\cancel{x - 4})}$$

cancel the same bracket

ANSWER $\frac{x + 4}{2x + 3}$

× **TRAP** You cannot cancel *terms*: $\frac{x + 4}{x + 6}$ is finished. Only a bracket that **multiplies** everything can go.

OPPOSITE BRACKETS $(3 - x) = -(x - 3)$. Take out the minus, then cancel.

IDEA 01

Practise: Simplifying: Cancel Brackets

My questions first, then selected practice.

Q1 DR ESLAM Golden rule first

CH 2 · TOPIC 6 3 marks

Simplify fully $\frac{x^2 - 9}{x^2 + x - 6}$.

Top: DOTS. Bottom: CC Rule. Then cancel.

Answer _____

Q2 DR ESLAM HCF on top

CH 2 · TOPIC 6 2 marks

Simplify fully $\frac{x^2 + 5x}{x^2 - 25}$.

Answer _____

SELECTED PRACTICE

Q3 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 2 marks

4 Simplify $\frac{4(x+5)}{x^2 + 2x - 15}$

(2 marks)

Q4 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 1 mark

14 Circle the expression that is equivalent to $\frac{3x^2}{6x^2 + 3}$

$\frac{x^2}{2x^2 + 3}$	$\frac{x^2}{6x^2 + 1}$	$\frac{x^2}{2x^2 + 1}$	$\frac{1}{2} + x^2$
------------------------	------------------------	------------------------	---------------------

(1 mark)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6 3 marks

3 Simplify $\frac{x^2 - 16}{2x^2 - 5x - 12}$

(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6 6 marks

6 (a) Simplify fully $\frac{x^2 + 3x - 4}{2x^2 - 5x + 3}$

(3 marks)

(b) Write $\frac{4}{x+2} + \frac{3}{x-2}$ as a single fraction in its simplest form.

(3 marks)

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6

3 marks

11 Simplify fully $\frac{x^5 - 4x^3}{3x - 6}$

(3 marks)**Q8****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6

2 marks

10 Simplify $\frac{2^n - 1}{4^n - 1}$

(2 marks)

02 Multiply and Divide: KFC

Multiplying: **cancel first**, then multiply. Dividing: turn it into multiplying.

WHEN YOU SEE → YOU DO

“**x**” between fractions

→ cancel first

“**÷**” between fractions

→ Keep – Flip – Change



WORKED MODEL • KFC, THEN CANCEL

$$\frac{3x+6}{x^2-3x-10} \div \frac{x+5}{x^3-25x}$$

1 $\frac{3x+6}{x^2-3x-10} \times \frac{x^3-25x}{x+5}$

KFC

2 $\frac{3(x+2)}{(x-5)(x+2)} \times \frac{x(x+5)(x-5)}{x+5}$

factorise everything

3 cancel $(x+2)$, $(x-5)$, $(x+5)$

cancel first

ANSWER $3x$

IDEA 02

Practise: Multiply and Divide: KFC

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Cancel first

CH 2 · TOPIC 6

3 marks

Simplify $\frac{x^2 - 4}{3x + 6} \times \frac{9}{x - 2}$.

Answer _____

Q10**DR ESLAM**

KFC

CH 2 · TOPIC 6

3 marks

Simplify $\frac{2x + 6}{x^2 - 1} \div \frac{x + 3}{x - 1}$.

Keep · Flip · Change, then factorise and cancel.

Answer _____

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Easy

CH 2 · TOPIC 6 4 marks

1 (a) Simplify $3e^7f \times 4e^2f$

(2 marks)

(b) Simplify fully $(27a^{12})^{\frac{2}{3}}$

(2 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 2 marks

12 Simplify $\frac{25a}{8} \times \frac{2a}{5}$

Give your answer as a single fraction in its simplest form.

(2 marks)

Q13**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6**3 marks**

- 2 Show that $\frac{1}{6x^2 + 7x - 5} \div \frac{1}{4x^2 - 1}$ simplifies to $\frac{ax + b}{cx + d}$ where a , b , c and d are integers.

(3 marks)**Q14****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6**4 marks**

- 5 Show that $\frac{3x + 6}{x^2 - 3x - 10} \div \frac{x + 5}{x^3 - 25x}$ simplifies to ax where a is an integer.

(4 marks)

PAST PAPERS

Q15 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2H • Q23**CH 2 • TOPIC 6** **4 marks**

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

03 Add and Subtract: the LCM

One fraction bar. The bottom is the LCM; each top is multiplied by what its bottom was missing.

WHEN YOU SEE → YOU DO

“as a single fraction”

→ LCM bottom, adjust the tops

“in its simplest form”

→ factorise the answer and cancel if you can

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

top × missing (x + 1) top × missing (x - 2)

1

LCM of the bottoms

Every different bracket, letter and number — once each (after factorising).

2

Adjust the tops

Multiply each top by what its bottom was missing.

3

Expand and simplify the top

This is Expanding Brackets. Watch the minus in front of the second top.

WORKED MODEL

1
$$\frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

one fraction bar

2
$$4x + 4 - 3x + 6 = x + 10$$

expand the top

ANSWER
$$\frac{x+10}{(x-2)(x+1)}$$

IDEA 03

Practise: Add and Subtract: the LCM

My questions first, then selected practice, then real past papers.

Q16 **DR ESLAM** Two brackets underneath

CH 2 · TOPIC 6 3 marks

Write $\frac{3}{x+2} + \frac{2}{x-1}$ as a single fraction.

Answer _____

Q17 **DR ESLAM** Numbers underneath · watch the minus

CH 2 · TOPIC 6 3 marks

Write $\frac{x+3}{4} - \frac{x-1}{6}$ as a single fraction.

Answer _____

SELECTED PRACTICE

Q18 PRACTICE Practice · Medium

CH 2 · TOPIC 6 3 marks

1 Write as a single fraction in its simplest form

$$\frac{2}{y+3} - \frac{1}{y-6}$$

(3 marks)

Q19 PRACTICE Practice · Medium

CH 2 · TOPIC 6 3 marks

8 Write $\frac{2x+1}{4} + \frac{x-2}{3}$ as a single fraction in its simplest form.

(3 marks)

Q20**PRACTICE**

Practice · Medium

CH 2 · TOPIC 6 2 marks

16 Express as a single fraction.

$$\frac{m+1}{n+1} - \frac{m}{n}$$

Simplify your answer.

(2 marks)

Q21**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 3 marks

9 Express $\frac{1}{9x^2 - 25} - \frac{1}{6x + 10}$ as a single fraction in its simplest form.

(3 marks)

PAST PAPERS

Q22 **PAST PAPER** Edexcel 4MA1 · January 2022 · Paper 1HR · Q12**CH 2 · TOPIC 6** **5 marks****12** (a) Expand and simplify $n(n - 4)(3n + 5)$

(b) Express

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

as a single fraction in its simplest form.

(Total for Question 12 is 5 marks)**Q23** **PAST PAPER** Edexcel 4MA1 · November 2023 · Paper 1H · Q11**CH 2 · TOPIC 6** **3 marks****11** Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.**(Total for Question 11 is 3 marks)**

Q24 **PAST PAPER** Edexcel 4MA1 · May 2024 · Paper 1H · Q12**CH 2 · TOPIC 6** **5 marks**12 (a) Factorise fully $6y^2 - 5y - 4$ (b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.**(Total for Question 12 is 5 marks)****Q25** **PAST PAPER** Edexcel 4MA1 · June 2025 · Paper 1H · Q12**CH 2 · TOPIC 6** **3 marks**12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.**(Total for Question 12 is 3 marks)**

04 Mixed Operations and Fractions Inside Fractions

BODMAS: brackets, then $\times$ and $\div$, and only then $+$ and $-$.

WHEN YOU SEE $\rightarrow$ YOU DO

" $4 - [\dots \div \dots]$ "

$\rightarrow$ BODMAS: bracket first, subtract last

"given $x = \dots$ and $y = \dots$, find x in terms of a "

$\rightarrow$ substitute, multiply top & bottom

1

Brackets and $\times \div$ first

Factorise, KFC, cancel — until one simple fraction is left.

2

Then $+$ $-$

Use the LCM with whatever is left. A whole term like 4 or $(x - 7)$ is a fraction over 1.

WORKED MODEL • JUN 2024 P2HR

$$4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$$

$$1 \quad (3x - 5) \times \frac{4x - 1}{(3x - 5)(x + 2)} = \frac{4x - 1}{x + 2}$$

KFC • factorise • cancel

$$2 \quad \frac{4(x + 2) - (4x - 1)}{x + 2}$$

now subtract

ANSWER $\frac{9}{x + 2}$

FRACTION INSIDE A FRACTION Substituting $y = \frac{5}{5a-2}$ into $\frac{5}{9y+5}$? Multiply the top and the bottom by the small denominator $(5a - 2)$ — the inner fraction disappears.

IDEA 04

Practise: Mixed Operations

My questions first, then selected practice, then real past papers.

Q26 **DR ESLAM** A whole number is a fraction over 1

CH 2 · TOPIC 6 2 marks

Write $2 - \frac{3}{x+1}$ as a single fraction.

Answer _____

Q27 **DR ESLAM** Fraction inside a fraction

CH 2 · TOPIC 6 3 marks

Given that $a = \frac{3}{x+1}$ and $x = \frac{2}{y}$, write a in terms of y as a single fraction.

Substitute, then multiply the top and bottom by y .

$a =$ _____

SELECTED PRACTICE

Q28**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 6 4 marks

1 $2 - \frac{x+2}{x-3} - \frac{x-6}{x+3}$ can be written as a single fraction in the form $\frac{ax+b}{x^2-9}$

where a and b are integers.

Work out the value of a and the value of b .

(4 marks)

Q29**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 6 4 marks

4 Write

$$4 - \left[(x+3) \div \frac{x^2 + 5x + 6}{x-2} \right]$$

as a single fraction in its simplest form.
You must show your working.

(4 marks)

Q30**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 6 4 marks

6 Show that $6 + \left[(x+5) \div \frac{x^2 + 3x - 10}{x-1} \right]$ simplifies to $\frac{ax-b}{cx-d}$ where a, b, c and d are integers.

(4 marks)

Q31**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 4 marks

16 Write $\frac{5}{4x^2 - 25} - (2x + 3) \div \left(\frac{4x^2 + 16x + 15}{7} \right)$ as a single fraction in its simplest form.

Show clear algebraic working

(4 marks)**PAST PAPERS****Q32****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q24

CH 2 · TOPIC 6 4 marks**24** Express

$$\left(\frac{4}{2x - 5} - \frac{3}{2x - 3} \right) \div \frac{9x - 4x^3}{6x^2 - 17x + 5}$$

as a single fraction in its simplest form.

(Total for Question 24 is 4 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q21

CH 2 • TOPIC 6 5 marks

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

.....

(Total for Question 21 is 5 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q21

CH 2 • TOPIC 6 4 marks

21 Write $\frac{25x^2-64}{5x^2-13x-6} \times \frac{x^2-8x+15}{5x+8} - (x-7)$

as a single fraction in its simplest form.
Show clear algebraic working.

.....

(Total for Question 21 is 4 marks)

Q35 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 1H • Q21**CH 2 • TOPIC 6** **4 marks**

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

Q36 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2HR • Q23**CH 2 • TOPIC 6** **3 marks**

23 Express $\left(\frac{20}{x^2-36} - \frac{2}{x-6}\right) \times \frac{1}{4-x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

Q37 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 2H · Q19

CH 2 · TOPIC 6 3 marks

$$19 \quad a = \frac{14}{3x-7} \quad x = \frac{7}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 19 is 3 marks)

Q38 PAST PAPER Edexcel 4MA1 · May 2023 · Paper 1HR · Q23

CH 2 · TOPIC 6 4 marks

$$23 \quad \text{Simplify } (x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$$

Give your answer in the form $\frac{ax^2}{bx+c}$ where a, b and c are integers.

 $\dots\dots\dots$

(Total for Question 23 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q26

CH 2 · TOPIC 6 4 marks

26 Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2H · Q20

CH 2 · TOPIC 6 3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x + 2}$ in the form $\frac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

Q41 PAST PAPER Edexcel 4MA1 · November 2024 · Paper 2H · Q23

CH 2 · TOPIC 6 4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

Q42 PAST PAPER Edexcel 4MA1 · November 2025 · Paper 1H · Q23

CH 2 · TOPIC 6 3 marks

23 $a = \frac{2x + 5}{1 - x}$ $x = \frac{5 - 2y}{3y}$

Write a in the form $\frac{m + ny}{p(y - 1)}$ where m , n and p are integers.

Show your working clearly.

$$a = \dots\dots\dots$$

(Total for Question 23 is 3 marks)

05 Solving Equations with Fractions

An equation lets you get rid of every denominator at once: multiply **every term** by the LCM.

WHEN YOU SEE → YOU DO

“solve” + fractions

→ × every term by the LCM

“give your answers to 3 s.f.”

→ ERCS → quadratic formula

a speed or a length

→ reject the negative answer

1

Clear the fractions

Multiply every term (the right-hand side too) by the LCM. Two fractions equal? Cross-multiply.

2

ERCS

Expand, Rearrange to = 0, Collect → $ax^2 + bx + c = 0$ (or a linear equation).

3

Solve and check

Calculator first. Reject any answer that makes a denominator zero — or a speed negative.

WORKED MODEL • JAN 2021 P2HR

$$\frac{5}{x+2} + \frac{3}{x(x+2)} = 2$$

1 $5x + 3 = 2x(x + 2)$

× LCM $x(x + 2)$

2 $2x^2 - x - 3 = 0$

ERCS

ANSWER $x = \frac{3}{2}$ or $x = -1$

CALCULATOR CHECK Put $x = 1.5$ back in: $\frac{5}{3.5} + \frac{3}{5.25} = 2$. Correct.

IDEA 05

Practise: Solving Equations with Fractions

My questions first, then selected practice, then real past papers.

Q43 DR ESLAM × LCM, linear

CH 2 · TOPIC 6 3 marks

Solve $\frac{x+1}{3} - \frac{x-2}{4} = 2$.

x = _____

Q44 DR ESLAM × LCM, then a quadratic

CH 2 · TOPIC 6 4 marks

Solve $\frac{3}{x-1} + \frac{2}{x+1} = 1$, giving your answers to 2 d.p.

× (x - 1)(x + 1), then ERCS, then the calculator.

x = _____

SELECTED PRACTICE

Q45

PRACTICE

Practice · Medium

CH 2 · TOPIC 6 4 marks

17 Solve $\frac{3x-2}{5} - \frac{3-4x}{2} = 2$

Show clear algebraic working.

(4 marks)

Q46

PRACTICE

Practice · Hard

CH 2 · TOPIC 6 3 marks

15 Solve

$$\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$$

Show clear algebraic working

(3 marks)

Q47 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 6 4 marks

13 Solve $\frac{x}{x+4} + \frac{7}{x-2} = 1$

You **must** show your working.

$X = \dots\dots\dots$

(4 marks)

Q48 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 6 6 marks

15 Solve this equation, giving your answers correct to 1 decimal place.

$$\frac{5}{x+2} + \frac{3}{x-3} = 2$$

$X = \dots\dots\dots$ or $X = \dots\dots\dots$

(6 marks)

PAST PAPERS

Q49**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q18

CH 2 • TOPIC 6 5 marks**18** Solve the equation

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

Show clear algebraic working.

.....

(Total for Question 18 is 5 marks)

Q50**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q24

CH 2 • TOPIC 6 5 marks

24 Solve $\frac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

(Total for Question 24 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--|--|
| 00 | Golden rule | Factorise every top and every bottom first. |
| 01 | Simplify | Cancel whole brackets only. $(3 - x) = -(x - 3)$. |
| 02 | $\times$ and $\div$ | Cancel first. Dividing: Keep – Flip – Change. |
| 03 | $+$ and $-$ | LCM bottom $\cdot$ adjust the tops $\cdot$ expand the top. |
| 04 | Mixed | BODMAS: $\times \div$ first, $+-$ last. Nested: multiply top & bottom. |
| 05 | Equations | $\times$ LCM every term $\rightarrow$ ERCS $\rightarrow$ solve $\rightarrow$ reject bad answers. |

BOTTOM LINE Factorise $\rightarrow$ cancel $\rightarrow$ KFC for $\div$ $\rightarrow$ LCM for $\pm$ $\rightarrow$ in an equation, multiply every term by the LCM.

Answers

Algebraic Fractions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SIMPLIFYING: CANCEL BRACKETS

Q1	Dr Eslam • Golden rule first	$\frac{x-3}{x-2}$
Q2	Dr Eslam • HCF on top	$\frac{x}{x-5}$
Q3	Practice • Medium	$\frac{4}{x-3}$
Q4	Practice • Medium	$\frac{x^2}{2x^2+1}$
Q5	Practice • Hard	$\frac{x+4}{2x+3}$
Q6	Practice • Hard	(a) $\frac{x+4}{2x-3}$ (b) $\frac{7x-2}{(x+2)(x-2)}$
Q7	Practice • Hard	$\frac{x^3(x+2)}{3}$
Q8	Practice • Very Hard	$\frac{1}{2^n+1}$

IDEA 02 • MULTIPLY AND DIVIDE: KFC

- Q9** Dr Eslam • Cancel first 3
- Q10** Dr Eslam • KFC $\frac{2}{x+1}$
- Q11** Practice • Easy (a) $12e^9f^2$ (b) $9a^8$
- Q12** Practice • Medium $\frac{5a^2}{4}$
- Q13** Practice • Very Hard $\frac{2x+1}{3x+5}$
- Q14** Practice • Very Hard $3x$
- Q15** Edexcel 4MA1 • June 2025 • Paper 2H • Q23 $p = 8$

IDEA 03 • ADD AND SUBTRACT: THE LCM

Q16	Dr Eslam • Two brackets underneath	$\frac{5x + 1}{(x - 1)(x + 2)}$
Q17	Dr Eslam • Numbers underneath • watch the minus	$\frac{x + 11}{12}$
Q18	Practice • Medium	$\frac{y - 15}{(y + 3)(y - 6)}$
Q19	Practice • Medium	$\frac{10x - 5}{12}$
Q20	Practice • Medium	$\frac{n - m}{n(n + 1)}$
Q21	Practice • Very Hard	$\frac{7 - 3x}{2(9x^2 - 25)}$
Q22	Edexcel 4MA1 • January 2022 • Paper 1HR • Q12	(a) $3n^3 - 7n^2 - 20n$ (b) $\frac{3x + 16}{4x}$
Q23	Edexcel 4MA1 • November 2023 • Paper 1H • Q11	$\frac{7x + 10}{12x}$
Q24	Edexcel 4MA1 • May 2024 • Paper 1H • Q12	(a) $(2y + 1)(3y - 4)$ (b) $\frac{31 - 14x}{12x}$
Q25	Edexcel 4MA1 • June 2025 • Paper 1H • Q12	$\frac{17x - 6}{12x}$

IDEA 04 • MIXED OPERATIONS

- Q26** Dr Eslam • A whole number is a fraction over 1 $\frac{2x-1}{x+1}$
- Q27** Dr Eslam • Fraction inside a fraction $\frac{3y}{y+2}$
- Q28** Practice • Very Hard $a = 4, b = -42$
- Q29** Practice • Very Hard $\frac{3x+10}{x+2}$
- Q30** Practice • Very Hard $\frac{7x-13}{x-2}$
- Q31** Practice • Very Hard $\frac{40-14x}{(2x+5)(2x-5)}$
- Q32** Edexcel 4MA1 • January 2020 • Paper 2H • Q24 $\frac{1-3x}{x(2x-3)^2}$
- Q33** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q21 $\frac{34-6x}{(x-5)(x-1)}$
- Q34** Edexcel 4MA1 • January 2021 • Paper 2H • Q21 $\frac{54}{5x+2}$
- Q35** Edexcel 4MA1 • May 2021 • Paper 1H • Q21 $x = \frac{5a-2}{5a+7}$
- Q36** Edexcel 4MA1 • June 2022 • Paper 2HR • Q23 $\frac{2}{(x+6)(x-6)}$
- Q37** Edexcel 4MA1 • June 2022 • Paper 2H • Q19 $a = \frac{3-4y}{2y-3}$
- Q38** Edexcel 4MA1 • May 2023 • Paper 1HR • Q23 $\frac{-7x^2}{4x+1}$
- Q39** Edexcel 4MA1 • June 2024 • Paper 2HR • Q26 $\frac{9}{x+2}$
- Q40** Edexcel 4MA1 • June 2024 • Paper 2H • Q26 $\frac{5-20y^2}{1+8y}$

IDEA 05 · SOLVING EQUATIONS WITH FRACTIONS

- Q43** Dr Eslam · × LCM, linear $x = 14$
- Q44** Dr Eslam · × LCM, then a quadratic $x = \frac{5 \pm \sqrt{33}}{2}$: 5.37 or -0.37
- Q45** Practice · Medium $x = \frac{3}{2}$
- Q46** Practice · Hard $y = 2.9$
- Q47** Practice · Very Hard $x = -12$
- Q48** Practice · Very Hard $x = 5.3$ or -0.3
- Q49** Edexcel 4MA1 · January 2021 · Paper 2HR · Q18 $x = \frac{3}{2}$ or $x = -1$
- Q50** Edexcel 4MA1 · November 2023 · Paper 2H · Q24 $x = \frac{7}{4}$

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Rearranging Formulas

One universal method: clean the formula, then isolate the subject. It works when the subject appears once, twice, or under a root.

THE TOPIC ON ONE MAP

Rearranging Formulas

01 Subject once

"make t the subject"

clean, then isolate

02 Powers and roots

$\sqrt{\quad}$ round a side: square

t^2 : $\pm\sqrt{\quad}$

03 Subject twice

collect $\rightarrow$ factorise $\rightarrow$ divide

04 Root and twice

$\sqrt{\frac{x+a}{x-b}}$

05 Quadratic in the subject

complete the square first

$f^{-1}(x)$

THE UNIVERSAL METHOD

PRELIMINARY

Root / power?
undo it

STAGE 1

Clean
brackets · fractions

Collect
subject one side

STAGE 2

Factorise
if it is twice

Divide
by the bracket

01 Subject Once: Clean, then Isolate

Undo what was done to the subject — in reverse order.

WHEN YOU SEE → YOU DO

“make t the subject” → clean, then collect and divide

“rearrange ... to make x the subject” → undo in reverse order

1

Clean

Multiply by any denominator. Expand a bracket if the subject is inside it.

2

Collect and divide

Subject term on one side, everything else on the other. Divide by what multiplies the subject.

WORKED MODEL • MAKE F THE SUBJECT

$$C = \frac{5(F - 32)}{9}$$

1 $9C = 5(F - 32)$

clean: $\times 9$

2 $9C = 5F - 160$

clean: expand

3 $5F = 9C + 160$

collect

ANSWER $F = \frac{9C + 160}{5}$

IDEA 01

Practise: Subject Once

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Collect, divide

CH 2 • TOPIC 7 2 marks

Make a the subject of $v = u + at$.

$a =$ _____

Q2 DR ESLAM Clean the fraction first

CH 2 • TOPIC 7 2 marks

Make x the subject of $y = \frac{x+4}{3}$.

$x =$ _____

SELECTED PRACTICE

Q3

PRACTICE

Practice • Easy

CH 2 • TOPIC 7

2 marks

- 1 Make h the subject of the formula

$$t = \frac{gh}{10}$$

(2 marks)

Q4

PRACTICE

Practice • Easy

CH 2 • TOPIC 7

2 marks

- 5 Make t the subject of the formula $p = at - d$

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 2 • TOPIC 7 2 marks

11 Rearrange $v = u + at$ to make t the subject of the formula.

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 2 • TOPIC 7 2 marks

12 Use the formula $s = ut + \frac{1}{2}at^2$.

Make a the subject of the formula.

 $a = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q3

CH 2 · TOPIC 7 8 marks

3 (a) Make a the subject of $d = g + 2ac$

.....

(b) Factorise fully $9ef - 12f$

.....

(c) Expand and simplify $(x + 2)(x - 5)$

.....

(d) Simplify fully $\frac{n^4 \times n^7}{n^5}$

.....

(Total for Question 3 is 8 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q6

CH 2 · TOPIC 7 5 marks

6 (a) Make c the subject of $A = \frac{c}{y} - 5z$

.....

(b) Write down the value of g^0

.....

(c) Factorise $x^2 - 11x + 24$

.....

(Total for Question 6 is 5 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q5

CH 2 · TOPIC 7 6 marks

5 (a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

(b) Make t the subject of the formula $p = at - d$

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

(c) work out the value of n .

(2)

$n =$

(Total for Question 5 is 6 marks)

02 Powers and Roots: the Preliminary Step

A root or a square around the subject comes off **first** — or **last**, whichever layer is outside.

WHEN YOU SEE → YOU DO

$\sqrt{\quad}$ around a whole side

→ square both sides first

subject squared, t^2

→ $\pm\sqrt{\quad}$ at the end

Whole side under $\sqrt{\quad}$ →

square both sides

Subject squared, t^2 →

$\pm\sqrt{\quad}$ both sides (at the end)

Cube root / cube →

cube / cube-root both sides

WORKED MODEL • ROOT OUTSIDE

$$m = \sqrt{\frac{k+1}{4}}$$

$$1 \quad m^2 = \frac{k+1}{4}$$

square both sides

$$2 \quad 4m^2 = k+1$$

× 4

ANSWER $k = 4m^2 - 1$

× **TRAP** Square-rooting gives $\pm$: from $y = 3p^2 - 4$, $p = \pm\sqrt{\frac{y+4}{3}}$.

IDEA 02

Practise: Powers and Roots

My questions first, then selected practice.

Q10 DR ESLAM Squared subject

CH 2 · TOPIC 7 2 marks

Make r the subject of $A = \pi r^2$.

Divide first, then square-root.

$r =$ _____

Q11 DR ESLAM Root in the middle

CH 2 · TOPIC 7 3 marks

Make h the subject of $T = 2\pi\sqrt{\frac{h}{g}}$.

$h =$ _____

SELECTED PRACTICE

Q12 PRACTICE Practice · Easy

CH 2 • TOPIC 7 2 marks

9 Make t the subject of $c = t^3 - 8v$

(2 marks)

Q13 PRACTICE Practice · Medium

CH 2 • TOPIC 7 3 marks

1 Make y the subject of the formula $p = \sqrt{\frac{x+y}{5}}$

(3 marks)

Q14 PRACTICE Practice · Medium

CH 2 · TOPIC 7 3 marks

4 Make p the subject of the formula $y = 3p^2 - 4$

(3 marks)

Q15 PRACTICE Practice · Medium

CH 2 · TOPIC 7 2 marks

9 Make t the subject of $s = \frac{1}{2} at^2$

(2 marks)

Q16 PRACTICE Practice · Medium

CH 2 · TOPIC 7 2 marks

13 Make t the subject of $s = \frac{1}{2} at^2$

(2 marks)

Q17 PRACTICE Practice · Hard

CH 2 · TOPIC 7 3 marks

13 Rearrange $y = \frac{1}{\sqrt{x+1}}$ to make x the subject.

(3 marks)

03 Subject Twice: Factorise

The key trick. When the subject is in two terms, collect them and take the subject out as a common factor.

WHEN YOU SEE → YOU DO

subject in two terms

→ collect → factorise → divide

1

Clean

× the denominator, then expand so the subject is free.

2

Collect

All subject terms on one side, everything else on the other.

3

Factorise and divide

Subject outside a bracket → divide by the bracket.

WORKED MODEL • NOV 2021 P1H

$$w = \frac{2f + 3}{8 - f}$$

1 $w(8 - f) = 2f + 3$

clean: × (8 - f)

2 $8w - wf = 2f + 3$

expand

3 $8w - 3 = 2f + wf$

collect f terms

4 $8w - 3 = f(2 + w)$

factorise

ANSWER $f = \frac{8w - 3}{w + 2}$

IDEA 03

Practise: Subject Twice: Factorise

My questions first, then selected practice, then real past papers.

Q18 DR ESLAM Twice, no fractions

CH 2 • TOPIC 7 2 marks

Make x the subject of $ax + b = cx + d$.

$x =$ _____

Q19 DR ESLAM Twice, with a fraction

CH 2 • TOPIC 7 3 marks

Make x the subject of $y = \frac{x+2}{x-3}$.

× $(x - 3)$ · expand · collect x · factorise · divide.

$x =$ _____

SELECTED PRACTICE

Q20**PRACTICE**

Practice • Medium

CH 2 • TOPIC 7 3 marks

2 Make t the subject of the formula $2(d - t) = 4t + 7$

(3 marks)

Q21**PRACTICE**

Practice • Hard

CH 2 • TOPIC 7 3 marks

2 Make v the subject of the formula $w = \frac{15(t - 2v)}{v}$

(3 marks)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

5 Make a the subject of the formula $p = \frac{3a+5}{4-a}$

(4 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 3 marks

8 Make m the subject of the formula $f = \frac{3m+4}{m-1}$

(3 marks)

PAST PAPERS

Q24 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 1H · Q17

CH 2 · TOPIC 7 8 marks

17 (a) Factorise $6y^2 - y - 5$

(2)

(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$ (c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

(Total for Question 17 is 8 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q14

CH 2 • TOPIC 7 7 marks

- 14 (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

(b) Make c the subject of $g = \frac{c + 3}{4 + c} - 7$

(Total for Question 14 is 7 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q15

CH 2 • TOPIC 7 3 marks

- 15 Make n the subject of the formula $x = \frac{3p + n}{3n - 4}$

(Total for Question 15 is 3 marks)

04 The Full Exam Question: Root and Twice

10 of the 16 past papers here are this one type. It is Idea 02 and Idea 03 joined together.

WHEN YOU SEE → YOU DO

$$\sqrt{\frac{\dots X \dots}{\dots X \dots}}$$

→ square → × denominator → collect → factorise

WORKED MODEL • JAN 2020 P2H

$$y = \sqrt{\frac{x+1}{x-4}}$$

$$1 \quad y^2 = \frac{x+1}{x-4}$$

preliminary: square

$$2 \quad y^2(x-4) = x+1$$

clean: × denominator

$$3 \quad xy^2 - 4y^2 = x+1$$

expand

$$4 \quad xy^2 - x = 4y^2 + 1$$

collect x terms

$$5 \quad x(y^2 - 1) = 4y^2 + 1$$

factorise

ANSWER $x = \frac{4y^2 + 1}{y^2 - 1}$

CHECK Pick $x = 5$: $y = \sqrt{6}$, so $y^2 = 6 \rightarrow \frac{4(6)+1}{6-1} = 5$. Correct.

IDEA 04

Practise: Root and Twice

My questions first, then selected practice, then real past papers.

Q27**DR ESAM**

The full exam type

CH 2 · TOPIC 7 4 marks

Make x the subject of $p = \sqrt{\frac{x+3}{x-1}}$.

$x =$ _____

SELECTED PRACTICE**Q28****PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

10 Make c the subject of the formula $p = \sqrt{\frac{ac+8}{3+c}}$

(4 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

12 Given that $n > 0$ make n the subject of the formula $y = \frac{n^2 + d}{n^2}$

(4 marks)

Q30**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

14 Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

(4 marks)

PAST PAPERS

Q31**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q16

CH 2 • TOPIC 7 4 marks

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

(Total for Question 16 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q14

CH 2 · TOPIC 7 7 marks

14 (a) Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

$a = \dots\dots\dots$

(b) Make c the subject of the formula $p = \sqrt{\frac{ac+8}{3+c}}$

$\dots\dots\dots$

(Total for Question 14 is 7 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q15

CH 2 · TOPIC 7 4 marks

15 Make t the subject of $n^2 = \frac{4d+t^3}{t^3}$

$\dots\dots\dots$

(Total for Question 15 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 • June 2023 • Paper 2HR • Q17

CH 2 • TOPIC 7 4 marks

17 Make x the subject of $y = \sqrt[3]{\frac{6+5x}{x+4}}$

(Total for Question 17 is 4 marks)

Q35 PAST PAPER Edexcel 4MA1 • June 2023 • Paper 2H • Q17

CH 2 • TOPIC 7 7 marks

17 (a) Expand and simplify $(x+6)(3x-2)(x+6)$

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

(Total for Question 17 is 7 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q15

CH 2 • TOPIC 7 7 marks

15 (a) Make g the subject of $e = \sqrt{\frac{7g+5}{11+2g}}$

(b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

(Total for Question 15 is 7 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q12

CH 2 · TOPIC 7 7 marks

12 (a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$

Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

(b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

(Total for Question 12 is 7 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q17

CH 2 · TOPIC 7 4 marks

17 Make k the subject of $p = \frac{8k^2+5}{7-3k^2}$

(Total for Question 17 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q16

CH 2 • TOPIC 7 4 marks

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q18

CH 2 • TOPIC 7 4 marks

18 Make y the subject of $p = \sqrt{\frac{y + w}{3y - t}}$

(Total for Question 18 is 4 marks)

05 Quadratic in the Subject: Complete the Square

The subject appears as x^2 **and** as x ? You cannot collect them. The first step is **completing the square** — then x appears only once.

WHEN YOU SEE → YOU DO

subject as x^2 and x

→ complete the square first

“find $f^{-1}(x)$ ” of a quadratic

→ complete the square, then $\pm\sqrt{\quad}$

1

Complete the square

List $a, b, c, \frac{b}{2a} \rightarrow a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$. (Chapter 2 • Topic 5.)

2

Now it is “subject once”

Move the number, divide by a , square-root with $\pm$, then move the last number.

WORKED MODEL • MAKE x THE SUBJECT

$$y = x^2 + 6x + 2$$

$$1 \quad y = (x + 3)^2 - 9 + 2 = (x + 3)^2 - 7$$

complete the square

$$2 \quad y + 7 = (x + 3)^2$$

move the number

$$3 \quad \pm\sqrt{y + 7} = x + 3$$

root with $\pm$

ANSWER $x = -3 \pm \sqrt{y + 7}$

× **TRAP** Trying to “factorise out x ” from $x^2 + 6x$ leaves $x(x + 6)$ — the subject is still there twice. Complete the square instead.

IDEA 05

Practise: Quadratic in the Subject

Questions written to practise exactly this idea.

Q41 DR ESLAM Complete the square first

CH 2 • TOPIC 7 3 marks

Make x the subject of $y = x^2 - 4x + 1$.

Step 1: $y = (x - 2)^2 - ?$ Step 2: move the number, then $\pm\sqrt{}$.

$x =$ _____

Q42 DR ESLAM With $a \neq 1$

CH 2 • TOPIC 7 4 marks

Make x the subject of $y = 2x^2 + 8x - 1$.

$x =$ _____

Q43 **DR ESLAM** Inverse of a quadratic

CH 2 • TOPIC 7 3 marks

$f(x) = x^2 + 10x$ for $x \geq -5$. Find $f^{-1}(x)$.

$f^{-1}(x) =$ _____

06 Where You Will Use It

Rearranging is the same skill as solving an equation — and it comes back in three later topics.

WHEN YOU SEE → YOU DO

“find $f^{-1}(x)$ ”

→ $y = f(x)$, make x the subject, swap

“solve” a linear equation

→ the same clean → isolate method

Linear equations

Same method, numbers instead of letters.

Inverse functions

$y = f(x) \rightarrow$ make x the subject $\rightarrow$ swap.

Rearranging

Simultaneous equations

Make x or y the subject, then substitute.

Formulas in science

$v = u + at$, $s = ut + \frac{1}{2}at^2$

WORKED MODEL • INVERSE FUNCTION

$$f(x) = \frac{x+2}{x-3} \quad \text{Find } f^{-1}(x).$$

$$1 \quad y(x-3) = x+2 \Rightarrow xy - x = 3y + 2$$

clean • collect

$$2 \quad x(y-1) = 3y+2 \Rightarrow x = \frac{3y+2}{y-1}$$

factorise • divide

ANSWER $f^{-1}(x) = \frac{3x+2}{x-1}$

IDEA 06

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q44 DR ESLAM Inverse function

CH 2 • TOPIC 7 3 marks

$$f(x) = \frac{2x + 1}{x - 4}. \text{ Find } f^{-1}(x).$$

Write $y = f(x)$, make x the subject, then swap x and y .

$$f^{-1}(x) = \underline{\hspace{5cm}}$$

Q45 DR ESLAM Same method, numbers

CH 2 • TOPIC 7 2 marks

Solve $5(x + 3) = 3x - 4$.

$$x = \underline{\hspace{5cm}}$$

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--|--|
| 00 | Preliminary | Root outside $\rightarrow$ square both sides. Subject squared $\rightarrow \pm\sqrt{\quad}$ at the end. |
| 01 | Stage 1: Clean | $\times$ every denominator. Expand the bracket holding the subject. |
| 02 | Stage 2: Isolate | Collect $\rightarrow$ Factorise (if twice) $\rightarrow$ Divide. |
| 03 | Exam type | Square $\rightarrow \times$ denominator $\rightarrow$ expand $\rightarrow$ collect $\rightarrow$ factorise $\rightarrow$ divide. |
| 04 | x^2 and x | Complete the square first — then x is there once. Root with $\pm$. |
| 05 | Check | Put a number in for the subject, work out the other letter, and go back. |

BOTTOM LINE Undo the root $\rightarrow$ clean $\rightarrow$ collect $\rightarrow$ factorise $\rightarrow$ divide.

Answers

Rearranging Formulas • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SUBJECT ONCE

- | | | |
|-----------|--|---|
| Q1 | Dr Eslam • Collect, divide | $a = \frac{v - u}{t}$ |
| Q2 | Dr Eslam • Clean the fraction first | $x = 3y - 4$ |
| Q3 | Practice • Easy | $h = \frac{10t}{g}$ |
| Q4 | Practice • Easy | $t = \frac{p + d}{a}$ |
| Q5 | Practice • Easy | $t = \frac{v - u}{a}$ |
| Q6 | Practice • Medium | $a = \frac{2(s - ut)}{t^2}$ |
| Q7 | Edexcel 4MA1 • January 2020 • Paper 1HR • Q3 | (a) $a = \frac{d - g}{2c}$ (b) $3f(3e - 4)$ (c) $x^2 - 3x - 10$ (d) n^6 |
| Q8 | Edexcel 4MA1 • January 2021 • Paper 1HR • Q6 | (a) $c = y(A + 5z)$ (b) 1 (c) $(x - 3)(x - 8)$ |
| Q9 | Edexcel 4MA1 • May 2021 • Paper 1H • Q5 | (a) $3x^2 + 4x$ (b) $t = \frac{p + d}{a}$ (c) $n = 8$ |

IDEA 02 • POWERS AND ROOTS

Q10 Dr Eslam • Squared subject

$$r = \sqrt{\frac{A}{\pi}}$$

Q11 Dr Eslam • Root in the middle

$$h = \frac{gT^2}{4\pi^2}$$

Q12 Practice • Easy

$$t = \sqrt[3]{c + 8v}$$

Q13 Practice • Medium

$$y = 5p^2 - x$$

Q14 Practice • Medium

$$p = \pm \sqrt{\frac{y + 4}{3}}$$

Q15 Practice • Medium

$$t = \sqrt{\frac{2s}{a}}$$

Q16 Practice • Medium

$$t = \sqrt{\frac{2s}{a}}$$

Q17 Practice • Hard

$$x = \frac{1 - y^2}{y^2}$$

IDEA 03 • SUBJECT TWICE: FACTORISE

- Q18** Dr Eslam • Twice, no fractions $x = \frac{d - b}{a - c}$
- Q19** Dr Eslam • Twice, with a fraction $x = \frac{3y + 2}{y - 1}$
- Q20** Practice • Medium $t = \frac{2d - 7}{6}$
- Q21** Practice • Hard $v = \frac{15t}{w + 30}$
- Q22** Practice • Hard $a = \frac{4p - 5}{p + 3}$
- Q23** Practice • Hard $m = \frac{f + 4}{f - 3}$
- Q24** Edexcel 4MA1 • November 2021 • Paper 1H • Q17
 (a) $(6y + 5)(y - 1)$ (b) $f = \frac{8w - 3}{w + 2}$ (c) $4(x - 1)^2 + 3$
- Q25** Edexcel 4MA1 • June 2022 • Paper 2H • Q14
 (a) $-2x^3 - x^2 + 43x + 60$ (b) $c = -\frac{4g + 25}{g + 6}$
- Q26** Edexcel 4MA1 • November 2023 • Paper 1H • Q15
 $n = \frac{3p + 4x}{3x - 1}$

IDEA 04 • ROOT AND TWICE

- Q27** Dr Eslam • The full exam type $x = \frac{p^2 + 3}{p^2 - 1}$
- Q28** Practice • Hard $c = \frac{8 - 3p^2}{p^2 - a}$
- Q29** Practice • Hard $n = \sqrt{\frac{d}{y - 1}}$
- Q30** Practice • Hard $e = \frac{dw^2 + g}{fw^2 - 1}$
- Q31** Edexcel 4MA1 • January 2020 • Paper 2H • Q16 $x = \frac{4y^2 + 1}{y^2 - 1}$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2H • Q14 (a) $a = 4$ (b) $c = \frac{8 - 3p^2}{p^2 - a}$
- Q33** Edexcel 4MA1 • January 2022 • Paper 2HR • Q15 $t = \sqrt[3]{\frac{4d}{n^2 - 1}}$
- Q34** Edexcel 4MA1 • June 2023 • Paper 2HR • Q17 $x = \frac{6 - 4y^3}{y^3 - 5}$
- Q35** Edexcel 4MA1 • June 2023 • Paper 2H • Q17 (a) $3x^3 + 34x^2 + 84x - 72$ (b) $e = \frac{dw^2 + g}{fw^2 - 1}$
- Q36** Edexcel 4MA1 • June 2024 • Paper 2H • Q15 (a) $g = \frac{5 - 11e^2}{2e^2 - 7}$ (b) $y < -4$ or $y > \frac{8}{3}$
- Q37** Edexcel 4MA1 • November 2024 • Paper 2H • Q12 (a) $x = \frac{1}{16}$ (b) $c = \frac{a + df^2}{f^2 - b}$
- Q38** Edexcel 4MA1 • June 2025 • Paper 2HR • Q17 $k = \sqrt{\frac{7p - 5}{3p + 8}}$
- Q39** Edexcel 4MA1 • June 2025 • Paper 2H • Q16 $t = \sqrt{\frac{7c - 3}{8c + 1}}$
- Q40** Edexcel 4MA1 • November 2025 $y = \frac{p^2t + w}{3p^2 - 1}$

IDEA 05 • QUADRATIC IN THE SUBJECT

Q41 Dr Eslam • Complete the square first

$$y = (x - 2)^2 - 3 \Rightarrow x = 2 \pm \sqrt{y + 3}$$

Q42 Dr Eslam • With $a \neq 1$

$$y = 2(x + 2)^2 - 9 \Rightarrow x = -2 \pm \sqrt{\frac{y + 9}{2}}$$

Q43 Dr Eslam • Inverse of a quadratic

$$f(x) = (x + 5)^2 - 25 \Rightarrow f^{-1}(x) = -5 + \sqrt{x + 25}$$

IDEA 06 • WHERE YOU WILL USE IT

Q44 Dr Eslam • Inverse function

$$\frac{4x + 1}{x - 2}$$

Q45 Dr Eslam • Same method, numbers

$$x = -9.5$$

DR ESLAM AHMED
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Algebraic Proof

Translate the words, expand and simplify, then factorise to show the property. Every proof is these three moves.

THE TOPIC ON ONE MAP

Algebraic Proof

01 The dictionary

$n, n + 1$ $2n$ $2n + 1$

02 Even, odd, multiple of k

$2(\dots)$ $8(\dots) + 2$

03 Square, cube, always positive

$(\dots)^2$ complete the square

04 Disprove

one counter-example

05 Proof by reasoning

consecutive integers primes

01 The Dictionary and the Three Moves

First turn the words into algebra. This table is your dictionary.

WHEN YOU SEE → YOU DO

“prove” / “show that ... for all integers n ”

→ use n — never test numbers

“consecutive”

→ $n, n + 1, n + 2$

“any two odd numbers”

→ two letters: $2a + 1, 2b + 1$

If the question says...

any integer

consecutive integers

an even number

an odd number

consecutive even / odd numbers

a multiple of k · one more than a multiple of k

a square number

you write...

n

$n, n + 1, n + 2$

$2n$

$2n + 1$

$2n, 2n + 2$ / $2n + 1, 2n + 3$

$kn \cdot kn + 1$

n^2

1

Translate

Use the dictionary. Two *different* odd numbers? Use two letters: $2a + 1, 2b + 1$.

2

Expand and simplify

Expand every bracket, collect like terms.

3

Factorise to prove

Make the answer *look like* what you must prove — then write one sentence of conclusion.

IDEA 01

Practise: The Dictionary and the Three Moves

My questions first, then selected practice.

Q1 DR ESLAM Use the dictionary

CH 2 · TOPIC 8 4 marks

Write an expression for (a) three consecutive integers (b) two consecutive even numbers (c) an odd number (d) a multiple of 3.

Q2 DR ESLAM Three moves

CH 2 · TOPIC 8 3 marks

Prove that the sum of any three consecutive integers is a multiple of 3.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 4 marks

12 Prove that the mean of any four consecutive even integers is an integer.

(4 marks)

Q4**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 5 marks

13 (a) Prove that the sum of four consecutive whole numbers is always even.

(3 marks)

(b) Give an example to show that the sum of four consecutive integers is **not** always divisible by 4.

(2 marks)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8

4 marks

3 Prove that , for all positive values of n ,

$$\frac{(n+2)^2 - (n+1)^2}{2n^2 + 3n} = \frac{1}{n}$$

(4 marks)**Q6****PRACTICE**

Practice · Hard

CH 2 · TOPIC 8

4 marks

4 Prove algebraically that the difference between the squares of any two consecutive integers is equal to the sum of these two integers.

(4 marks)

Q7**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 8 5 marks

9 (a) Here are the first four terms of a sequence of fractions.

$$\frac{1}{1} \quad \frac{2}{3} \quad \frac{3}{5} \quad \frac{4}{7}$$

The numerators of the fractions form the sequence of whole numbers 1 2 3 4 ...

The denominators of the fractions form the sequence of odd numbers 1 3 5 7 ...

Write down an expression, in terms of n , for the n th term of this sequence of fractions.

(2 marks)

(b) Using algebra, prove that when the square of any odd number is divided by 4 the remainder is 1

(3 marks)

Q8**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 8 7 marks**12 (a)**

The diagram shows a cross placed on a number grid.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

 L is the product of the left and right numbers of the cross. T is the product of the top and bottom numbers of the cross. M is the middle number of the cross.Show that when $M = 35$, $L - T = 99$.**(2 marks)****(b)** Prove that, for any position of the cross on the number grid above, $L - T = 99$.**(5 marks)**

02 Even, Odd, Multiple of k

The last line decides the mark: take out the right factor, then say why.

WHEN YOU SEE → YOU DO

"always even"	→ $2(\dots)$
"always odd"	→ $2(\dots) + 1$
"a multiple of 8"	→ $8(\dots)$
"remainder is 2 when divided by 8"	→ $8(\dots) + 2$

Even	$2 \times (\text{integer})$	take out 2
Odd	$2 \times (\text{integer}) + 1$	take out 2, leave +1
Multiple of k	$k \times (\text{integer})$	take out k

WORKED MODEL • MULTIPLE OF 8

Prove that $(2n + 3)^2 - (2n + 1)^2$ is a multiple of 8.

1 $(4n^2 + 12n + 9) - (4n^2 + 4n + 1)$ expand

2 $8n + 8$ simplify

3 $8(n + 1)$ factorise to prove

ANSWER $8(n + 1)$ — a multiple of 8, since $n + 1$ is an integer

× **TRAP** Showing it works for $n = 1, 2, 3$ is **not** a proof. Use n .

IDEA 02

Practise: Even, Odd, Multiple of k

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Multiple of 8

CH 2 · TOPIC 8 3 marks

Prove that $(n + 5)^2 - (n + 1)^2$ is a multiple of 8 for all integers n .

Q10 DR ESLAM Odd

CH 2 · TOPIC 8 3 marks

Prove that the product of two consecutive odd numbers is always odd.

Q11 DR ESLAM Multiple of 4

CH 2 · TOPIC 8 2 marks

Prove that the sum of two consecutive odd numbers is a multiple of 4.

SELECTED PRACTICE

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 8 3 marks

1 Prove algebraically that

$$(2n + 1)^2 - (2n + 1) \text{ is an even number}$$

for all positive integer values of n .

(3 marks)

Q13 **PRACTICE** Practice · Medium

CH 2 · TOPIC 8 4 marks

3 n is an integer greater than 1Prove algebraically that $n^2 - 2 - (n - 2)^2$ is always an even number.

(4 marks)

Q14 **PRACTICE** Practice · Hard

CH 2 · TOPIC 8 4 marks

2 Prove that the square of an odd number is always 1 more than a multiple of 4

(4 marks)

Q15 **PRACTICE** Practice · Hard

CH 2 · TOPIC 8 3 marks

12 c is a positive integer.

Prove that $\frac{6c^3 + 30c}{3c^2 + 15}$ is an even number.

(3 marks)

PAST PAPERS

Q16 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2HR · Q25

CH 2 · TOPIC 8 3 marks

25 N is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

Q17 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2H · Q17

CH 2 · TOPIC 8 3 marks

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

(Total for Question 17 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q15

CH 2 · TOPIC 8**4 marks**

15 Prove algebraically that the product of any two odd numbers is always an odd number.

(Total for Question 15 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q17

CH 2 · TOPIC 8**3 marks**

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

(Total for Question 17 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q15

CH 2 · TOPIC 8**3 marks**

- 15 Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

(Total for Question 15 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2H · Q18

CH 2 · TOPIC 8**3 marks**

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

(Total for Question 18 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q17

CH 2 · TOPIC 8**3 marks**

- 17 Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

(Total for Question 17 is 3 marks)

03 Square, Cube, Always Positive

To prove a square number, factorise into $(\quad)^2$. To prove “always positive”, complete the square.

WHEN YOU SEE → YOU DO

“is a square number”

→ factorise to $(\dots)^2$

“is a cube number”

→ simplify to $(\dots)^3$

“always positive”

→ complete the square

WORKED MODEL • A SQUARE NUMBER

Prove that $35 + (3x + 1)^2 - 2x(4x - 3)$ is a square number.

1 $35 + 9x^2 + 6x + 1 - 8x^2 + 6x$

expand

2 $x^2 + 12x + 36$

simplify

ANSWER $(x + 6)^2$ — a square number

WORKED MODEL • ALWAYS POSITIVE

Prove that $x^2 + x + 1$ is always positive.

1 $\left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + 1$

complete the square

2 $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

simplify

ANSWER $\left(x + \frac{1}{2}\right)^2 \geq 0 \Rightarrow \text{expression} \geq \frac{3}{4} > 0$

IDEA 03

Practise: Square, Cube, Always Positive

My questions first, then selected practice.

Q23 DR ESLAM Factorise to a square

CH 2 · TOPIC 8 3 marks

Prove that $n(n + 2) + 1$ is always a square number.

Q24 DR ESLAM Complete the square

CH 2 · TOPIC 8 3 marks

Prove that $x^2 - 6x + 10 > 0$ for all values of x .

Complete the square first.

SELECTED PRACTICE

Q25

PRACTICE

Practice · Medium

CH 2 · TOPIC 8 4 marks

8 x is an integer.

Prove that $35 + (3x + 1)^2 - 2x(4x - 3)$ is a square number.

(4 marks)

Q26

PRACTICE

Practice · Hard

CH 2 · TOPIC 8 4 marks

8 n is the middle integer of three consecutive positive integers.

The three integers are multiplied to give a product.

n is then added to the product.

Prove that the result is a cube number.

(4 marks)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8 4 marks

- 9 Expressions for consecutive triangular numbers are

$$\frac{n(n+1)}{2} \text{ and } \frac{(n+1)(n+2)}{2}$$

Prove that the sum of two consecutive triangular numbers is always a square number.

(4 marks)**Q28****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 3 marks

- 5 The product of two consecutive positive integers is added to the larger of the two integers.

Prove that the result is always a square number.

(3 marks)

Q29**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 4 marks

10 The table gives information about the first six terms of a sequence of numbers.

Term number	1	2	3	4	5	6
Term of sequence	$\frac{1 \times 2}{2}$	$\frac{2 \times 3}{2}$	$\frac{3 \times 4}{2}$	$\frac{4 \times 5}{2}$	$\frac{5 \times 6}{2}$	$\frac{6 \times 7}{2}$

Prove algebraically that the sum of any two consecutive terms of this sequence is always a square number.

(4 marks)**Q30****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 3 marks

11 Prove that $x^2 + x + 1$ is always positive.

(3 marks)

04 Disprove: One Counter-Example

To show a statement is **false**, one example is enough — but you must show the working that it fails.

WHEN YOU SEE → YOU DO

“show that ... is wrong” / “is Ali correct?”

→ one counter-example, with working

“always / sometimes / never true”

→ test positive, negative, zero

1

Try small values

Put $n = 1, 2, 3, \dots$ into the expression until one fails.

2

Show it fails

Not prime? Write its factors: $91 = 7 \times 13$. Not even? Show the number.

WORKED MODEL • “ALWAYS PRIME”?

$k = n^2 + 9n + 1$ is prime for $n = 1$ to 9?

1 $n = 6: 36 + 54 + 1 = 91$

try values

ANSWER $91 = 7 \times 13$ — not prime, so the statement is wrong

ALWAYS / SOMETIMES / NEVER Test a positive, a negative, and zero. One case each way = “sometimes”.

IDEA 04

Practise: Disprove: One Counter-Example

My questions first, then selected practice.

Q31 DR ESLAM Counter-example

CH 2 · TOPIC 8 2 marks

Sam says $n^2 + n + 11$ is prime for every positive integer n . Show that Sam is wrong.

Q32 DR ESLAM One example is enough

CH 2 · TOPIC 8 1 mark

Show that the statement “the sum of two prime numbers is always even” is false.

SELECTED PRACTICE

Q33**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 2 marks

6 $E = n^2 + n + 5$

Ali thinks that the value of E will be a prime number for any whole number value of n .

Is Ali correct?

You must give a reason for your answer.

(2 marks)

Q34**PRACTICE**

Practice • Medium

CH 2 • TOPIC 8 4 marks

7 p is a positive number. n is a negative number.

For each statement, tick the correct box.

	Always true	Sometimes true	Never true
$p + n$ is positive	☐	☐	☐
$p - n$ is positive	☐	☐	☐
$p^2 + n^2$ is positive	☐	☐	☐
$p^3 \div n^3$ is positive	☐	☐	☐

(4 marks)

Q35**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 1 mark

9 Which of these is a correct identity?

Circle your answer.

$x + 4x \equiv 5x$	$6x \equiv 18$	$2x + 1 \equiv 7$	$7x + 9 \equiv x$
--------------------	----------------	-------------------	-------------------

(1 mark)

Q36**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 3 marks

10

$$k = n^2 + 9n + 1$$

Mo says, “ k will be a prime number for all integer values of n from 1 to 9”

Show that Mo is wrong.

You **must** show that your value of k is **not** prime.

(3 marks)

Q37**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 1 mark

11 Tick whether the following statement is true or false.

Give a reason for your answer.

When n is a positive integer, the value of $2n$ is **always** a factor of the value of $20n$.

True ☐ False ☐

(1 mark)

05 Proof by Reasoning

Some proofs need a sentence, not just algebra: consecutive numbers, factors, primes.

WHEN YOU SEE → YOU DO

“prime”

→ a product of two factors > 1 is not prime

$n^3 - n, n(n+1)$

→ consecutive integers → even / multiple of 6

Two consecutive integers

one of them is even → the product is even

Three consecutive integers

one is even, one is a multiple of 3 → product is a multiple of 6

A product of two factors, both > 1

cannot be prime

$a^2 - b^2 = (a+b)(a-b)$ **is prime**

so $a - b = 1$: a and b are consecutive

WORKED MODEL • MULTIPLE OF 6

Prove that $n^3 - n$ is always a multiple of 6.

1 $n(n^2 - 1) = (n - 1)n(n + 1)$

factorise

ANSWER a multiple of 6

WHY $(n - 1), n, (n + 1)$ are three consecutive integers: one is even and one is a multiple of 3.

IDEA 05

Practise: Proof by Reasoning

My questions first, then selected practice.

Q38 **DR ESLAM** Square of an even number

CH 2 · TOPIC 8 2 marks

Prove that the square of any even number is a multiple of 4.

Q39 **DR ESLAM** Consecutive integers

CH 2 · TOPIC 8 2 marks

Prove that $n^2 + n$ is always even.

Factorise, then think about consecutive numbers.

SELECTED PRACTICE

Q40

PRACTICE

Practice · Hard

CH 2 · TOPIC 8

4 marks

6 (a) Show that $x(x - 1)(x + 1) = x^3 - x$

(1 mark)

(b) Prove that the difference between a whole number and the cube of this number is always a multiple of 6

(3 marks)

Q41

PRACTICE

Practice · Hard

CH 2 · TOPIC 8

2 marks

11 $a^2 - b^2 \equiv (a + b)(a - b)$

a and b are positive whole numbers with $a > b$

$a^2 - b^2$ is a **prime** number.

Why are a and b consecutive numbers?

(2 marks)

Q42**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8**3 marks**

- 13** The lengths of the sides of a right-angled triangle are all integers. Prove that if the lengths of the two shortest sides are even, then the length of the third side must also be even.

(3 marks)**Q43****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8**3 marks**

- 1** (i) Factorise $2t^2 + 5t + 2$.

[2]

- (ii) t is a positive whole number.

The expression $2t^2 + 5t + 2$ can never have a value that is a prime number.
Explain why.

[1]

(3 marks)

Q44 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 8 6 marks

3 Here are the first five terms of an arithmetic sequence.

7 13 19 25 31

Prove that the difference between the squares of any two terms of the sequence is always a multiple of 24.

(6 marks)

Q45 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 8 2 marks

4 Given that n can be any integer such that $n > 1$, prove that $n^2 - n$ is never an odd number.

(2 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------------------|--|
| 01 | Three moves | Translate (dictionary) → expand & simplify → factorise to prove. |
| 02 | Even / odd / k | $2() \cdot 2() + 1 \cdot k()$ — then one sentence. |
| 03 | Square • positive | Factorise to $()^2$. Always positive → complete the square. |
| 04 | Disprove | One counter-example, with the working that shows it fails. |
| 05 | Reasoning | Consecutive numbers, factors, primes — explain in words. |

BOTTOM LINE Translate → expand → simplify → factorise to prove → write the conclusion.

Answers

Algebraic Proof • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE DICTIONARY AND THE THREE MOVES

- Q1** Dr Eslam • Use the dictionary (a) $n, n + 1, n + 2$ (b) $2n, 2n + 2$ (c) $2n + 1$ (d) $3n$
- Q2** Dr Eslam • Three moves $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$
- Q3** Practice • Medium $\text{mean} = \frac{8n + 12}{4} = 2n + 3$, an integer
- Q4** Practice • Medium (a) $4n + 6 = 2(2n + 3)$ (b) $1 + 2 + 3 + 4 = 10$
- Q5** Practice • Hard $\frac{2n + 3}{n(2n + 3)} = \frac{1}{n}$
- Q6** Practice • Hard $(n + 1)^2 - n^2 = 2n + 1 = n + (n + 1)$
- Q7** Practice • Very Hard (a) $\frac{n}{2n - 1}$ (b) $4(n^2 + n) + 1$
- Q8** Practice • Very Hard (a) $1224 - 1125 = 99$ (b) $(n^2 - 1) - (n^2 - 100) = 99$

IDEA 02 • EVEN, ODD, MULTIPLE OF K

Q9	Dr Eslam • Multiple of 8	$8n + 24 = 8(n + 3)$
Q10	Dr Eslam • Odd	$(2n + 1)(2n + 3) = 4n^2 + 8n + 3 = 2(2n^2 + 4n + 1) + 1$
Q11	Dr Eslam • Multiple of 4	$(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$
Q12	Practice • Medium	$4n^2 + 2n = 2(2n^2 + n), \text{ even}$
Q13	Practice • Medium	$4n - 6 = 2(2n - 3), \text{ even}$
Q14	Practice • Hard	$4n^2 + 4n + 1 = 4(n^2 + n) + 1$
Q15	Practice • Hard	$\frac{6c(c^2 + 5)}{3(c^2 + 5)} = 2c$
Q16	Edexcel 4MA1 • January 2020 • Paper 2HR • Q25	$4N, \text{ with } N = 5k: 20k$
Q17	Edexcel 4MA1 • January 2020 • Paper 2H • Q17	$2n + 1, \text{ odd}$
Q18	Edexcel 4MA1 • January 2021 • Paper 2HR • Q15	$2(2ab + a + b) + 1$
Q19	Edexcel 4MA1 • May 2021 • Paper 1H • Q17	$16n + 16 = 8(2n + 2)$
Q20	Edexcel 4MA1 • January 2022 • Paper 1HR • Q15	$2n^2 + 4n + 4 = 2(n + 1)^2 + 2$
Q21	Edexcel 4MA1 • January 2022 • Paper 2H • Q18	$8(n^2 + 2n + 1) + 2$
Q22	Edexcel 4MA1 • June 2025 • Paper 1HR • Q17	$(4n + 8)^2 - (4n)^2 = 64(n + 1)$

IDEA 03 • SQUARE, CUBE, ALWAYS POSITIVE

- Q23** Dr Eslam • Factorise to a square $n^2 + 2n + 1 = (n + 1)^2$
- Q24** Dr Eslam • Complete the square $(x - 3)^2 + 1 \geq 1 > 0$
- Q25** Practice • Medium $x^2 + 12x + 36 = (x + 6)^2$
- Q26** Practice • Hard $(n - 1)n(n + 1) + n = n^3$
- Q27** Practice • Hard $(n + 1)^2$
- Q28** Practice • Very Hard $n(n + 1) + (n + 1) = (n + 1)^2$
- Q29** Practice • Very Hard $(n + 1)^2$
- Q30** Practice • Very Hard $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$

IDEA 04 • DISPROVE: ONE COUNTER-EXAMPLE

- Q31** Dr Eslam • Counter-example $n = 10: 121 = 11 \times 11$, not prime
- Q32** Dr Eslam • One example is enough $2 + 3 = 5$, which is odd
- Q33** Practice • Medium No: $n = 5$ gives $35 = 5 \times 7$
- Q34** Practice • Medium sometimes • always • always • never
- Q35** Practice • Medium $x + 4x \equiv 5x$
- Q36** Practice • Medium $n = 6: k = 91 = 7 \times 13$
- Q37** Practice • Medium True: $20n = 10 \times 2n$

IDEA 05 • PROOF BY REASONING

- Q38** Dr Eslam · Square of an even number
 $(2n)^2 = 4n^2 = 4(n^2)$
- Q39** Dr Eslam · Consecutive integers
 $n(n + 1)$: two consecutive integers, one of them even
- Q40** Practice · Hard
(b) $(n - 1)n(n + 1)$: one even, one a multiple of 3
prime $\Rightarrow a - b = 1$
- Q41** Practice · Hard
 $c^2 = 4(m^2 + n^2)$ even, so c even
- Q42** Practice · Hard
(i) $(2t + 1)(t + 2)$ (ii) both factors ≥ 3
- Q43** Practice · Very Hard
 $12(a - b)(3a + 3b + 1)$; one factor even
- Q44** Practice · Very Hard
 $n(n - 1)$: consecutive, so even
- Q45** Practice · Very Hard



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Solving Linear Equations

One method for every linear equation: clean it first, then isolate the letter. Brackets, fractions, letters on both sides — same two stages.

THE TOPIC ON ONE MAP

Solving Linear Equations

01 — Brackets and x on both sides

$$3(x - 2) = x + 7$$

$$5(4 - x) = 7 - 3x$$

$$4x - 13 = 17 + 8x$$

spot the mistake

02 — One fraction

$$\frac{5x - 3}{4} = 2x + 3$$

$$p = \frac{3p - 5}{10}$$

$$\frac{2f}{3} = 4f - 17$$

$$\frac{4(8x - 2)}{3x} = 10$$

03 — Two or more fractions

$$\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$$

$$\frac{x + 2}{3x} + \frac{x - 2}{2x} = 3$$

$$3(x - 4) + \frac{x + 2}{5} = 6$$

04 — Hidden linear equations

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$fg(x) = f(x)$$

the x^2 cancels

IN THE EXAM Almost every Paper 1 and Paper 2 has one: part (b) or (c), 3 marks, “Show clear algebraic working”.

01 Clean, then Isolate

Every linear equation — however it looks — is solved in two stages. **Clean** removes the obstacles; **Isolate** gets x alone.

WHEN YOU SEE → YOU DO

“Solve” + one letter, no x^2

→ linear: Clean → Isolate

a bracket

→ expand it first (multiply **every** term inside)

x on both sides

→ move the **smaller** x term

“Show clear algebraic working”

→ write every line — trial and error scores 0

Clean

Collect

Add / Subtract

Divide

ZONE A

Brackets

Expand.

$$5(x + 3) \rightarrow 5x + 15$$

ZONE B

One fraction

Multiply both sides by the bottom.

$$\frac{5x - 3}{4} = \dots \rightarrow \times 4$$

ZONE C

Two fractions

Multiply **every term** by the LCM.

bottoms 3 and 2 $\rightarrow \times 6$

1

Clean

Expand brackets, remove every denominator. Now there are no brackets, no fractions.

2

Collect

x terms on one side, numbers on the other. Change the sign when a term crosses “=”.

3

Add / Subtract, then Divide

Combine each side to one term, then divide by the number in front of x .

WORKED MODEL · BRACKETS AND x ON BOTH SIDESSolve $5(x + 3) = 3x - 4$.

1 $5x + 15 = 3x - 4$

clean: expand

2 $5x - 3x = -4 - 15$

collect (move $3x$)

3 $2x = -19$

add / subtract

ANSWER $x = -9.5$

× **TRAP** $5(x + 2) = 5x + 2$ is wrong — the 5 multiplies **both** terms: $5x + 10$.

IDEA 01

Practise: Clean, then Isolate

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Expand first

CH 2 · TOPIC 9 3 marks

Solve $4(x - 3) = 2x + 6$

X = _____

Q2 DR ESLAM Minus before a bracket

CH 2 · TOPIC 9 3 marks

Solve $7 - 2(x + 1) = 3x$

X = _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 5 marks

1 (a) Solve $3(x - 2) = x + 7$

(3 marks)

(b) Solve $\frac{2 - y}{5} = 1$

(2 marks)

Q4**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 3 marks

5 Solve $5x - 6 = 3(x - 1)$

(3 marks)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 2 marks

10 Solve $5(x + 3) = 3x - 4$ Show clear algebraic working.

 $x =$

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 7 marks

4 (a) Solve $5(f - 3) = f + 10$

(3 marks)

(b) Solve $\frac{h + 7}{3} + \frac{2h - 1}{2} = \frac{5}{6}$

(4 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q3

CH 2 · TOPIC 9 6 marks

- 3 (a) Simplify $h^7 \times h^2$

.....

$$G = c^2 - 4c$$

- (b) Find the value of G when $c = -5$

$$G = \dots\dots\dots$$

- (c) Solve $\frac{5x - 3}{4} = 2x + 3$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(Total for Question 3 is 6 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q3

CH 2 · TOPIC 9 6 marks

3 (a) Simplify $(p^3)^5$

.....

(b) Expand and simplify $2n(4n + 3) + n(n - 4)$

.....

(2)

(c) Solve $\frac{2x + 5}{3} = 4 - x$

Show clear algebraic working.

$x =$

(Total for Question 3 is 6 marks)

02 One Fraction: Multiply Both Sides

One denominator? Multiply **both sides** by it. Put the other side in a bracket first — that is where marks are lost.

WHEN YOU SEE → YOU DO

$$\frac{\square}{4} = \square$$

→ ×4 both sides, other side in a bracket

x in the denominator: $\frac{\square}{3x} = 10$

→ still multiply both sides: ×3x

“Solve $p = \frac{3p - 5}{10}$ ”

→ $10p = 3p - 5$

Bracket the other side

Multiply up

Expand

Collect

Divide

WORKED MODEL • MAY 2024 P1H

Solve $3n - 4 = \frac{5n + 6}{3}$.

1 $3(3n - 4) = 5n + 6$

clean: ×3, bracket

2 $9n - 12 = 5n + 6$

clean: expand

3 $9n - 5n = 6 + 12$

collect

4 $4n = 18$

add / subtract

ANSWER $n = 4.5$

× **TRAP** $3n - 4 = \frac{5n + 6}{3} \rightarrow 3n - 12 = 5n + 6$ is wrong: **every** term on that side is multiplied — $9n - 12$.

CALCULATOR CHECK Put your answer back in both sides (CALC button). Both sides equal? Done.

IDEA 02

Practise: One Fraction: Multiply Both Sides

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

One fraction

CH 2 · TOPIC 9 3 marks

Solve $\frac{3x + 1}{4} = x - 2$

X = _____

Q10**DR ESLAM**

Fraction with a minus

CH 2 · TOPIC 9 3 marks

Solve $5 - \frac{x - 1}{3} = 2x$

X = _____

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Easy

CH 2 · TOPIC 9 3 marks

8 Solve $y = \frac{2y + 1}{5}$

Show clear algebraic working.

$y =$
(3 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 9 3 marks

2 Solve $\frac{4(8x - 2)}{3x} = 10$

(3 marks)

Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 9 3 marks

6 Solve $\frac{5 - x}{2} = 2x - 7$

(3 marks)

Q14**PRACTICE**

Practice · Medium

CH 2 · TOPIC 9 3 marks

13 Solve $\frac{2x + 5}{6} = 2x - 5$

Show clear algebraic working.

(3 marks)

PAST PAPERS

Q15**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q7

CH 2 • TOPIC 9 9 marks7 (a) Expand and simplify $(m - 8)(m + 5)$

.....

(b) Factorise fully $5y + 20y^2$

.....

(c) Simplify $(p^2 + 3)^0$

.....

(1)

(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

 $x =$ **(Total for Question 7 is 9 marks)**

Q16**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q5

CH 2 · TOPIC 9 5 marks

5 (a) Factorise fully $15y^4 + 20uy^3$

.....
(2)

(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

$x =$

(Total for Question 5 is 5 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q3

CH 2 · TOPIC 9 5 marks3 (a) Solve $4y + 5 > 12$
(2)(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

 $x =$ **(Total for Question 3 is 5 marks)**

Q18 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 1H · Q5**CH 2 · TOPIC 9** **5 marks**5 (a) Expand and simplify $(n - 6)(n + 4)$

(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 5 is 5 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q6

CH 2 · TOPIC 9 8 marks

6 (a) Solve $p = \frac{3p - 5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots$$

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots (1)$$

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots$$

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots$$

(Total for Question 6 is 8 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q6

CH 2 · TOPIC 9 8 marks

6 (a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$$f = \dots\dots\dots$$

(b) Simplify $(e + 12)^0$ where $e > 0$

$$\dots\dots\dots$$

(1)

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

$$\dots\dots\dots$$

(d) Factorise fully $20x^5y + 12x^3y^4$

$$\dots\dots\dots$$

(Total for Question 6 is 8 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q6

CH 2 · TOPIC 9 5 marks6 (a) Expand and simplify $(m + 5)(m - 8)$ (b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

 $n =$ **(Total for Question 6 is 5 marks)**

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q1

CH 2 · TOPIC 9 5 marks

- 1 (a) Factorise fully $18c - 45cd$

.....
(2)

- (b) Solve $\frac{5 - 2x}{6} = 3x - 4$

Show clear algebraic working.

$x =$

(Total for Question 1 is 5 marks)

03 Two or More Fractions: LCM

Several denominators? Find the LCM and multiply **every** term — including the number on the right.

WHEN YOU SEE → YOU DO

$$\frac{\square}{3} - \frac{\square}{2} = 13$$

→ LCM = 6: $2(\dots) - 3(\dots) = 78$

a minus in front of a fraction

→ bracket it: $-3(3 - 2x) = -9 + 6x$

x in the bottoms: 3x, 2x

→ LCM = 6x

a term with no fraction

→ it gets multiplied too

Find LCM

Multiply every term

Cancel

Expand

Collect

Divide

WORKED MODEL • JUNE 2022 P2H

Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$.

1 $2(4x+5) - 3(3-2x) = 78$

clean: $\times 6$

2 $8x + 10 - 9 + 6x = 78$

expand (watch the -)

3 $14x + 1 = 78 \Rightarrow 14x = 77$

collect

ANSWER $x = 5.5$

× **TRAP** Forgetting to multiply the right-hand side: $2(4x+5) - 3(3-2x) = 13$ loses the method mark. It is $13 \times 6 = 78$.

IDEA 03

Practise: Two or More Fractions: LCM

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM LCM of 2 and 5

CH 2 · TOPIC 9 4 marks

Solve $\frac{x+3}{2} - \frac{x-1}{5} = 4$

X = _____

Q24 DR ESLAM Every term

CH 2 · TOPIC 9 3 marks

Solve $\frac{x}{3} + \frac{x}{4} = 14$

X = _____

SELECTED PRACTICE

Q25

PRACTICE

Practice · Hard

CH 2 · TOPIC 9 3 marks

1 Solve $\frac{4x - 1}{5} + \frac{x + 4}{2} = 3$

(3 marks)

Q26

PRACTICE

Practice · Hard

CH 2 · TOPIC 9 3 marks

3 Solve $\frac{x + 2}{3x} + \frac{x - 2}{2x} = 3$

(3 marks)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

6 Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

a =

(3 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 4 marks

10 Solve the equation.

$$3(x-4) + \frac{x+2}{5} = 6$$

x =

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q15

CH 2 • TOPIC 9 7 marks

15 (a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$x =$

(b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

.....

(Total for Question 15 is 7 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q15

CH 2 · TOPIC 9 6 marks

15 (a) Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$

Show clear algebraic working.

$$a = \dots\dots\dots$$

(4)

(b) Express $\left(\frac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

(Total for Question 15 is 6 marks)

04 Hidden Linear Equations

Some equations *look* quadratic. Expand everything — if the x^2 terms cancel, it was linear all along.

WHEN YOU SEE → YOU DO

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

→ expand both sides — the $4x^2$ cancels

$$fg(x) = f(x)$$

→ build $fg(x)$, then the x^2 cancels

“What mistake did he make?”

→ find the first wrong line, say what should be there

WORKED MODEL • THE x^2 CANCELS

Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$.

$$1 \quad 4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

clean: expand both

$$2 \quad 20x - 4x = -3 - 25$$

$4x^2$ cancels; collect

$$3 \quad 16x = -28$$

add / subtract

ANSWER $x = -\frac{7}{4}$

WHERE YOU WILL USE IT Rearranging formulas (Topic 7), simultaneous equations (Topic 12), forming equations (Topic 13) all end in a linear equation.

IDEA 04

Practise: Hidden Linear Equations

My questions first, then selected practice.

Q31 DR ESAM The square cancels

CH 2 · TOPIC 9 3 marks

Solve $(x + 4)^2 = x^2 + 40$

$x =$ _____

SELECTED PRACTICE**Q32** PRACTICE Practice · Easy

CH 2 · TOPIC 9 1 mark

6 Steve is asked to solve the equation $5(x + 2) = 47$

Here is his working.

$$5(x + 2) = 47$$

$$5x + 2 = 47$$

$$5x = 45$$

$$x = 9$$

Steve's answer is wrong.

What mistake did he make?

(1 mark)

Q33**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

7 Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

 $x = \dots\dots\dots$ **(3 marks)****Q34****PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

9 The functions f and g are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

Solve the equation $fg(x) = f(x)$

Show clear algebraic working.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.



- 01 Clean, then Isolate** Expand brackets → collect (move the smaller x) → add/subtract → divide.
- 02 One fraction** Bracket the other side, multiply both sides by the bottom.
- 03 Two fractions** LCM × **every** term, the right-hand side too. Bracket after a minus.
- 04 Hidden linear** Expand everything; the x^2 cancels.

BOTTOM LINE Clean → Collect → Add / Subtract → Divide → check with CALC.

Answers

Solving Linear Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CLEAN, THEN ISOLATE

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam • Expand first | $4x - 12 = 2x + 6; x = 9$ |
| Q2 | Dr Eslam • Minus before a bracket | $7 - 2x - 2 = 3x; x = 1$ |
| Q3 | Practice • Easy | (a) $x = 6.5$ (b) $y = -3$ |
| Q4 | Practice • Easy | $x = 1.5$ |
| Q5 | Practice • Easy | $x = -9.5$ |
| Q6 | Practice • Hard | (a) $f = \frac{25}{4}$ (b) $h = -\frac{3}{4}$ |
| Q7 | Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q3 | (a) h^9 (b) 45 (c) $x = -5$ |
| Q8 | Edexcel 4MA1 • November 2024 • Paper 1H • Q3 | (a) p^{15} (b) $9n^2 + 2n$ (c) $x = \frac{7}{5}$ |

IDEA 02 • ONE FRACTION: MULTIPLY BOTH SIDES

- Q9** Dr Eslam · One fraction $3x + 1 = 4x - 8; x = 9$
- Q10** Dr Eslam · Fraction with a minus $15 - (x - 1) = 6x; x = \frac{16}{7}$
- Q11** Practice · Easy $y = \frac{1}{3}$
- Q12** Practice · Medium $x = 4$
- Q13** Practice · Medium $x = 3.8$
- Q14** Practice · Medium $x = 3.5$
- Q15** Edexcel 4MA1 · January 2020 · Paper 1H · Q7 (a) $m^2 - 3m - 40$ (b) $5y(1 + 4y)$ (c) 1 (d) $x = 3$
- Q16** Edexcel 4MA1 · January 2021 · Paper 1H · Q5 (a) $5y^3(3y + 4u)$ (b) $x = \frac{11}{4}$
- Q17** Edexcel 4MA1 · January 2022 · Paper 2HR · Q3 (a) $y > \frac{7}{4}$ (b) $x = \frac{3}{8}$
- Q18** Edexcel 4MA1 · June 2022 · Paper 1H · Q5 (a) $n^2 - 2n - 24$ (b) $x = \frac{7}{5}$
- Q19** Edexcel 4MA1 · June 2022 · Paper 2HR · Q6 (a) $p = -\frac{5}{7}$ (b) 1 (c) $\frac{y^2}{2x}$ (d) $5cd^2(2c^2 + 3d^2)$
- Q20** Edexcel 4MA1 · June 2024 · Paper 2HR · Q6 (a) $f = 5.1$ (b) 1 (c) $3a^3h^4$ (d) $4x^3y(5x^2 + 3y^3)$
- Q21** Edexcel 4MA1 · May 2024 · Paper 1H · Q6 (a) $m^2 - 3m - 40$ (b) $n = 4.5$
- Q22** Edexcel 4MA1 · June 2025 · Paper 2HR · Q1 (a) $9c(2 - 5d)$ (b) $x = 1.45$

IDEA 03 · TWO OR MORE FRACTIONS: LCM

- Q23** Dr Eslam · LCM of 2 and 5 $5(x + 3) - 2(x - 1) = 40; x = \frac{23}{3}$
- Q24** Dr Eslam · Every term $4x + 3x = 168; x = 24$
- Q25** Practice · Hard $x = \frac{12}{13}$
- Q26** Practice · Hard $x = -\frac{2}{13}$
- Q27** Practice · Hard $a = 4$
- Q28** Practice · Hard $x = 5.5$
- Q29** Edexcel 4MA1 · June 2022 · Paper 2H · Q15 (a) $x = 5.5$ (b) $-2.5 \leq y \leq 6$
- Q30** Edexcel 4MA1 · June 2025 · Paper 1HR · Q15 (a) $a = 18.5$ (b) $3y^{-\frac{1}{2}}$

IDEA 04 · HIDDEN LINEAR EQUATIONS

- Q31** Dr Eslam · The square cancels $x^2 + 8x + 16 = x^2 + 40; x = 3$
- Q32** Practice · Easy He wrote $5x + 2$: it should be $5x + 10$ (so $x = 7.4$)
- Q33** Practice · Hard $x = -\frac{7}{4}$
- Q34** Practice · Hard $(x - 10)^2 + 6 = x^2 + 6; x = 5$



DR ESLAM AHMED
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Solving Quadratics

Stage 1: ERCS — make it = 0. Stage 2: let the calculator decide — clean answers factorise, messy answers go in the formula.

THE TOPIC ON ONE MAP

Solving Quadratics

01 — ERCS: make it = 0

$$x(3x - 9) = 4$$

$$x^2 - 6x + 15 = 3x - 5$$

$$\frac{5}{4x + 1} = \frac{2x}{x^2 + 3}$$

ratios

02 — Clean answers: factorise

$$x^2 - 5x - 36 = 0$$

$$5x^2 + 7x + 2 = 0$$

"Solve by factorising"

03 — Messy answers: the formula

"to 2 decimal places"

"to 3 s.f."

$$3x^2 + 4x - 12 = 0$$

find the error

04 — Square root and completing the square

$$3x^2 = 75$$

$$9(x + 3)^2 = 4$$

"in the form $a \pm \sqrt{b}$ "

"hence solve"

05 — Shapes and hidden quadratics

area $\Rightarrow$ "show that"

$$6x^4 + 7x^2$$

indices $\Rightarrow$ quadratic

ratio $x : y$

IN THE EXAM Solving a quadratic is inside about one question in five: graphs, shapes, simultaneous equations.

01 ERCS: Make It Equal Zero

You can only solve a quadratic when it is $ax^2 + bx + c = 0$. **ERCS** gets you there — then the calculator takes over.

WHEN YOU SEE → YOU DO

a bracket: $x(3x - 9) = 4$

→ Expand

terms on both sides

→ Rearrange: everything to one side, $= 0$

fractions or a ratio $a : b = c : d$

→ cross-multiply: $ad = bc$

“in the form” $ax^2 + bx + c = 0$

→ Collect — this line is a mark by itself

Expand

Rearrange ($= 0$)

Collect

Solve

ZONE A

Answers are whole numbers or fractions

Factorise (Idea 02).

ZONE B

Answers are long decimals

Quadratic formula (Idea 03).

ZONE C

No x term, or a bracket squared

Square root both sides — $\pm!$ (Idea 04)

ZONE D

“in the form” $(x + a)^2 + b$ / “ $a \pm \sqrt{b}$ ”

Complete the square (Idea 04).

CALCULATOR CHECK **MODE 5 3** (equation, degree 2). Type a , b , c . The answers tell you which zone you are in.

WORKED MODEL • RATIO

$2x - 1 : x - 4 = 16x + 1 : 2x - 1$. Find the possible values of x .

1 $(2x - 1)^2 = (x - 4)(16x + 1)$

cross-multiply

2 $4x^2 - 4x + 1 = 16x^2 - 63x - 4$

Expand

3 $12x^2 - 59x - 5 = 0$

Rearrange + Collect

4 $(12x + 1)(x - 5) = 0$

Solve (clean answers)

ANSWER $x = 5$ or $x = -\frac{1}{12}$

IDEA 01

Practise: ERCS: Make It Equal Zero

My questions first, then selected practice.

Q1 **DR ESLAM** Expand, then = 0

CH 2 · TOPIC 10 3 marks

Solve $x(x + 3) = 10$

$x =$ _____

SELECTED PRACTICE

Q2 **PRACTICE** Practice · Medium

CH 2 · TOPIC 10 3 marks

13 (a) Write $x(3x - 9) = 4$ in the form $ax^2 + bx + c = 0$ where a , b and c are integers.

(1 mark)

(b) Solve $x(3x - 9) = 4$

Give your answers to 2 decimal places.

(2 marks)

Q3**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 4 marks

18 Solve.

$$x^2 - 6x + 15 = 3x - 5$$

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots$$

(4 marks)

Q4**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 4 marks

4 Solve $5x^2 = 10x + 4$

Give your answers to 2 decimal places.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

5 marks**3** Given that

$$2x - 1 : x - 4 = 16x + 1 : 2x - 1$$

find the possible values of x .**(5 marks)****Q6****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

4 marks**5** Given that

$$x^2 : (3x + 5) = 1 : 2$$

find the possible values of x .**(4 marks)**

Q7**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

5 marks

10 Solve $\frac{5}{4x+1} = \frac{2x}{x^2+3}$

Give your solutions to 3 significant figures. You **must** show your working.

(5 marks)

02 Clean Answers: Factorise

Calculator gives whole numbers or fractions? Then it factorises. Turn each answer into a bracket.

WHEN YOU SEE → YOU DO

“Solve by factorising”

→ brackets are **required**

calculator: $x = 9, x = -4$

→ $(x - 9)(x + 4) = 0$

calculator: $x = -\frac{2}{5}$

→ $(5x + 2)$ — the bottom goes in front of x

“the only solution is $x = 5$ ”

→ $(x - 5)^2 = 0$

MODE 5 3

Answers to brackets

Write = 0

Each bracket = 0

WORKED MODEL · JANUARY 2020 P1HR

Solve $x^2 - 5x - 36 = 0$. Show clear algebraic working.

1 calculator: $x = 9, x = -4$

MODE 5 3

2 $(x - 9)(x + 4) = 0$

answers → brackets

3 $x - 9 = 0$ or $x + 4 = 0$

each bracket = 0

ANSWER $x = 9$ or $x = -4$

× **TRAP** Writing only the calculator answers scores 0 for “show clear algebraic working”. The brackets line is the method mark.

IDEA 02

Practise: Clean Answers: Factorise

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Answers to brackets

CH 2 · TOPIC 10 3 marks

Solve $x^2 - 7x + 12 = 0$. Show clear algebraic working.

$x =$ _____

SELECTED PRACTICE**Q9** PRACTICE Practice · Medium

CH 2 · TOPIC 10 3 marks

10 (a) Factorise $x^2 + 2x - 24$.

(2 marks)

(b) Hence solve $x^2 + 2x - 24 = 0$

[1]

(1 mark)

Q10**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

14 Solve $x^2 - x - 12 = 0$

(3 marks)

Q11**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 3 marks

7 Solve by factorisation.

$$5x^2 + 7x + 2 = 0$$

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

Q12**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 3 marks

10 Solve by factorisation $10r^2 - 23r + 9 = 0$.

 $r = \dots\dots\dots$ or $r = \dots\dots\dots$

(3 marks)

PAST PAPERS**Q13****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q6

CH 2 · TOPIC 10 3 marks

6 Solve $x^2 - 5x - 36 = 0$
Show clear algebraic working.

.....

(Total for Question 6 is 3 marks)

Q14 **PAST PAPER** Edexcel 4MA1 · January 2020 · Paper 2H · Q7**CH 2 · TOPIC 10** **5 marks**7 (a) Solve the inequality $2x + 7 > 4$ (b) Solve $x^2 - 3x - 40 = 0$
Show clear algebraic working.**(Total for Question 7 is 5 marks)****Q15** **PAST PAPER** Edexcel 4MA1 · January 2021 · Paper 2HR · Q9**CH 2 · TOPIC 10** **3 marks**9 Solve $x^2 - 21x + 20 = 0$
Show your working clearly.**(Total for Question 9 is 3 marks)**

03 Messy Answers: The Quadratic Formula

Calculator gives long decimals? It does not factorise. Substitute into the formula and **show** the substitution.

WHEN YOU SEE → YOU DO

“correct to 2 decimal places / 3 s.f.”

→ quadratic formula

b is negative

→ $-b$ becomes positive: $-(-4) = 4$

“find the error in her working”

→ check: $-b, b^2$ in brackets, divide **everything** by $2a$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Write a b c

Substitute

Calculator

Round

WORKED MODEL • 2 DECIMAL PLACES

Solve $3x^2 + 4x - 12 = 0$, giving your solutions to 2 d.p.

1 $a = 3, b = 4, c = -12$

write a, b, c

2 $x = \frac{-4 \pm \sqrt{4^2 - 4(3)(-12)}}{2(3)} = \frac{-4 \pm \sqrt{160}}{6}$

substitute

ANSWER $x = 1.44$ or $x = -2.77$

× **TRAP** The line under the fraction is under **everything**: $-b$ is divided by $2a$ too. $8 \pm \frac{\sqrt{40}}{4}$ is wrong.

IDEA 03

Practise: Messy Answers: The Formula

My questions first, then selected practice.

Q16 DR ESLAM Formula

CH 2 · TOPIC 10 3 marks

Solve $x^2 - 4x - 3 = 0$, giving your answers to 2 decimal places.

$x =$ _____

SELECTED PRACTICE

Q17 PRACTICE Practice · Medium

CH 2 · TOPIC 10 3 marks

2 Solve the equation $3x^2 + 4x - 12 = 0$

Give your solutions correct to 2 decimal places.

(3 marks)

Q18**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

4 Solve $3x^2 - x - 1 = 0$

Give your solutions correct to 2 decimal places.

(3 marks)

Q19**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

7 Solve $3x^2 - 4x - 2 = 0$

Give your solutions correct to 3 significant figures.

(3 marks)

Q20**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 **3 marks**

8 Solve $3x^2 - 5x - 1 = 0$

Give your solutions correct to 3 significant figures.

(3 marks)

Q21**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 **4 marks**

17 Solve this equation algebraically.

Give your solutions correct to 2 decimal places.

$$3x^2 + 8x - 5 = 0$$

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(4 marks)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 6 marks

6 (a) Here are two pieces of work.

For each one, describe the error made and give the complete correct solution.

Question:

Solve by factorisation.

$$3x^2 - 2x - 5 = 0$$

Solution:

$$(3x + 5)(x - 1) = 0$$

$$\text{Therefore } x = -5/3 \text{ or } x = 1$$

Error: Correct solution:

(3 marks)

Question: Solve, giving your answers correct to 3 significant figures.

$$2x^2 - 8x + 3 = 0$$

Solution:

$$x = -(-8) \pm \frac{\sqrt{(-8)^2 - 4 \times 2 \times 3}}{2 \times 2}$$

$$\text{Therefore } x = 6.42 \text{ or } x = 9.58$$

(b)

Error: Correct solution:

(3 marks)

04 Square Root and Completing the Square

No x term, or a bracket squared already? Square-root both sides — and never forget the $\pm$. Asked for a form? Complete the square.

WHEN YOU SEE → YOU DO

$3x^2 = 75$ (no x term)

→ $x^2 = 25$, $x = \pm 5$

$9(x+3)^2 = 4$, “without expanding”

→ $(x+3)^2 = \frac{4}{9}$, $x+3 = \pm \frac{2}{3}$

“in the form $a \pm \sqrt{b}$ ” / “surd form”

→ complete the square

“use your answer to part (a)” / “hence”

→ use the completed square form

Isolate the square

Square root ($\pm$)

Solve both

WORKED MODEL · SURD FORM

Solve $x^2 - 6x - 8 = 0$. Give your answer as $a \pm \sqrt{b}$.

1 $(x-3)^2 - 9 - 8 = 0$

complete the square

2 $(x-3)^2 = 17$

isolate the square

3 $x-3 = \pm\sqrt{17}$

square root, $\pm$

ANSWER $x = 3 \pm \sqrt{17}$

× **TRAP** Liz wrote $x^2 = 25 \Rightarrow x = 5$. She lost $x = -5$. Square root always gives **two** answers.

IDEA 04

Practise: Square Root and Completing the Square

My questions first, then selected practice.

Q23 DR ESLAM Square root

CH 2 · TOPIC 10 2 marks

Solve $2x^2 = 50$

$x =$ _____

Q24 DR ESLAM Bracket squared

CH 2 · TOPIC 10 2 marks

Solve $(x + 3)^2 = 16$ without expanding.

$x =$ _____

SELECTED PRACTICE

Q25**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10

2 marks

1 Solve $(x - 2)^2 = 3$

Give your solutions correct to 3 significant figures.

(2 marks)

Q26**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10

1 mark

6 Liz is asked to solve the equation $3x^2 + 8 = 83$

Here is her working.

$$3x^2 + 8 = 83$$

$$3x^2 = 75$$

$$x^2 = 25$$

$$x = 5$$

Explain what is wrong with Liz's answer.

(1 mark)

Q27**PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 2 marks

19 Solve.

$$3x^2 = 75$$

 $x = \dots\dots\dots$

(2 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 3 marks

1 Solve $x^2 - 6x - 8 = 0$ Write your answer in the form $a \pm \sqrt{b}$ where a and b are integers.

(3 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 10

7 marks

9 (i) Write $x^2 + 4x - 16$ in the form $(x + a)^2 - b$.

[3]

(ii) Solve the equation $x^2 + 4x - 16 = 0$.

Give your answers in surd form as simply as possible.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

(7 marks)

Q30**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

5 marks

9 (a) Express $7 - 4x - x^2$ in the form $p - (x + q)^2$ where p and q are constants.

(2 marks)

(b) Use your answer to part (a) to solve the equation $7 - 4(y + 3) - (y + 3)^2 = 0$

Give your solutions in the form $e \pm \sqrt{f}$ where e and f are integers.

(3 marks)

05 Shapes and Hidden Quadratics

The quadratic is hidden inside an area, a ratio, an index law or a power of x^2 . Build the equation, then ERCS.

WHEN YOU SEE → YOU DO

“the area is 351 cm^2 . Show that ...”

→ area formula = 351, then ERCS

“work out the value of x ” (a length)

→ reject the negative answer

$6x^4 + 7x^2$

→ let $u = x^2$: $6u^2 + 7u$

$6x^2 = 7xy + 20y^2$, “find $x : y$ ”

→ factorise: $(2x - 5y)(3x + 4y) = 0$

WORKED MODEL · TRAPEZIUM

Parallel sides $x - 4$ and $x + 5$, height $2x$. Area = 351. Find x .

1 $\frac{1}{2}(x - 4 + x + 5)(2x) = 351$

area formula

2 $2x^2 + x - 351 = 0$

Expand + Collect

3 $(2x + 27)(x - 13) = 0$

clean answers

ANSWER $x = 13$ ($x = -13.5$ is a length < 0 : reject)

WHERE YOU WILL USE IT Quadratic inequalities (Topic 11), non-linear simultaneous equations (Topic 12), forming equations (Topic 13).

IDEA 05

Practise: Shapes and Hidden Quadratics

My questions first, then selected practice.

Q31 DR ESLAM Rectangle

CH 2 · TOPIC 10 4 marks

A rectangle is x cm wide and $(x + 4)$ cm long. Its area is 60 cm^2 . Find the width.

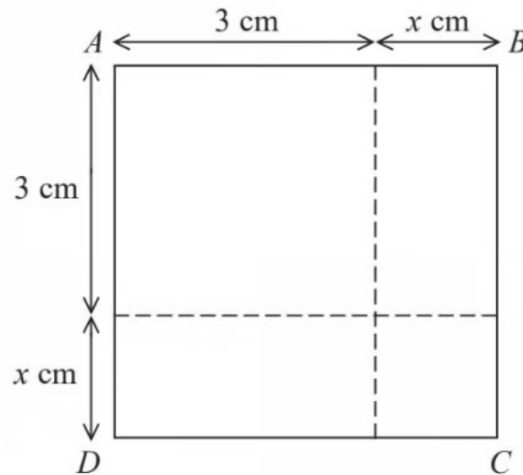
width = _____

SELECTED PRACTICE

Q32 PRACTICE Practice · Hard

CH 2 · TOPIC 10 3 marks

3



The area of square $ABCD$ is 10 cm^2 .

Show that $x^2 + 6x = 1$

(3 marks)

Q33**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10 5 marks

2 (a) Solve $2x^2 + 9x - 7 = 0$

Give your solutions correct to 3 significant figures.

(3 marks)

(b) Solve $\frac{2}{y^2} + \frac{9}{y} - 7 = 0$

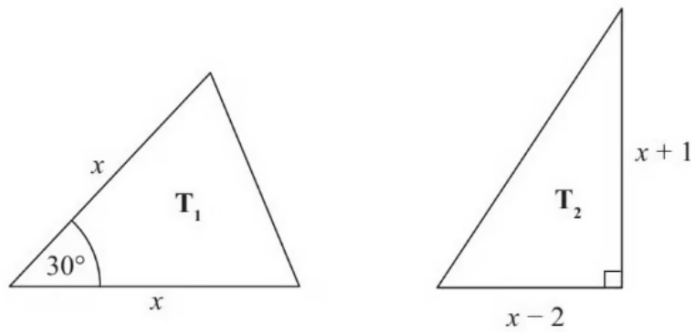
Give your solutions correct to 3 significant figures.

(2 marks)

Q34 PRACTICE Practice · Very Hard

CH 2 · TOPIC 10 5 marks

6

Diagram **NOT**
accurately drawn

The lengths of the sides are in centimetres.

The area of triangle T_1 is equal to the area of triangle T_2 .

Work out the value of x , giving your answer in the form $a + \sqrt{b}$ where a and b are integers.

(5 marks)

Q35 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 10 5 marks

7 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(5 marks)

Q36 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 10 6 marks

11 $y = 6x^4 + 7x^2$ and $x = \sqrt{w+1}$

Find the value of w when $y = 10$. Show your working.

$w = \dots\dots\dots$

(6 marks)

Q37**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10 3 marks

12 $6x^2 = 7xy + 20y^2$ where $x > 0$ and $y > 0$

Find the ratio $x:y$

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.



- | | | |
|-----------|----------------------|--|
| 01 | ERCS | Expand, Rearrange to = 0, Collect. Ratios and fractions: cross-multiply. |
| 02 | Clean answers | Factorise: answers → brackets → each bracket = 0. |
| 03 | Messy answers | Formula; show the substitution; divide <i>everything</i> by $2a$. |
| 04 | Root / CTS | No x term → square root, $\pm$. " $a \pm \sqrt{b}$ " → complete the square. |
| 05 | Hidden | Area, ratio, $u = x^2$ — build it, ERCS, reject impossible answers. |

BOTTOM LINE ERCS → MODE 5 3 → clean: factorise · messy: formula · a form asked: complete the square.

Answers

Solving Quadratics · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · ERCS: MAKE IT EQUAL ZERO

- | | | |
|-----------|-----------------------------|--|
| Q1 | Dr Eslam · Expand, then = 0 | $x^2 + 3x - 10 = 0$; $x = 2$ or $x = -5$ |
| Q2 | Practice · Medium | (a) $3x^2 - 9x - 4 = 0$ (b) $x = 3.39$ or -0.39 |
| Q3 | Practice · Medium | $x^2 - 9x + 20 = 0$; $x = 4$ or 5 |
| Q4 | Practice · Hard | $5x^2 - 10x - 4 = 0$; $x = 2.34$ or -0.34 |
| Q5 | Practice · Very Hard | $12x^2 - 59x - 5 = 0$; $x = 5$ or $-\frac{1}{12}$ |
| Q6 | Practice · Very Hard | $2x^2 - 3x - 5 = 0$; $x = 2.5$ or -1 |
| Q7 | Practice · Very Hard | $3x^2 + 2x - 15 = 0$; $x = 1.93$ or -2.59 |

IDEA 02 · CLEAN ANSWERS: FACTORISE

- Q8** Dr Eslam · Answers to brackets $(x - 3)(x - 4) = 0$; $x = 3$ or $x = 4$
- Q9** Practice · Medium (a) $(x + 6)(x - 4)$ (b) $x = -6$ or 4
- Q10** Practice · Medium $(x - 4)(x + 3) = 0$; $x = 4$ or -3
- Q11** Practice · Hard $(5x + 2)(x + 1) = 0$; $x = -\frac{2}{5}$ or -1
- Q12** Practice · Hard $(2r - 1)(5r - 9) = 0$; $r = \frac{1}{2}$ or $\frac{9}{5}$
- Q13** Edexcel 4MA1 · January 2020 · Paper 1HR · Q6 $(x - 9)(x + 4) = 0$; $x = 9$ or -4
- Q14** Edexcel 4MA1 · January 2020 · Paper 2H · Q7 (a) $x > -1.5$ (b) $(x - 8)(x + 5) = 0$; $x = 8$ or -5
- Q15** Edexcel 4MA1 · January 2021 · Paper 2HR · Q9 $(x - 1)(x - 20) = 0$; $x = 1$ or 20

IDEA 03 · MESSY ANSWERS: THE FORMULA

- Q16** Dr Eslam · Formula $x = \frac{4 \pm \sqrt{28}}{2}$; 4.65 or -0.65
- Q17** Practice · Medium $x = 1.44$ or -2.77
- Q18** Practice · Medium $x = 0.77$ or -0.43
- Q19** Practice · Medium $x = 1.72$ or -0.387
- Q20** Practice · Medium $x = 1.85$ or -0.180
- Q21** Practice · Medium $x = 0.52$ or -3.19
- Q22** Practice · Hard (a) wrong brackets: $(3x - 5)(x + 1)$, $x = \frac{5}{3}$ or -1
 (b) 8 not divided by 4: $x = \frac{8 \pm \sqrt{40}}{4} = 3.58$ or 0.419

IDEA 04 · SQUARE ROOT AND COMPLETING THE SQUARE

- Q23** Dr Eslam · Square root $x^2 = 25; x = \pm 5$
- Q24** Dr Eslam · Bracket squared $x + 3 = \pm 4; x = 1 \text{ or } x = -7$
- Q25** Practice · Medium $x = 2 \pm \sqrt{3}; 3.73 \text{ or } 0.268$
- Q26** Practice · Medium she forgot $x = -5$ ($x = \pm 5$)
- Q27** Practice · Medium $x = \pm 5$
- Q28** Practice · Hard $x = 3 \pm \sqrt{17}$
- Q29** Practice · Hard (i) $(x + 2)^2 - 20$ (ii) $x = -2 \pm 2\sqrt{5}$
- Q30** Practice · Very Hard (a) $p = 11, q = 2$ (b) $y = -5 \pm \sqrt{11}$

IDEA 05 · SHAPES AND HIDDEN QUADRATICS

- Q31** Dr Eslam · Rectangle $x^2 + 4x - 60 = 0; (x + 10)(x - 6) = 0; \text{ width } 6 \text{ cm}$
- Q32** Practice · Hard $(x + 3)^2 = 10 \Rightarrow x^2 + 6x + 9 = 10 \Rightarrow x^2 + 6x = 1$
- Q33** Practice · Very Hard (a) $x = 0.676 \text{ or } -5.18$ (b) $y = 1.48 \text{ or } -0.193$
- Q34** Practice · Very Hard $\frac{1}{4}x^2 = \frac{1}{2}(x + 1)(x - 2); x = 1 + \sqrt{5}$
- Q35** Practice · Very Hard $3n^2 - 14n - 5 = 0; n = 5 \text{ or } -\frac{1}{3}$
- Q36** Practice · Very Hard $u = w + 1 = x^2; 6u^2 + 7u - 10 = 0, u = \frac{5}{6}; w = -\frac{1}{6}$
- Q37** Practice · Very Hard $(2x - 5y)(3x + 4y) = 0; x : y = 5 : 2$



DR ESLAM AHMED
eliteigcse.com

Solving Inequalities

An inequality is an equation with one extra rule: multiply or divide by a negative — flip the sign.
Quadratic? Sketch the curve and read the answer.

THE TOPIC ON ONE MAP

Solving Inequalities

01 — Number lines and integers

$$-2 \leq n < 3: \text{list } n$$

read the diagram

$$\text{draw } y \leq 1$$

02 — Solve a linear inequality

$$7t - 8 < 2t + 7$$

$$3 - 4x \leq 11 \text{ (flip!)}$$

$$4 > 11 - \frac{x}{3}$$

is Martha correct?

03 — Double inequalities

$$-7 \leq 2x - 3 < 5$$

$$-4 \leq 2y < 6, \text{ integers}$$

two conditions at once

04 — Quadratic inequalities

$$6x^2 - 7x - 20 > 0$$

$$4x^2 + 4x - 15 < 0$$

read it from a graph

05 — Problems with inequalities

$$\text{area} < 75 \text{ cm}^2$$

linear + quadratic, list n

greatest value of $y - x$

IN THE EXAM Paper 1 starts with an easy number line; Paper 2 ends with a quadratic inequality for 3–4 marks.

01 Number Lines and Integers

Open circle: the number is **not** included. Filled circle: it **is**. Integers: list every whole number in the gap.

WHEN YOU SEE → YOU DO

$<$ or $>$

→ open circle •

$\leq$ or $\geq$

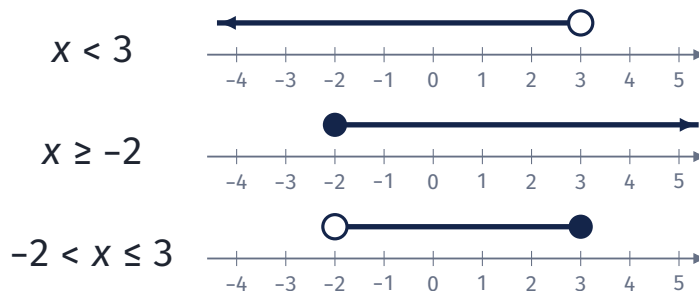
→ filled circle •

“ n is an integer ... write down all the values”

→ whole numbers only — check both ends

“write down the inequality shown”

→ read both circles: $-3 < x \leq 4$



WORKED MODEL • INTEGERS

$-2 \leq n < 3$, n is an integer. Write down all the values of n .

1 -2 is in ($\leq$); 3 is out ($<$)

check the ends

ANSWER $-2, -1, 0, 1, 2$

× **TRAP** Forgetting 0, or including the end with the $<$ sign. Say every integer out loud.

IDEA 01

Practise: Number Lines and Integers

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** List the integers

CH 2 • TOPIC 11 2 marks

$-3 \leq 2n + 1 < 7$, n is an integer. Write down all the values of n .

SELECTED PRACTICE

Q2

PRACTICE

Practice • Easy

CH 2 • TOPIC 11 5 marks

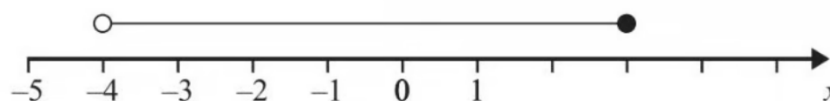
2 (a) n is an integer.

$$-1 \leq n < 4$$

List the possible values of n .

(2 marks)

(b)



Write down the inequality shown in the diagram.

(2 marks)

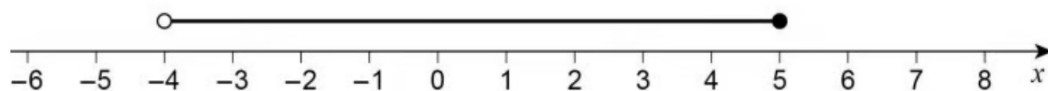
(c) Solve $y - 2 > 5$

(1 mark)

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 11

1 mark**13** Circle the inequality shown by the diagram.

$-4 \leq x < 5$	$-4 \leq x \leq 5$	$-4 < x < 5$	$-4 < x \leq 5$
-----------------	--------------------	--------------	-----------------

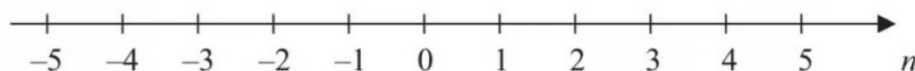
(1 mark)**Q4****PRACTICE**

Practice · Medium

CH 2 · TOPIC 11

4 marks**1 (a)** $-2 < n \leq 3$

Represent this inequality on the number line.

**(2 marks)****(b)** Solve the inequality $8x - 3 \geq 6x + 4$ **(2 marks)**

Q5**PRACTICE** Practice · Medium

CH 2 · TOPIC 11 5 marks

16 (a) Solve $5x + 6 \geq 3x + 15$

(3 marks)

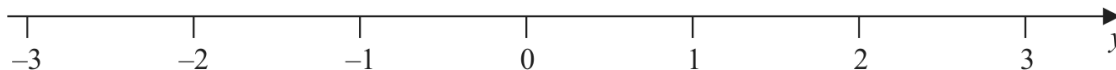
(b) Write down the inequality represented by the number line.

(2 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 · January 2021 · Paper 1HR · Q3**CH 2 · TOPIC 11** **4 marks**

- 3 (a) On the number line, show the inequality $-2 \leq y < 1$



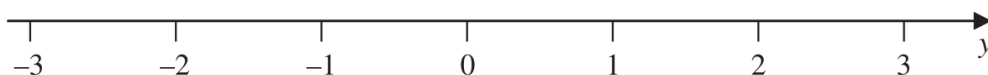
n is an integer.

- (b) Write down all the values of n that satisfy $-3.4 < n \leq 2$

.....
(Total for Question 3 is 4 marks)

Q7**PAST PAPER**

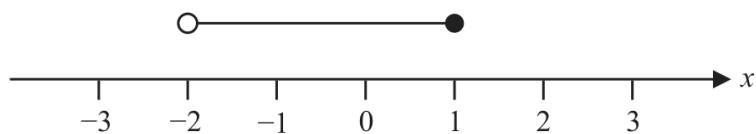
Edexcel 4MA1 • January 2022 • Paper 1HR • Q1

CH 2 • TOPIC 11 3 marks**You must write down all the stages in your working.****1** n is an integer.(a) Write down all the values of n such that $-2 \leq n < 3$ (b) On the number line, represent the inequality $y \leq 1$ **(Total for Question 1 is 3 marks)**

Q8 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 1HR • Q6

CH 2 • TOPIC 11 4 marks

6



(a) Write down the inequality shown on the number line.

.....

(b) Solve the inequality $7a - 5 \leq 3a + 28$
Show clear algebraic working.

.....

(Total for Question 6 is 4 marks)

.....

02 Solve a Linear Inequality

Solve it exactly like an equation — Clean, Collect, Divide. One extra rule: $\times$ or $\div$ by a negative flips the sign.

WHEN YOU SEE → YOU DO

“Solve the inequality”

→ same steps as an equation, keep the sign

dividing by -4

→ flip: $\leq$ becomes $\geq$

“largest integer that satisfies”

→ solve, then take the integer below

“Is her solution correct?”

→ solve it yourself, then compare circle and direction

Clean

Collect

Divide

Negative? Flip

WORKED MODEL • JUNE 2022 P1HR

Solve $3 - 4x \leq 11$.

1 $-4x \leq 8$

collect

2 $x \geq \frac{8}{-4}$

$\div$ negative: flip

ANSWER $x \geq -2$

AVOID THE FLIP Move the x term to the side where it stays positive: $3 - 4x \leq 11 \Rightarrow 3 - 11 \leq 4x \Rightarrow -2 \leq x$.

IDEA 02

Practise: Solve a Linear Inequality

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Keep the sign

CH 2 • TOPIC 11

2 marks

Solve $3x + 2 < 17$

Q10**DR ESLAM**

Flip the sign

CH 2 • TOPIC 11

2 marks

Solve $4 - 3x \geq 13$

SELECTED PRACTICE

Q11

PRACTICE

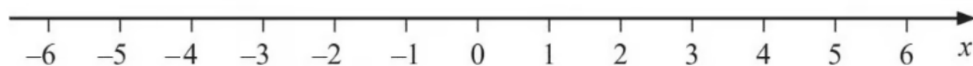
Practice • Easy

CH 2 • TOPIC 11

5 marks

1 (a) Solve $14n > 11n + 6$

(2 marks)

(b) On the number line below, show the set of values of x for which $-2 < x + 3 \leq 4$ 

(3 marks)

Q12

PRACTICE

Practice • Easy

CH 2 • TOPIC 11

2 marks

15 Solve the inequality.

$$3x - 2 > 10$$

(2 marks)

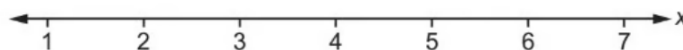
Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 11 4 marks

18 Solve $3x + 4 < 19$.

Show your solution on the number line.

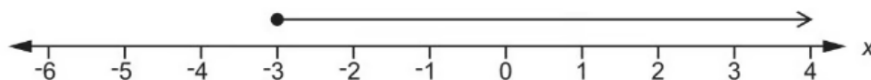


(4 marks)

Q14**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 4 marks

12 Martha's solution to the inequality $8x + 5 \leq 3x - 10$ is shown on the number line.

Is her solution correct?

Explain your reasoning.

(4 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 2H • Q5

CH 2 • TOPIC 11 7 marks

- 5 (a) Expand and simplify
- $4x(2x + 5) - 3x(2x - 3)$

(2)

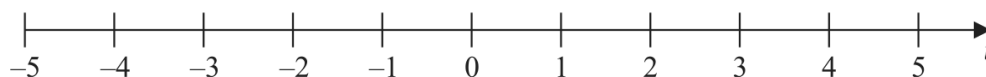
Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

- (b) work out the value of
- n
- .

 $n =$

- (c) (i) Solve the inequality
- $7t - 8 < 2t + 7$

- (ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(Total for Question 5 is 7 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q9

CH 2 · TOPIC 11 2 marks

9 Solve the inequality $3 - 4x \leq 11$

(Total for Question 9 is 2 marks)

03 Double Inequalities

Three parts: left $\leq$ middle $<$ right. Whatever you do, do it to **all three parts** until x is alone in the middle.

WHEN YOU SEE → YOU DO

$$-7 \leq 2x - 3 < 5$$

→ +3 to all three, then $\div 2$ all three

“ y is an integer”

→ solve, then list

two separate inequalities

→ solve each, draw both, keep the **overlap**

WORKED MODEL • NOVEMBER 2021 P1H

Solve $-7 \leq 2x - 3 < 5$ and show it on a number line.

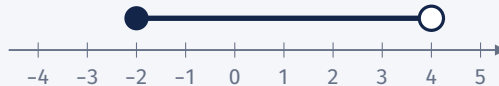
1 $-4 \leq 2x < 8$

+3 to all three

2 $-2 \leq x < 4$

$\div 2$ to all three

ANSWER $-2 \leq x < 4$



IDEA 03

Practise: Double Inequalities

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM All three parts

CH 2 · TOPIC 11 3 marks

Solve $-5 < 3x + 1 \leq 10$

SELECTED PRACTICE**Q18** PRACTICE Practice · Medium

CH 2 · TOPIC 11 4 marks

4 (a) Solve $6x + 4 > x + 17$

(2 marks)

(b) n is an integer with $-5 < 2n \leq 6$

Write down all the values of n

(2 marks)

Q19**PRACTICE**

Practice • Medium

CH 2 • TOPIC 11 **3 marks**

12 x is an integer.

$$-4 < x \leq 2$$

and

$$2 \leq x + 3 < 9$$

Work out all the possible values of x .

(3 marks)

Q20**PRACTICE**

Practice • Hard

CH 2 • TOPIC 11 3 marks

11 Here are two inequalities.

$$-2 \leq x \leq 3$$

$$9 \leq x + y \leq 11$$

x and y are integers.

Work out the **greatest** possible value of $y - x$

(3 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 1H · Q7

CH 2 · TOPIC 11 4 marks

7 $-4 \leq 2y < 6$

 y is an integer.(a) Write down all the possible values of y .

.....

(b) Solve the inequality $7t - 3 \leq 2t + 31$

Show your working clearly.

.....

(Total for Question 7 is 4 marks)

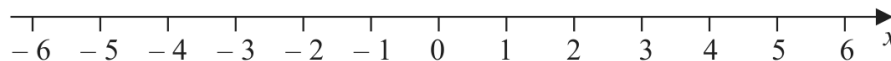
Q22**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q9

CH 2 • TOPIC 11 5 marks

- 9 (i) Solve the inequalities $-7 \leq 2x - 3 < 5$

- (ii) On the number line, represent the solution set to part (i)



(Total for Question 9 is 5 marks)

04 Quadratic Inequalities: Sketch and Read

Never solve it like an equation. Find the two roots, sketch the U, then read the answer: > 0 is **outside**, < 0 is **between**.

WHEN YOU SEE → YOU DO

$x^2 \dots > 0$ ("greater than")

→ outside: $x < a$ or $x > b$

$x^2 \dots < 0$ ("less than")

→ between: $a < x < b$

"Show clear algebraic working"

→ show the factorised form and the sketch

terms on both sides

→ ERCS first: make it $\dots > 0$ with $+x^2$

Make it > 0 or < 0

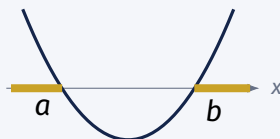
Roots (MODE 5 3)

Sketch the U

Read the region

ZONE A

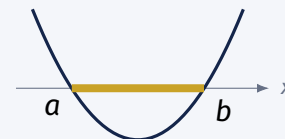
> 0 or ≥ 0 : **OUTSIDE**



$x < a$ or $x > b$ (two answers)

ZONE B

< 0 or ≤ 0 : **BETWEEN**



$a < x < b$ (one answer)

WORKED MODEL • JUNE 2024 P2HR

Solve $6x^2 - 7x - 20 > 0$.

1 $(3x + 4)(2x - 5) = 0 \Rightarrow x = -\frac{4}{3}, \frac{5}{2}$

roots

2 > 0 : the parts of the U above the axis

sketch: outside

ANSWER $x < -\frac{4}{3}$ or $x > \frac{5}{2}$

× **TRAP** $x^2 > 9$ is **not** $x > 3$ only. It is $x < -3$ or $x > 3$ — sketch, do not square-root.

IDEA 04

Practise: Quadratic Inequalities

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Between

CH 2 • TOPIC 11 3 marks

Solve $x^2 - x - 6 < 0$

Q24 DR ESLAM Outside

CH 2 • TOPIC 11 3 marks

Solve $x^2 - 4x > 0$

SELECTED PRACTICE

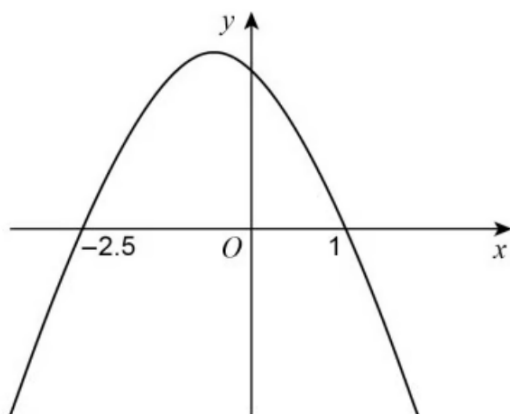
Q25**PRACTICE**

Practice · Easy

CH 2 · TOPIC 11

1 mark

- 14 Here is a sketch of $y = f(x)$ where $f(x)$ is a quadratic function. The graph intersects the x -axis where $x = -2.5$ and $x = 1$



Circle the solution of $f(x) > 0$

$x < -2.5$ or $x > 1$	$x > -2.5$ or $x > 1$
$-2.5 < x < 1$	$x > -2.5$ or $x < 1$

(1 mark)

Q26**PRACTICE**

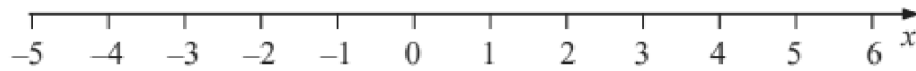
Practice · Medium

CH 2 · TOPIC 11 4 marks

22 (a) Solve $x^2 + 2x > 6x + 5$

(3 marks)

(b) Represent your solution set to part (a) on the number line below.



(1 mark)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 4 marks

4 Solve the inequality $4x^2 - 5x - 6 > 0$

(4 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 3 marks

14 Find the interval for which $x^2 - 7x + 10 \leq 0$.

$$\dots\dots\dots \leq x \leq \dots\dots\dots$$

(3 marks)

PAST PAPERS**Q29****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q19

CH 2 · TOPIC 11 6 marks

19 (a) Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(b) Solve the inequality $5y^2 - 17y \leq 40$

$$\dots\dots\dots$$

(Total for Question 19 is 6 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q20

CH 2 · TOPIC 11**4 marks**

- 20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

.....

(Total for Question 20 is 4 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q20

CH 2 · TOPIC 11**3 marks**

- 20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

.....

(Total for Question 20 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q21

CH 2 · TOPIC 11 3 marks

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q19

CH 2 · TOPIC 11 3 marks

19 Solve the inequality $4x^2 + 4x - 15 < 0$

Show clear algebraic working.

(Total for Question 19 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 · November 2025 · Paper 1H · Q25**CH 2 · TOPIC 11** **3 marks**

- 25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

.....

(Total for Question 25 is 3 marks)

05 Problems with Inequalities

Build the inequality from the words, solve it, then use common sense: a length must be positive.

WHEN YOU SEE → YOU DO

“the area is less than 75 cm^2 ”

→ area formula < 75 , then Idea 04

a side $x - 1$

→ also need $x - 1 > 0$, so $x > 1$

“integer such that ... and ...”

→ solve both, list the integers in the overlap

“greatest possible value of $y - x$ ”

→ make y big and x small

WORKED MODEL • JANUARY 2022 P2H

Rectangle $(2x + 3)$ by $(x - 1)$. Area < 75 . Find the range of x .

1 $(2x + 3)(x - 1) < 75$

area < 75

2 $2x^2 + x - 78 < 0$

ERCS

3 $(2x + 13)(x - 6) < 0 \Rightarrow -6.5 < x < 6$

between

4 side $x - 1 > 0 \Rightarrow x > 1$

lengths positive

ANSWER $1 < x < 6$

IDEA 05

Practise: Problems with Inequalities

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q35**

PRACTICE

Practice • Hard

CH 2 • TOPIC 11

5 marks

6 n is an integer such that $3n + 2 \leq 14$ and $\frac{6n}{n^2 + 5} > 1$

Find all the possible values of n .

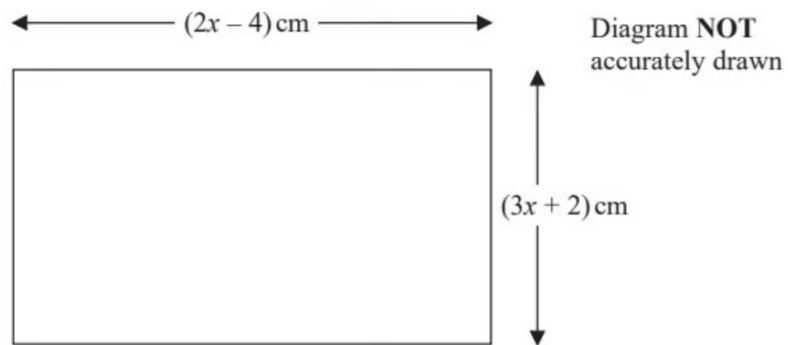
(5 marks)

Q36**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 5 marks

- 9 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

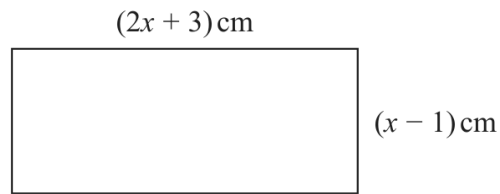
(5 marks)

PAST PAPERS

Q37 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 2H • Q22

CH 2 • TOPIC 11 5 marks

22 Here is a rectangle.

Diagram **NOT**
accurately drawnGiven that the area of the rectangle is less than 75 cm^2 find the range of possible values of x

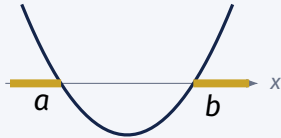
(Total for Question 22 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

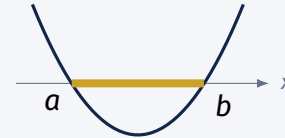
ZONE A

> 0 : **OUTSIDE**



ZONE B

< 0 : **BETWEEN**



01 Number lines

$<$, $>$ open circle; $\leq$, $\geq$ filled. List every integer, check both ends.

02 Linear

Like an equation; $\times$ or $\div$ by a negative flips the sign.

03 Double

Do it to all three parts. Two inequalities: keep the overlap.

04 Quadratic

Roots $\rightarrow$ sketch the U $\rightarrow > 0$ outside, < 0 between.

05 Problems

Build it, solve it, lengths > 0 .

BOTTOM LINE Solve like an equation $\rightarrow$ negative? flip $\rightarrow$ quadratic? roots, sketch, read.

Answers

Solving Inequalities • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • NUMBER LINES AND INTEGERS

- Q1** Dr Eslam • List the integers $-2 \leq n < 3$: $-2, -1, 0, 1, 2$
- Q2** Practice • Easy (a) $-1, 0, 1, 2, 3$ (b) $-4 < x \leq 3$ (c) $y > 7$
- Q3** Practice • Easy $-4 < x \leq 5$
- Q4** Practice • Medium (a) open at -2 , filled at 3 (b) $x \geq 3.5$
- Q5** Practice • Medium (a) $x \geq 4.5$ (b) $2 \leq x < 5$
- Q6** Edexcel 4MA1 • January 2021 • Paper 1HR • Q3 (a) filled at -2 , open at 1 (b) $-3, -2, -1, 0, 1, 2$
- Q7** Edexcel 4MA1 • January 2022 • Paper 1HR • Q1 (a) $-2, -1, 0, 1, 2$ (b) filled at 1 , arrow left
- Q8** Edexcel 4MA1 • June 2025 • Paper 1HR • Q6 (a) $-2 < x \leq 1$ (b) $a \leq 8.25$

IDEA 02 • SOLVE A LINEAR INEQUALITY

- Q9** Dr Eslam • Keep the sign $x < 5$
- Q10** Dr Eslam • Flip the sign $-3x \geq 9; x \leq -3$
- Q11** Practice • Easy (a) $n > 2$ (b) $-5 < x \leq 1$: open at -5 , filled at 1
- Q12** Practice • Easy $x > 4$
- Q13** Practice • Medium $x < 5$: open at 5 , arrow left
- Q14** Practice • Hard No: $x \leq -3$, the arrow should point left
- Q15** Edexcel 4MA1 • January 2021 • Paper 2H • Q5 (a) $2x^2 + 29x$ (b) $n = 14$ (c)(i) $t < 3$ (ii) open at 3 , arrow left
- Q16** Edexcel 4MA1 • June 2022 • Paper 1HR • Q9 $x \geq -2$

IDEA 03 • DOUBLE INEQUALITIES

- Q17** Dr Eslam • All three parts $-6 < 3x \leq 9; -2 < x \leq 3$
- Q18** Practice • Medium (a) $x > 2.6$ (b) $-2, -1, 0, 1, 2, 3$
- Q19** Practice • Medium $-1, 0, 1, 2$
- Q20** Practice • Hard $x = -2, y = 13: 15$
- Q21** Edexcel 4MA1 • May 2021 • Paper 1H • Q7 (a) $-2, -1, 0, 1, 2$ (b) $t \leq 6.8$
- Q22** Edexcel 4MA1 • November 2021 • Paper 1H • Q9 (i) $-2 \leq x < 4$ (ii) filled at -2 , open at 4

IDEA 04 • QUADRATIC INEQUALITIES

- Q23** Dr Eslam • Between $(x - 3)(x + 2) < 0; -2 < x < 3$
- Q24** Dr Eslam • Outside $x(x - 4) > 0; x < 0 \text{ or } x > 4$
- Q25** Practice • Easy $-2.5 < x < 1$
- Q26** Practice • Medium (a) $x < -1 \text{ or } x > 5$
- Q27** Practice • Hard $x < -\frac{3}{4} \text{ or } x > 2$
- Q28** Practice • Hard $2 \leq x \leq 5$
- Q29** Edexcel 4MA1 • January 2020 • Paper 2HR • Q19 (a) $x = 3$ (b) $-1.6 \leq y \leq 5$
- Q30** Edexcel 4MA1 • June 2024 • Paper 2HR • Q20 $x < -\frac{4}{3} \text{ or } x > \frac{5}{2}$
- Q31** Edexcel 4MA1 • November 2024 • Paper 2H • Q20 $(10x + 21)(x - 1) < 0; -2.1 < x < 1$
- Q32** Edexcel 4MA1 • June 2025 • Paper 2HR • Q21 $x < -1.5 \text{ or } x > 5$
- Q33** Edexcel 4MA1 • June 2025 • Paper 2H • Q19 $-2.5 < x < 1.5$
- Q34** Edexcel 4MA1 • November 2025 • Paper 1H • Q25 $x < -4 \text{ or } x > 3.5$

IDEA 05 • PROBLEMS WITH INEQUALITIES

- Q35** Practice • Hard $n \leq 4 \text{ and } 1 < n < 5: 2, 3, 4$
- Q36** Practice • Hard $6x^2 - 11x - 35 < 0, -\frac{5}{3} < x < \frac{7}{2}; \text{ with } x > 2:$
 $2 < x < 3.5$
 $1 < x < 6$
- Q37** Edexcel 4MA1 • January 2022 • Paper 2H • Q22



DR ESLAM AHMED
eliteigcse.com

Simultaneous Equations

Two equations, two unknowns. Both linear? Eliminate. One has a square? Substitute. Calculator first, always.

THE TOPIC ON ONE MAP

Simultaneous Equations

01 Linear: elimination

$$3x + 5y = 6, \quad 7x - 5y = -11$$

$$\text{decimals: } 3x + 5y = 3.1$$

letters a, c

02 Forming them from words

apples and pears

tickets

ratio, sequence

03 Graphs: the crossing point

"by drawing another line"

estimate from a grid

04 Linear and quadratic: substitution

$$x^2 + y^2 = 41, \quad 2x + y = 3$$

"intersect at A and B"

05 Show that: tangent, one point, midpoint

"is a tangent"

"exactly one point"

midpoint of PQ

ZONE A

Both linear

Eliminate (Idea 01)

ZONE B

One has x^2 or xy

Substitute (Idea 04)

ZONE C

Words

Name the letters (Idea 02)

ZONE D

A graph

Read the crossing (Idea 03)

01 Linear Pairs: Elimination

Get the answer on the calculator first. Then make one letter match, and remove it: **Same Sign → Subtract, Different Sign → Add.**

WHEN YOU SEE → YOU DO

two linear equations

→ calculator first (MODE 5 1), then eliminate

+5y and -5y (different signs)

→ ADD

6y and 6y (same signs)

→ SUBTRACT

nothing matches

→ multiply one or both equations first

Calculator

Match

Add or Subtract

Solve

Substitute back

WORKED MODEL • ELIMINATION

Solve $5x + 2y = 11$ and $4x - 3y = 18$.

1 calculator: $x = 3, y = -2$

MODE 5 1

2 $15x + 6y = 33$ and $8x - 6y = 36$

match: $\times 3, \times 2$

3 $23x = 69 \Rightarrow x = 3$

different signs: add

4 $5(3) + 2y = 11 \Rightarrow 2y = -4$

substitute back

ANSWER $x = 3, y = -2$

× **TRAP** Subtracting a negative: $6y - (-6y) = 12y$. Bracket the second equation when you subtract.

IDEA 01

Practise: Linear Pairs: Elimination

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Different signs: add

CH 2 · TOPIC 12 3 marks

Solve $3x + 2y = 16$ and $5x - 2y = 8$

Q2 **DR ESLAM** Match first

CH 2 · TOPIC 12 3 marks

Solve $2x + 3y = 12$ and $5x + 2y = 19$

SELECTED PRACTICE

Q3**PRACTICE**

Practice • Easy

CH 2 • TOPIC 12

3 marks

1 Solve the simultaneous equations

$$4x + y = 25$$

$$x - 3y = 16$$

(3 marks)

Q4**PRACTICE**

Practice • Easy

CH 2 • TOPIC 12

3 marks

7 Solve the simultaneous equations

$$x + 2y = -0.5$$

$$3x - y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(3 marks)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 12

4 marks**4** Solve the simultaneous equations

$$4x + 7y = 1$$

$$3x + 10y = 15$$

(4 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12

5 marks**15** Given that

$$m \begin{pmatrix} 4 \\ 1 \end{pmatrix} + n \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

find the value of m and the value of n . $m = \dots\dots\dots$ $n = \dots\dots\dots$ **(5 marks)**

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q7

CH 2 • TOPIC 12 3 marks

7 Solve the simultaneous equations

$$\begin{aligned}3x + 5y &= 6 \\ 7x - 5y &= -11\end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(Total for Question 7 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q12

CH 2 • TOPIC 12**4 marks****12** Solve the simultaneous equations

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q12

CH 2 • TOPIC 12 4 marks**12** Solve the simultaneous equations

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q12

CH 2 • TOPIC 12 4 marks

- 12 Solve the simultaneous equations $2x + 7y = 17$
 $5x + 3y = -1$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 12 is 4 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q8

CH 2 • TOPIC 12 3 marks

- 8 Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

 $a = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 8 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q10

CH 2 • TOPIC 12 3 marks**10** Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 10 is 3 marks)**

Q13**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q14

CH 2 • TOPIC 12 4 marks**14** Solve the simultaneous equations

$$3x - 5y = 25$$

$$4x + 3y = 14$$

Show clear algebraic working.

 $x =$ $y =$ **(Total for Question 14 is 4 marks)**

Q14 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 2H · Q10CH 2 · TOPIC 12 **3 marks****10** Solve the simultaneous equations

$$\begin{aligned}7x + 3y &= 3 \\ 3x - y &= 7\end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(Total for Question 10 is 3 marks)**Q15** **PAST PAPER** Edexcel 4MA1 · May 2023 · Paper 1H · Q9CH 2 · TOPIC 12 **3 marks****9** Solve the simultaneous equations

$$\begin{aligned}2x + 9y &= 14.5 \\ 7x + 3y &= 8\end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(Total for Question 9 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q10

CH 2 • TOPIC 12 3 marks**10** Solve the simultaneous equations

$$6x + 4y = 1$$

$$3x + 5y = 8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 10 is 3 marks)**Q17****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q12

CH 2 • TOPIC 12 4 marks**12** Solve the simultaneous equations

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Q18 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2HR • Q3CH 2 • TOPIC 12 **3 marks**

- 3 Solve the simultaneous equations

$$3x + 2y = 10$$

$$3x - 4y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 3 is 3 marks)**Q19** **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2H • Q8CH 2 • TOPIC 12 **3 marks**

- 8 Solve the simultaneous equations

$$3x + 5y = 8$$

$$4x + y = -3.5$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q2

CH 2 • TOPIC 12 3 marks**2** Solve the simultaneous equations

$$5x + y = 11$$

$$3x - y = 9$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 2 is 3 marks)

02 Forming Them from Words

Two facts, two unknowns. Choose a letter for each unknown, write one equation per sentence, then Idea 01.

WHEN YOU SEE → YOU DO

“5 apples and 3 pears cost \$1.96”

→ $5a + 3p = 1.96$

“work out the cost of ...”

→ answer the question asked, with units

a ratio $x : y = 7 : 4$

→ $4x = 7y$

a sequence, a probability, a vector

→ each given fact = one equation

Name the letters

One sentence = one equation

Eliminate

Answer the question

WORKED MODEL • NOVEMBER 2021 P1H

5 apples and 3 pears cost \$1.96. 3 apples and 2 pears cost \$1.22. Find the cost of 10 apples and 10 pears.

1 $5a + 3p = 1.96, 3a + 2p = 1.22$

name the letters

2 $10a + 6p = 3.92, 9a + 6p = 3.66 \Rightarrow a = 0.26$

same sign: subtract

3 $p = 0.22$

substitute back

ANSWER $10(0.26) + 10(0.22) = \$4.80$

IDEA 02

Practise: Forming Them from Words

My questions first, then selected practice, then real past papers.

Q21 **DR ESLAM** Cakes and drinks

CH 2 • TOPIC 12 4 marks

2 cakes and 1 drink cost £5. 1 cake and 3 drinks cost £10. Find the cost of one cake and one drink.

SELECTED PRACTICE

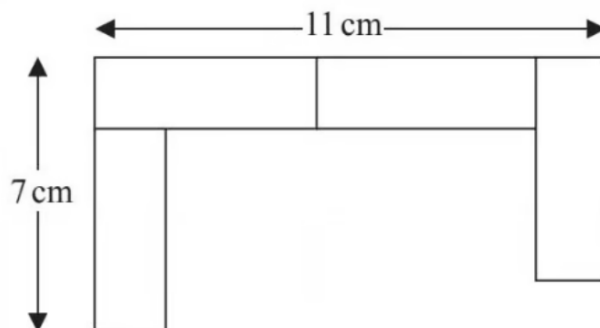
Q22**PRACTICE**

Practice • Medium

CH 2 • TOPIC 12

4 marks

7 A pattern is made using identical rectangular tiles.



Find the total area of the pattern.

(4 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12 4 marks

12 A linear sequence starts

$$a + 2b \quad a + 6b \quad a + 10b \quad \dots\dots\dots \dots\dots\dots$$

The 2nd term has value 8

The 5th term has value 44

Work out the values of a and b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(4 marks)

Q24**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 12 5 marks

4 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.

A counter is taken at random from the bag. The probability that the counter is green is

$$\frac{6}{13}$$

Find the number of red counters and the number of green counters that were in the bag originally.

(5 marks)

Q25**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 6 marks

18 (a) Each week Dan drives two routes, route X and route Y.

One week he drives route X three times and route Y twice. He drives a total of 134 miles that week.

Another week he drives route X twice and route Y five times. He drives a total of 203 miles that week.

Find the length of each route.

route X = miles

route Y = miles

(5 marks)

(b) State an assumption that has been made in answering part **(a)**.

[1]

(1 mark)

PAST PAPERS

Q26 **PAST PAPER** Edexcel 4MA1 • November 2021 • Paper 1H • Q6CH 2 • TOPIC 12 **5 marks**

- 6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$.....

(Total for Question 6 is 5 marks)

03 Graphs: The Crossing Point

Each equation is a line. The solution is where the lines **cross** — read x and y from that point.

WHEN YOU SEE → YOU DO

“by drawing another line on the grid”

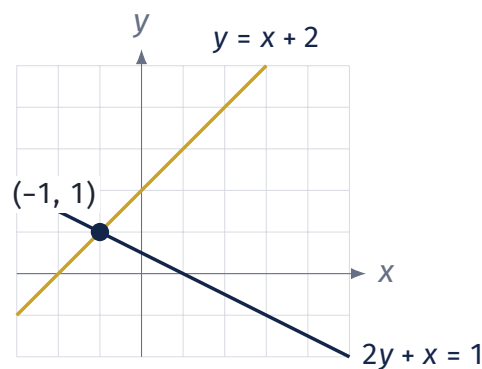
→ draw it with two points, read the crossing

“circle the coordinates of P ”

→ solve the two equations

“estimate”

→ read to the nearest small square



WORKED MODEL • NOVEMBER 2023 P2H

Solve $y - x - 2 = 0$ and $2y + x = 1$ by drawing another line.

1 $y = x + 2$: points $(0, 2)$ and $(2, 4)$

draw the new line

2 the lines cross at $(-1, 1)$

read the crossing

ANSWER $x = -1, y = 1$

IDEA 03

Practise: Graphs: The Crossing Point

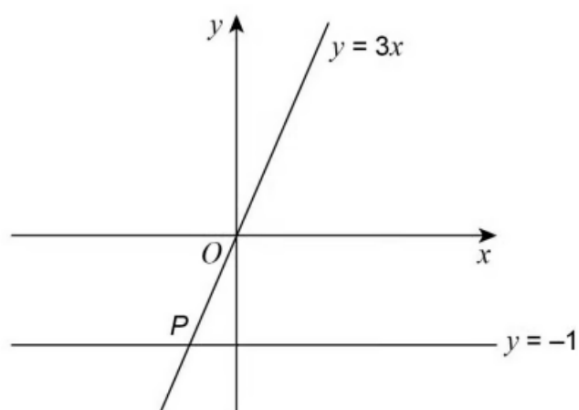
My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q27****PRACTICE**

Practice · Easy

CH 2 · TOPIC 12 1 mark

- 9 Two straight lines intersect at point P .

Circle the coordinates of P .

$(-3, -1)$	$\left(-1, -\frac{1}{3}\right)$	$(-1, -3)$	$\left(-\frac{1}{3}, -1\right)$
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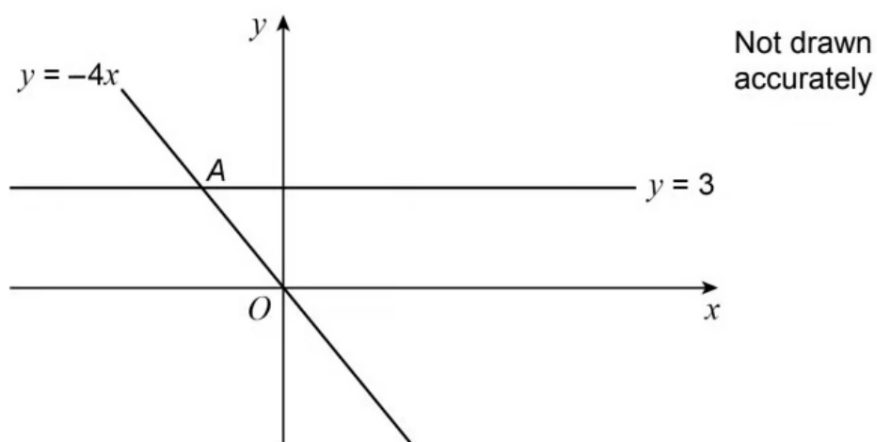
(1 mark)

Q28**PRACTICE**

Practice · Easy

CH 2 · TOPIC 12 1 mark

10 Two straight lines intersect at point A .



Circle the coordinates of A .

$(-\frac{3}{4}, 3)$	$(-4, 3)$	$(-12, 3)$	$(-\frac{4}{3}, 3)$
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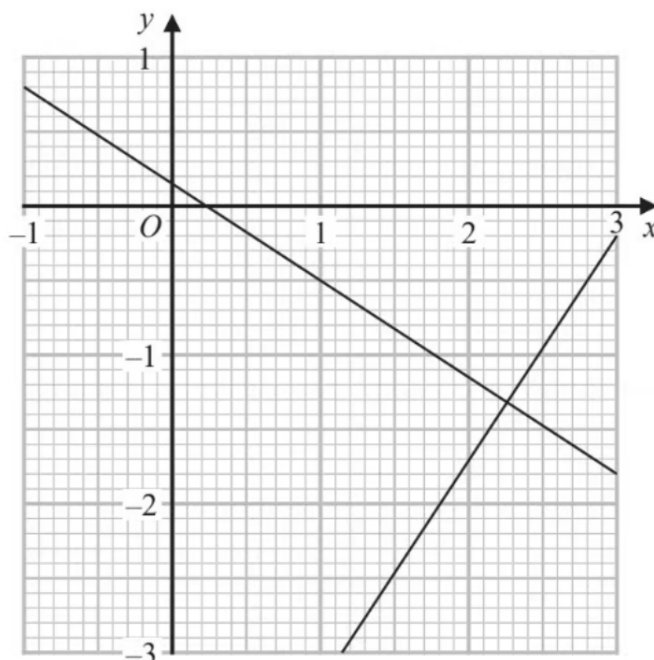
(1 mark)

Q29**PRACTICE**

Practice · Medium

CH 2 · TOPIC 12 2 marks

- 6 The graphs with equations $3y + 2x = \frac{1}{2}$ and $2y - 3x = -\frac{113}{12}$ have been drawn on the grid below.



Using the graphs, find estimates of the solutions of the simultaneous equations

$$3y + 2x = \frac{1}{2}$$

$$2y - 3x = -\frac{113}{12}$$

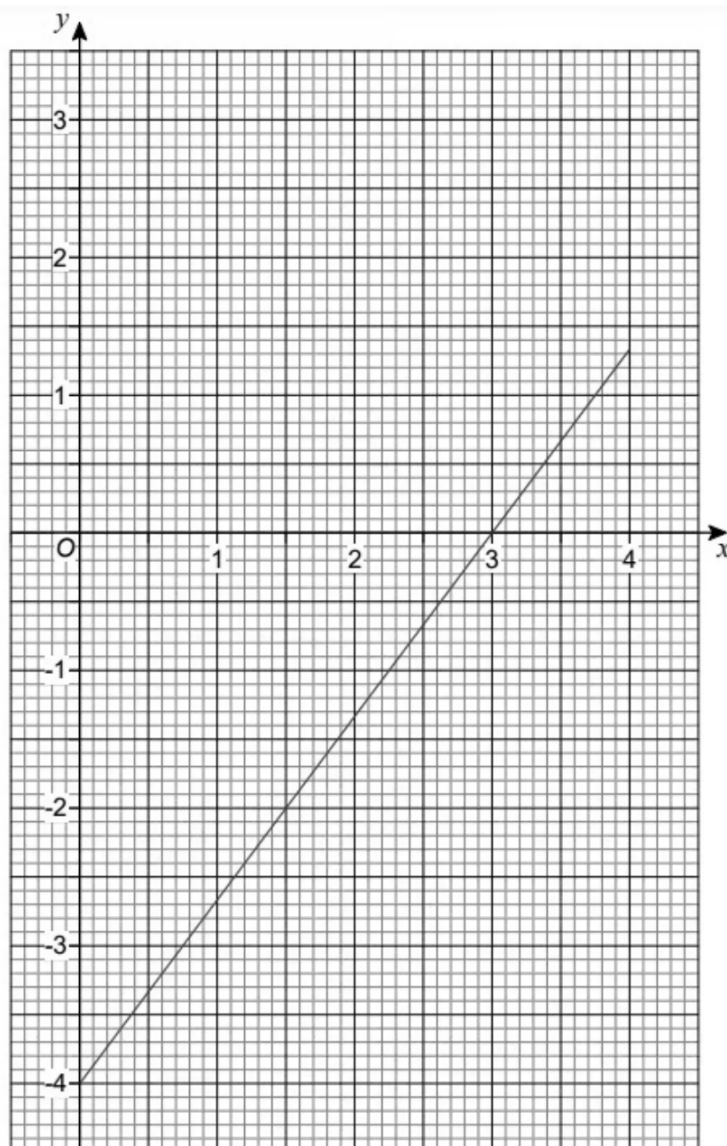
(2 marks)

Q30**PRACTICE**

Practice · Medium

CH 2 · TOPIC 12 3 marks

12 Here is the graph of $4x - 3y = 12$ for values of x from 0 to 4



By drawing a second graph on the grid, work out an approximate solution to the simultaneous equations

$$4x - 3y = 12 \text{ and } 3x + 2y = 6$$

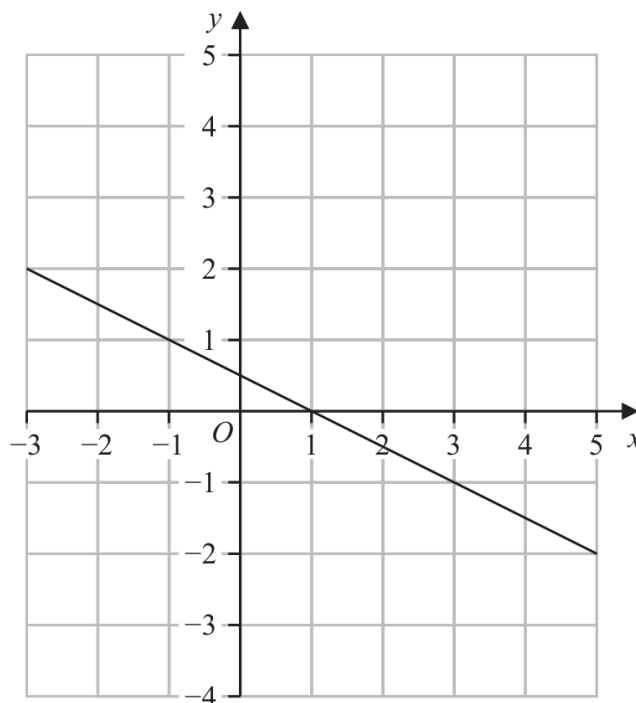
(3 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • November 2023 • Paper 2H • Q14

CH 2 • TOPIC 12 3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$

$$2y + x = 1$$

 $x = \dots\dots\dots$
 $y = \dots\dots\dots$

(Total for Question 14 is 3 marks)

04 Linear and Quadratic: Substitution

One equation has x^2 , y^2 or xy ? You cannot eliminate. Make one letter the subject of the **linear** one and substitute.

WHEN YOU SEE → YOU DO

$x^2 + y^2 = \dots$ and a straight line

→ substitute the line into the circle

“intersect at the points A and B”

→ two pairs of answers

$x - 6y = 5$

→ use the letter with coefficient 1: $x = 6y + 5$

“Show clear algebraic working”

→ write the substitution line and the quadratic = 0

Rearrange the line

Substitute

ERCS

Solve

Pair the answers

WORKED MODEL • JUNE 2025 P1HR

Solve $x^2 + y^2 = 41$ and $2x + y = 3$.

1 $y = 3 - 2x$

rearrange the line

2 $x^2 + (3 - 2x)^2 = 41$

substitute

3 $5x^2 - 12x - 32 = 0 \Rightarrow (x - 4)(5x + 8) = 0$

ERCS, solve

4 $x = 4 \Rightarrow y = -5; x = -1.6 \Rightarrow y = 6.2$

pair the answers

ANSWER (4, -5) and (-1.6, 6.2)

× **TRAP** $(3 - 2x)^2$ is not $9 + 4x^2$. Expand it: $9 - 12x + 4x^2$.

IDEA 04

Practise: Linear and Quadratic: Substitution

My questions first, then selected practice, then real past papers.

Q32 DR ESLAM Line and circle

CH 2 · TOPIC 12 5 marks

Solve $y = x + 1$ and $x^2 + y^2 = 25$

SELECTED PRACTICE

Q33 PRACTICE Practice · Hard

CH 2 · TOPIC 12 5 marks

2 Solve the equations

$$x^2 + y^2 = 36$$

$$x = 2y + 6$$

(5 marks)

Q34**PRACTICE**

Practice · Hard

CH 2 · TOPIC 12 5 marks

5 Solve algebraically the simultaneous equations

$$x^2 + y^2 = 25$$

$$y - 2x = 5$$

(5 marks)**Q35****PRACTICE**

Practice · Hard

CH 2 · TOPIC 12 5 marks

18 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B

Show clear algebraic working.

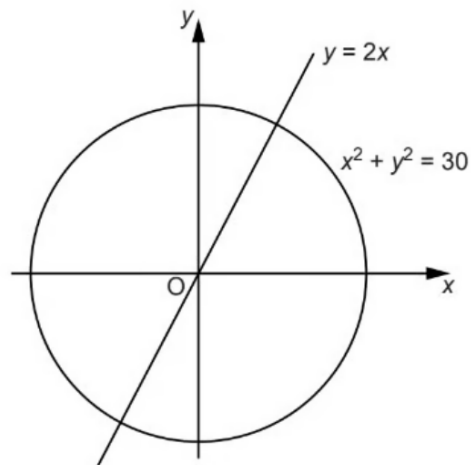
(5 marks)

Q36**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 5 marks

- 17** Find the exact coordinates of the two intersections of the line $y = 2x$ and the circle $x^2 + y^2 = 30$



Not to scale

(5 marks)

PAST PAPERS

Q37**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q17

CH 2 • TOPIC 12 5 marks

17 Solve the simultaneous equations

$$x - 6y = 5$$

$$xy - 2y^2 = 6$$

Show clear algebraic working.

(Total for Question 17 is 5 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q19

CH 2 • TOPIC 12 5 marks**19** Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

Q41 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 2HR · Q21**CH 2 · TOPIC 12** **5 marks****21** Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Q42 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2H • Q21**CH 2 • TOPIC 12** **5 marks****21** Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

.....

(Total for Question 21 is 5 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q22

CH 2 • TOPIC 12**5 marks****22** Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Q44**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q22

CH 2 • TOPIC 12 5 marks**22** Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q22

CH 2 • TOPIC 12**5 marks****22** Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

.....

(Total for Question 22 is 5 marks)

Q46**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q22

CH 2 • TOPIC 12**5 marks****22** Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$

$$y = x + 2$$

Show clear algebraic working.

.....

(Total for Question 22 is 5 marks)

05 Show That: Tangent, One Point, Midpoint

After you substitute, the quadratic tells the story: two roots = two points, **one repeated root = a tangent**, no roots = they never meet.

WHEN YOU SEE → YOU DO

“prove the line is a tangent to the circle”

→ substitute; show $(\dots)^2 = 0$ or $b^2 - 4ac = 0$

“exactly one point of intersection”

→ same: one repeated root

“the midpoint of PQ ”

→ find P and Q , average the coordinates

“the exact length of AB ”

→ Pythagoras on the two points

ZONE 1

$$b^2 - 4ac > 0$$

two points: the line cuts

ZONE 2

$$b^2 - 4ac = 0$$

one point: a **tangent**

ZONE 3

$$b^2 - 4ac < 0$$

no points: they miss

WORKED MODEL • TANGENT

Prove that $x - 2y = 10$ is a tangent to $x^2 + y^2 = 20$.

1 $x = 2y + 10$

rearrange

2 $(2y + 10)^2 + y^2 = 20 \Rightarrow 5y^2 + 40y + 80 = 0$

substitute, ERCS

3 $y^2 + 8y + 16 = 0 \Rightarrow (y + 4)^2 = 0$

one repeated root

ANSWER one point $(2, -4)$, so the line is a tangent

IDEA 05

Practise: Show That: Tangent, One Point, Mid-point

My questions first, then selected practice.

Q47 DR ESLAM One point

CH 2 · TOPIC 12 4 marks

Show that the line $y = 2x + 1$ meets the curve $y = x^2 - 2x + 5$ at exactly one point, and find it.

SELECTED PRACTICE**Q48** PRACTICE Practice · Very Hard

CH 2 · TOPIC 12 5 marks

- 2 Prove algebraically that the straight line with equation $x - 2y = 10$ is a tangent to the circle with equation $x^2 + y^2 = 20$

(5 marks)

Q49**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12

6 marks

7 The straight line **L** has equation $x - y = 3$

The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points **P** and **Q**.

Find the coordinates of the midpoint of **PQ**.

Show clear algebraic working.

(6 marks)

Q50**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12

6 marks

- 10** The curve with equation $x^2 - x + y^2 = 10$ and the straight line with equation $x - y = -4$ intersect at the points A and B .

Work out the exact length of AB .

Show your working clearly and give your answer in the form $\frac{\sqrt{a}}{2}$ where a is an integer.

(6 marks)

Q51**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12 8 marks

11 (a) The line $y = 3x + p$ and the circle $x^2 + y^2 = 53$ intersect at points A and B .

p is a positive integer.

Show that the x -coordinates of points A and B satisfy the equation

$$10x^2 + 6px + p^2 - 53 = 0$$

(3 marks)

(b) The coordinates of A are $(2, 7)$

Work out the coordinates of B .

You **must** show your working.

(5 marks)

Q52**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 12

4 marks

13 A curve has equation $y = 4x^2 + 5x + 3$

A line has equation $y = x + 2$

Show that the curve and the line have **exactly** one point of intersection.

Do **not** use a graphical method.

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE A Both linear Eliminate	ZONE B x^2, xy Substitute	ZONE C Words Name letters	ZONE D Graph Read crossing
--	--	---	--

- | | | |
|-----------|---------------------|---|
| 01 | Elimination | Calculator first. Match; same sign subtract, different sign add; substitute back. |
| 02 | From words | One sentence = one equation. Answer the question asked. |
| 03 | Graphs | Solution = the crossing point. |
| 04 | Substitution | Line → subject, substitute, ERCS, solve, pair the answers. |
| 05 | Show that | One repeated root = tangent. Midpoint: average. Length: Pythagoras. |

BOTTOM LINE Calculator first → linear: eliminate • quadratic: substitute
→ always give x **and** y, paired.

Answers

Simultaneous Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR PAIRS: ELIMINATION

Q1	Dr Eslam • Different signs: add	$8x = 24: x = 3, y = 3.5$
Q2	Dr Eslam • Match first	$x = 3, y = 2$
Q3	Practice • Easy	$x = 7, y = -3$
Q4	Practice • Easy	$x = 4.5, y = -2.5$
Q5	Practice • Medium	$x = -5, y = 3$
Q6	Practice • Hard	$m = -2, n = 4$
Q7	Edexcel 4MA1 • January 2020 • Paper 2HR • Q7	$x = -0.5, y = 1.5$
Q8	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q12	$x = 4, y = -3$
Q9	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q12	$x = 1.5, y = -2.5$
Q10	Edexcel 4MA1 • January 2021 • Paper 2HR • Q12	$x = -2, y = 3$
Q11	Edexcel 4MA1 • May 2021 • Paper 2H • Q8	$a = 2.25, c = -0.625$
Q12	Edexcel 4MA1 • January 2022 • Paper 2H • Q10	$x = 0.45, y = 0.35$
Q13	Edexcel 4MA1 • June 2022 • Paper 1HR • Q14	$x = 5, y = -2$
Q14	Edexcel 4MA1 • June 2022 • Paper 2H • Q10	$x = 1.5, y = -2.5$
Q15	Edexcel 4MA1 • May 2023 • Paper 1H • Q9	$x = 0.5, y = 1.5$
Q16	Edexcel 4MA1 • June 2024 • Paper 2H • Q10	$x = -1.5, y = 2.5$
Q17	Edexcel 4MA1 • May 2024 • Paper 1HR • Q12	$x = -1.2, y = 4.8$
Q18	Edexcel 4MA1 • June 2025 • Pa-	$x = 4, y = -1$

IDEA 02 • FORMING THEM FROM WORDS

- Q21** Dr Eslam • Cakes and drinks $2c + d = 5, c + 3d = 10$: cake £1, drink £3
- Q22** Practice • Medium $2L + W = 11, L + W = 7$: $L = 4, W = 3$; $4 \times 12 = 48$ cm²
- Q23** Practice • Hard $a = -10, b = 3$
- Q24** Practice • Very Hard 12 red, 9 green
- Q25** Practice • Very Hard (a) $X = 24, Y = 31$ miles (b) each route is the same length every time
- Q26** Edexcel 4MA1 • November 2021
• Paper 1H • Q6 apple \$0.26, pear \$0.22: \$4.80

IDEA 03 • GRAPHS: THE CROSSING POINT

- Q27** Practice • Easy $\left(-\frac{1}{3}, -1\right)$
- Q28** Practice • Easy $\left(-\frac{3}{4}, 3\right)$
- Q29** Practice • Medium $x \approx 2.25, y \approx -1.33$
- Q30** Practice • Medium $x \approx 2.5, y \approx -0.7$ (exact $\frac{42}{17}, -\frac{12}{17}$)
- Q31** Edexcel 4MA1 • November 2023
• Paper 2H • Q14 $x = -1, y = 1$

IDEA 04 • LINEAR AND QUADRATIC: SUBSTITUTION

- Q32** Dr Eslam • Line and circle $2x^2 + 2x - 24 = 0$: (3, 4) and (-4, -3)
- Q33** Practice • Hard (6, 0) and (-3.6, -4.8)
- Q34** Practice • Hard (0, 5) and (-4, -3)
- Q35** Practice • Hard (1, 2) and $(\frac{23}{7}, \frac{6}{7})$
- Q36** Practice • Very Hard $(\sqrt{6}, 2\sqrt{6})$ and $(-\sqrt{6}, -2\sqrt{6})$
- Q37** Edexcel 4MA1 • January 2021 • Paper 1H • Q19 (-7, 4) and (-2, 1.5)
- Q38** Edexcel 4MA1 • January 2021 • Paper 1HR • Q17 (-7, -2) and (9.5, 0.75)
- Q39** Edexcel 4MA1 • May 2021 • Paper 2H • Q19 (3, -3) and (-0.6, 4.2)
- Q40** Edexcel 4MA1 • January 2022 • Paper 1H • Q19 (1, -1) and (0.8, -1.4)
- Q41** Edexcel 4MA1 • June 2022 • Paper 2HR • Q21 (-7, -5) and $(\frac{7}{3}, -\frac{1}{3})$
- Q42** Edexcel 4MA1 • June 2023 • Paper 2H • Q21 (2, 1) and $(-\frac{16}{7}, -\frac{3}{7})$
- Q43** Edexcel 4MA1 • June 2024 • Paper 2HR • Q22 (1.5, 3.5) and (-0.6, -2.8)
- Q44** Edexcel 4MA1 • November 2024 • Paper 2H • Q22 (-1, 1) and (-1.5, 0.5)
- Q45** Edexcel 4MA1 • June 2025 • Paper 1HR • Q22 (4, -5) and (-1.6, 6.2)
- Q46** Edexcel 4MA1 • June 2025 • Paper 1H • Q22 (-3, -1) and (-0.5, 1.5)

IDEA 05 • SHOW THAT: TANGENT, ONE POINT, MIDPOINT

- Q47** Dr Eslam · One point
 $x^2 - 4x + 4 = 0, (x - 2)^2 = 0$: one point (2, 5)
- Q48** Practice · Very Hard
 $5y^2 + 40y + 80 = 0, (y + 4)^2 = 0$: one point (2, -4), a tangent
 $P(2, -1), Q(-3, -6)$: midpoint (-0.5, -3.5)
- Q49** Practice · Very Hard
 $A(-2, 2), B(-1.5, 2.5)$: $AB = \frac{\sqrt{2}}{2}$
- Q50** Practice · Very Hard
(a) substitute $y = 3x + p$ (b) $p = 1$: $B(-2.6, -6.8)$
- Q51** Practice · Very Hard
 $4x^2 + 4x + 1 = 0, (2x + 1)^2 = 0$: one point
- Q52** Practice · Very Hard

Forming and Solving Equations

Turn the words or the diagram into one equation, solve it, then answer the question that was asked.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Forming and Solving Equations

01 Words into an equation

three tins: total 137 ages “think of a number” ratio of savings

02 Angles

triangle = 180° parallelogram is AC a diameter?

03 Perimeter and area (linear)

perimeter = 32 same perimeter tiles, floor plans

04 It becomes a quadratic

area = 351: “show that” Pythagoras volume, probability

05 Two unknowns

tickets, teas and coffees boys and girls

WHICH EQUATION? Find what is **equal**:

ZONE A A total

add the parts =
total

ZONE B Perimeter

add every side

ZONE C Angles

180° , 360° , 90°

ZONE D Area

multiply $\rightarrow$ often
 x^2

01 Words into an Equation

Give the smallest unknown the letter x . Write every other amount in terms of x . Then find the sentence that says what they are **equal** to.

WHEN YOU SEE → YOU DO

“4 times the number ...”, “7 fewer than ...”

→ $4x$, $x - 7$

“the total is 137”

→ add all the expressions = 137

“twice as old”, “3 years younger”

→ $2y$, $y - 3$

“work out the number of buttons in tin C”

→ solve, then **substitute** — x is not the answer

Choose x

Expressions

Equation

Solve

Answer the question

WORKED MODEL · JANUARY 2020 P2HR

Tin A: x buttons. B has 4 times as many. C has 7 fewer than A. Total 137. Find tin C.

1 $A = x$, $B = 4x$, $C = x - 7$

expressions

2 $x + 4x + x - 7 = 137$

total

3 $6x = 144 \Rightarrow x = 24$

solve

ANSWER $C = 24 - 7 = 17$ buttons

× **TRAP** Stopping at $x = 24$. Read the last line of the question again — it asked for tin C.

IDEA 01

Practise: Words into an Equation

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Think of a number

CH 2 • TOPIC 13 2 marks

I think of a number, multiply it by 3 and subtract 7. The answer is 20. Find the number.

Q2 **DR ESLAM** Ages

CH 2 • TOPIC 13 3 marks

Sam is 4 years older than Ben. Their ages add to 30. How old is Sam?

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 13

3 marks

- 2 Kalinda buys x packs of currant buns and y boxes of iced buns.

There are 6 currant buns in a pack of currant buns.

There are 8 iced buns in a box of iced buns.

Kalinda buys a total of T buns.

Write down a formula for T in terms of x and y .

(3 marks)

Q4**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 **5 marks**

- 1 Julie and Liam write down the same number.

Julie multiplies the number by 5 and then adds 4 to the result. She writes down her answer.

Liam subtracts the number from 10

He writes down his answer.

Julie's answer is two thirds of Liam's answer.

Work out the number that Julie and Liam started with.

You must show your working.

(5 marks)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

3 marks

3 Becky has some marbles.

Chris has two times as many marbles as Becky.

Dan has seven more marbles than Chris.

They have a total of 57 marbles.

Dan says,

"If I give some marbles to Becky, each of us will have the same number of marbles."

Is Dan correct?

You must show how you get your answer.

(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 4 marks

17 Sundip and Emma have some money.

The ratio of Sundip's money to Emma's money is 3 : 5.

Emma spends £450 of her money.

The ratio of Sundip's money to Emma's money is now 2 : 3.

Find how much money Sundip has.

£

(4 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q3

CH 2 · TOPIC 13**4 marks**

- 3 Three tins, A , B and C , each contain buttons.

Tin A contains x buttons.

Tin B contains 4 times the number of buttons that tin A contains.

Tin C contains 7 fewer buttons than tin A .

The total number of buttons in the three tins is 137

Work out the number of buttons in tin C .

.....

(Total for Question 3 is 4 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q2

CH 2 · TOPIC 13**6 marks**

- 2 (a) Factorise fully $12pq - 18p$

There are 56 metal bars in a box.
Each metal bar is gold or silver or zinc.

y metal bars are gold.

$(3y + 7)$ metal bars are silver.

$(2y - 5)$ metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.
Show clear algebraic working.

(Total for Question 2 is 6 marks)

02 Angles

Angles in a diagram with x in them: find the angle fact that links them, write the equation, solve.

WHEN YOU SEE → YOU DO

a triangle

→ angles add to 180°

around a point / a quadrilateral

→ 360°

a right angle, a right-angled triangle

→ the other two add to 90°

parallel lines, a parallelogram

→ co-interior add to 180° ; opposite angles equal

“is AC a diameter?”

→ show the angle = 180°

WORKED MODEL · TRIANGLE

A triangle has angles 30° , $(4x + 10)^\circ$ and $(x + 20)^\circ$. Find x .

1 $30 + 4x + 10 + x + 20 = 180$

triangle = 180°

2 $5x + 60 = 180 \Rightarrow 5x = 120$

collect

ANSWER $x = 24$

ANSWER THE QUESTION “the smallest angle”, “the largest angle”: substitute x into every angle, then choose.

IDEA 02

Practise: Angles

My questions first, then selected practice.

Q9 DR ESLAM Triangle

CH 2 · TOPIC 13 2 marks

The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find the largest angle.

SELECTED PRACTICE

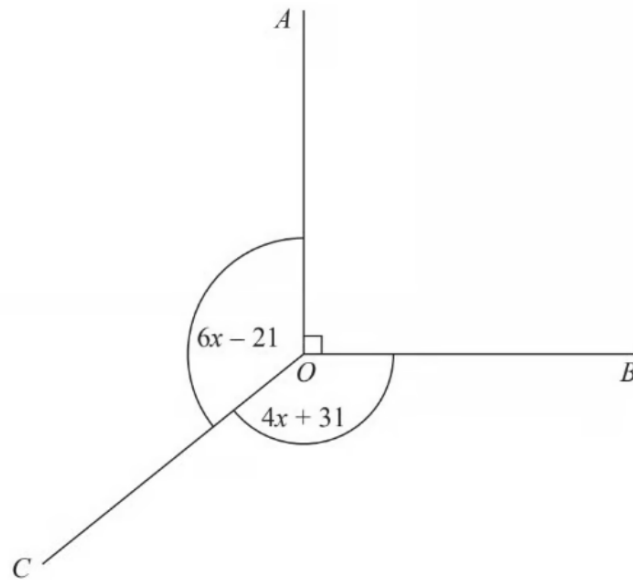
Q10

PRACTICE

Practice · Easy

CH 2 · TOPIC 13 3 marks

4

Diagram **NOT**
accurately drawn

In the diagram, all angles are in degrees.

Angle AOB is a right angle.

Angle $AOC = \text{Angle } BOC$.

Work out the value of x .

(3 marks)

Q11**PRACTICE**

Practice · Easy

CH 2 · TOPIC 13 4 marks

6 The diagram shows a triangle.

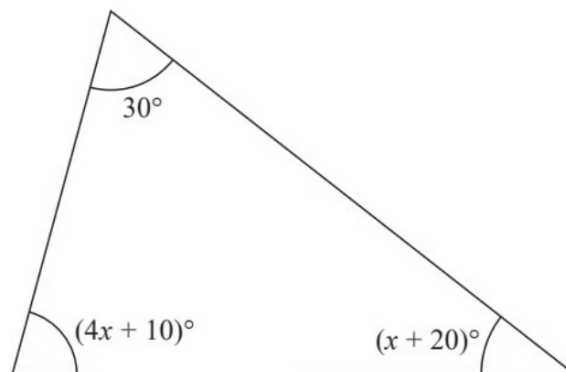


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

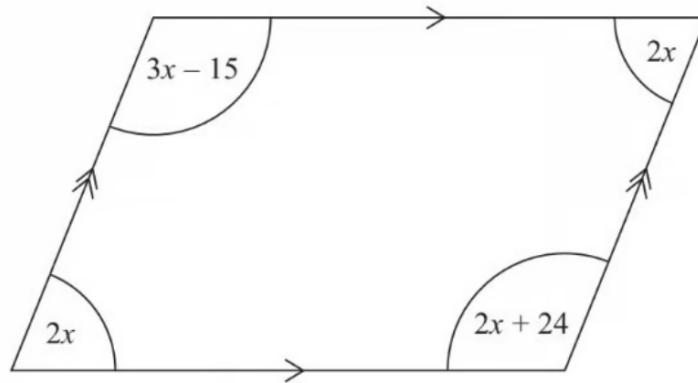
(4 marks)

Q12**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 3 marks

7

Diagram **NOT**
accurately drawn

The diagram shows a parallelogram.

The sizes of the angles, in degrees, are

$$2x$$

$$3x - 15$$

$$2x$$

$$2x + 24$$

Work out the value of x .

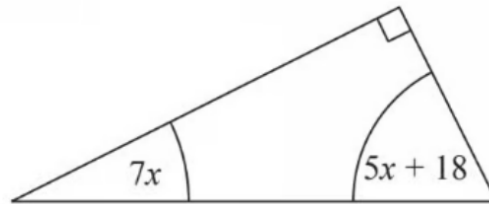
(3 marks)

Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 **3 marks**

9 The diagram shows a right-angled triangle.



All the angles are in degrees.

Work out the size of the smallest angle of the triangle.

(3 marks)

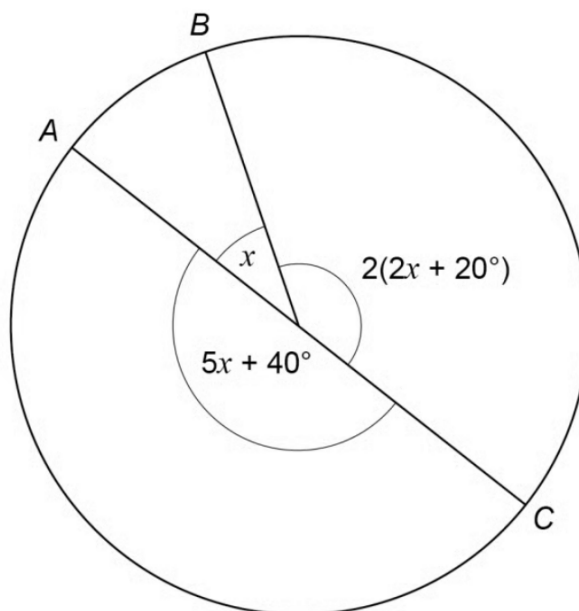
Q14 PRACTICE Practice · Medium

CH 2 · TOPIC 13 3 marks

15 A , B and C are three points on a circle.

The radii from A , B and C are shown.

Not drawn
accurately



Is AC a diameter of the circle?
You **must** show your working.

(3 marks)

Q15**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

16 $ABCD$ is a trapezium.

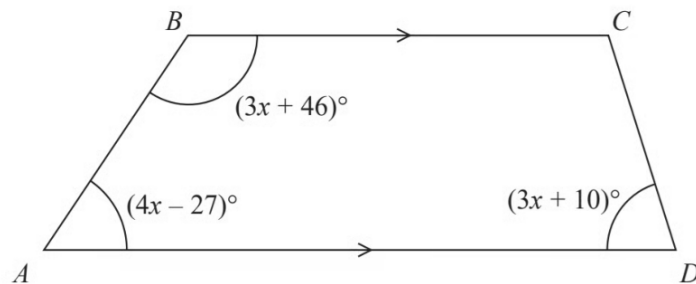


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

(4 marks)

03 Perimeter and Area (Linear)

Perimeter: add every side — including the sides with no label. Two shapes with the same perimeter or area: write both, make them equal.

WHEN YOU SEE → YOU DO

“the perimeter is 32”

→ add **all** sides = 32

a rectangle with $2x + 6$ and $5x - 9$ on opposite sides

→ opposite sides equal

“the two shapes have the same perimeter”

→ perimeter A = perimeter B

tiles, floor plans

→ find the missing lengths first

Label every side

Add or multiply

Equation

Solve

WORKED MODEL · NOVEMBER 2021 P2H

Rectangle A: 4 by $(5 - x)$. Rectangle B: 5 by $(2x - 1)$. Area B is twice area A. Find x .

1 $5(2x - 1) = 2 \times 4(5 - x)$

area B = $2 \times$ area A

2 $10x - 5 = 40 - 8x$

expand

3 $18x = 45$

collect

ANSWER $x = 2.5$

IDEA 03

Practise: Perimeter and Area (Linear)

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Perimeter

CH 2 • TOPIC 13 3 marks

A rectangle is $(x + 3)$ cm by $(2x - 1)$ cm. Its perimeter is 34 cm. Work out its area.

SELECTED PRACTICE

Q17 PRACTICE Practice · Easy

CH 2 · TOPIC 13 3 marks

- 1 The diagram shows a rectangle.

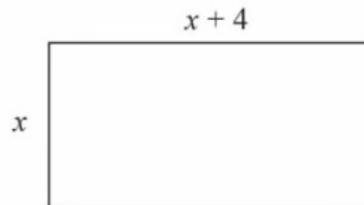


Diagram **NOT**
accurately drawn

All measurements are given in centimetres.

The perimeter of the rectangle is 45 cm.

Work out the value of x .

(3 marks)

Q18 PRACTICE Practice · Medium

CH 2 · TOPIC 13 5 marks

10 ABC is a triangle.

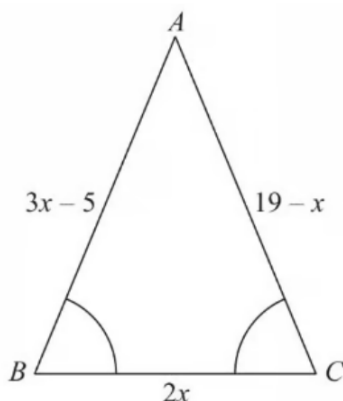


Diagram **NOT**
accurately drawn

Angle ABC = angle BCA .

The length of side AB is $(3x - 5)$ cm.

The length of side AC is $(19 - x)$ cm.

The length of side BC is $2x$ cm.

Work out the perimeter of the triangle.

Give your answer as a number of centimetres.

(5 marks)

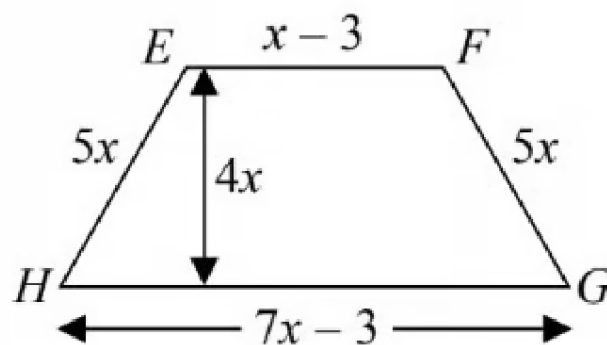
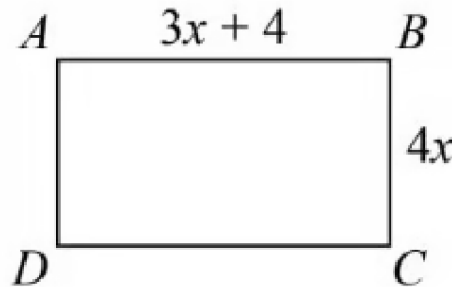
Q19**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 5 marks

5 $ABCD$ is a rectangle.

$EFGH$ is a trapezium.



All measurements are in centimetres.

The perimeters of these two shapes are the same.

Work out the area of the rectangle.

(5 marks)

Q20**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

5 marks

18 The diagram shows a hexagon $ABCDEF$

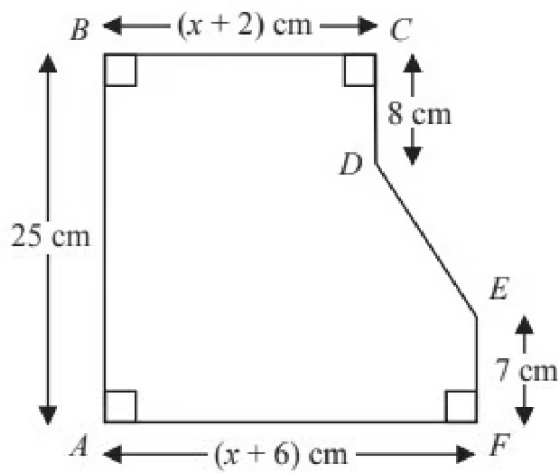


Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon $ABCDEF$ is 258 cm^2

Work out the value of x

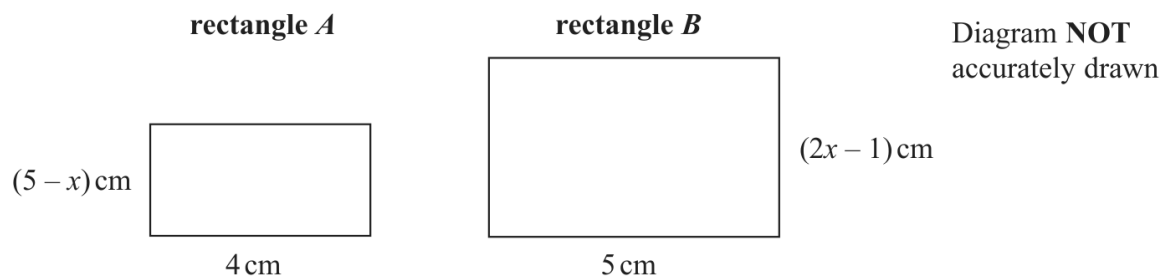
(5 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 2H · Q2

CH 2 · TOPIC 13 4 marks

- 2 Here are two rectangles, rectangle A and rectangle B .



The area of rectangle B is twice the area of rectangle A .

Work out the value of x .
Show your working clearly.

$x =$

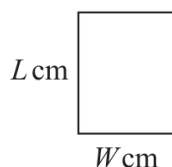
(Total for Question 2 is 4 marks)

Q22**PAST PAPER**

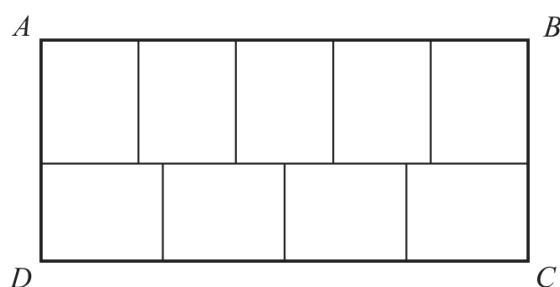
Edexcel 4MA1 · November 2021 · Paper 1H · Q5

CH 2 · TOPIC 13 5 marks

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

(Total for Question 5 is 5 marks)

04 It Becomes a Quadratic

Multiply two lengths with x in both and you get x^2 . ERCS, solve — then **reject** the answer that makes a length negative.

WHEN YOU SEE → YOU DO

area of a shape with x in two sides

→ a quadratic: ERCS, then solve

“Show that $2x^2 + x - 351 = 0$ ”

→ start from the formula, end at exactly that line

a right-angled triangle with x sides

→ Pythagoras: $a^2 + b^2 = c^2$

two answers, one negative

→ a length cannot be negative: say why you reject it

Formula

Expand

= 0

Solve

Reject

WORKED MODEL · NOVEMBER 2025 P2H

$ax^2 + 4x + c = 0$ has solutions $x = \frac{-4 \pm 2\sqrt{39}}{10}$. Find a and c .

1 $2a = 10 \Rightarrow a = 5$

compare the bottom

2 $\sqrt{16 - 4ac} = 2\sqrt{39} = \sqrt{156}$

compare the root

3 $16 - 20c = 156 \Rightarrow c = -7$

solve

ANSWER $a = 5, c = -7$

× **TRAP** In Kate's box question, $x = -40$ is a width: it is impossible. Only $x = 35$.

IDEA 04

Practise: It Becomes a Quadratic

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Area to quadratic

CH 2 · TOPIC 13 4 marks

A rectangle is x cm wide and $(x + 3)$ cm long. Its area is 40 cm^2 . Find x .

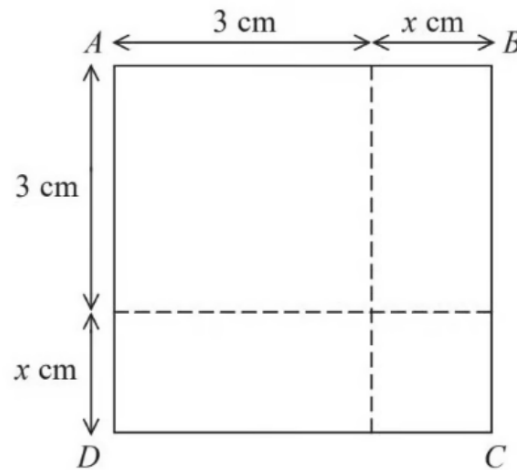
$x =$ _____

SELECTED PRACTICE

Q24 PRACTICE Practice · Easy

CH 2 · TOPIC 13 3 marks

5



The area of square $ABCD$ is 10 cm^2 .

Show that $x^2 + 6x = 1$

(3 marks)

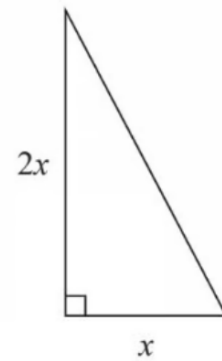
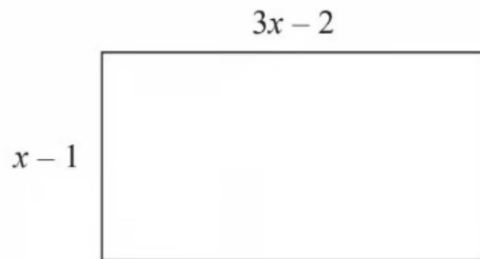
Q25**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

5 marks

6 Here is a rectangle and a right-angled triangle.



All measurements are in centimetres.

The area of the rectangle is greater than the area of triangle.

Find the set of possible values of x .

(5 marks)

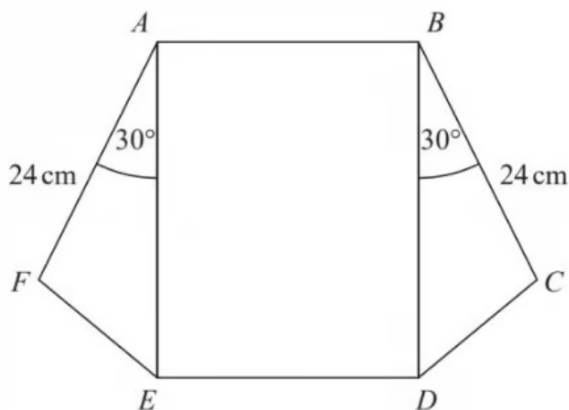
Q26**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13

4 marks

- 5 The diagram shows a rectangle, $ABDE$, and two congruent triangles, AFE and BCD .



area of rectangle $ABDE$ = area of triangle AFE + area of triangle BCD

$$AB : AE = 1 : 3$$

Work out the length of AE .

.....cm

(4 marks)

Q27**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13 4 marks

11 A sphere has radius $2x$ cm

A cone has

radius $3x$ cm

perpendicular height h cm

The sphere and the cone have the same volume.

Work out radius of cone : perpendicular height of cone

Give your answer in the form $a : b$ where a and b are integers.

(4 marks)

PAST PAPERS

Q28**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q20

CH 2 · TOPIC 13 3 marks

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c

Show your working clearly.

$a =$

$c =$

(Total for Question 20 is 3 marks)

05 Two Unknowns

Two unknown amounts and two facts? Use two letters, write two equations, and solve them simultaneously (Topic 12).

WHEN YOU SEE → YOU DO

“3 adult tickets and 1 child ticket cost £30”

→ $3a + c = 30$

“if 5 more boys and 8 more girls join ...”

→ write the new amounts, then the fact

“half as many boys as girls”

→ $2(\text{boys}) = \text{girls}$

WORKED MODEL · TICKETS

3 adults and 1 child cost £30. 1 adult and 3 children cost £22. Find each price.

1 $3a + c = 30, a + 3c = 22$

two letters, two facts

2 $9a + 3c = 90 \Rightarrow 8a = 68$

×3, subtract

ANSWER adult £8.50, child £4.50

WHERE YOU WILL USE IT Every problem-solving question on the paper — geometry, probability, ratio — ends with “form an equation and solve it”.

IDEA 05

Practise: Two Unknowns

My questions first, then selected practice.

Q29**DR ESLAM**

Pens and pencils

CH 2 · TOPIC 13

4 marks

3 pens and 2 pencils cost £2.60. 1 pen and 2 pencils cost £1.40. Find the cost of one pen.

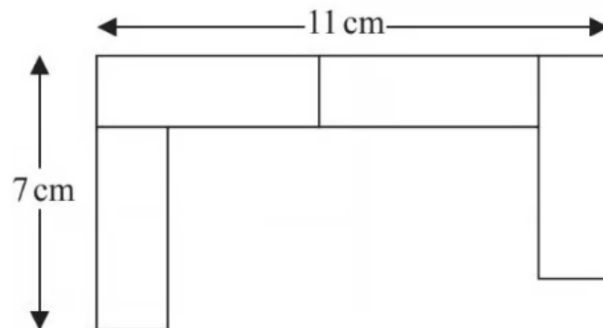
SELECTED PRACTICE

Q30**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

- 2 A pattern is made using identical rectangular tiles.



Find the total area of the pattern.

(4 marks)

Q31**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

- 6 3 kg of potatoes and 4 kg of carrots have a total cost of 440p.
4 kg of potatoes and 3 kg of carrots have a total cost of 470p.

Work out the total cost of 1 kg of potatoes and 1 kg of carrots.

(4 marks)**Q32****PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 4 marks

- 7 A cinema sells adult tickets and child tickets.

The total cost of 3 adult tickets and 1 child ticket is £30

The total cost of 1 adult ticket and 3 child tickets is £22

Work out the cost of an adult ticket and the cost of a child ticket.

(4 marks)

Q33**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

4 marks

- 8 3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

(4 marks)

Q34**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 5 marks

13 (a) In a chess club, there are x boys and y girls.

If 5 more boys and 8 more girls join, there would be half as many boys as girls.

Show that $y = 2x + 2$

(2 marks)

(b) If instead,

10 more boys and 1 more girl join, there would be the same number of boys and girls.

Work out x and y .

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(3 marks)**

Q35**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13 5 marks

4 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{6}{13}$

Find the number of red counters and the number of green counters that were in the bag originally.

(5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

Choose x

Expressions

Equation

Solve

Answer the question

- | | | |
|-----------|-------------------------|--|
| 01 | Words | Smallest unknown = x ; add the parts = total; answer what was asked. |
| 02 | Angles | 180° triangle, 360° point/quadrilateral, parallel lines. |
| 03 | Perimeter / area | Add every side; same perimeter $\rightarrow$ make them equal. |
| 04 | Quadratic | $x \times x$ gives x^2 : ERCS, solve, reject negative lengths. |
| 05 | Two unknowns | Two letters, two equations, eliminate. |

BOTTOM LINE What is **equal** to what? $\rightarrow$ write it $\rightarrow$ solve $\rightarrow$ substitute back and answer the question.

Answers

Forming and Solving Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • WORDS INTO AN EQUATION

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam • Think of a number | $3x - 7 = 20; x = 9$ |
| Q2 | Dr Eslam • Ages | $x + x + 4 = 30; \text{ Ben 13, Sam 17}$ |
| Q3 | Practice • Easy | $T = 6x + 8y$ |
| Q4 | Practice • Medium | $5n + 4 = \frac{2}{3}(10 - n); n = \frac{8}{17}$ |
| Q5 | Practice • Hard | Becky 10, Chris 20, Dan 27: equal shares need 19 each, but Chris has 20 — No |
| Q6 | Practice • Hard | £2700 |
| Q7 | Edexcel 4MA1 • January 2020 • Paper 2HR • Q3 | $6x - 7 = 137, x = 24: 17 \text{ buttons}$ |
| Q8 | Edexcel 4MA1 • November 2023 • Paper 1H • Q2 | (a) $6p(2q - 3)$ (b) $6y + 2 = 56, y = 9: 13 \text{ zinc bars}$ |

IDEA 02 • ANGLES

- Q9** Dr Eslam · Triangle $6x = 180, x = 30: 90^\circ$
- Q10** Practice · Easy $6x - 21 = 4x + 31; x = 26$
- Q11** Practice · Easy $x = 24$
- Q12** Practice · Medium $x = 39$
- Q13** Practice · Medium $12x + 18 = 90, x = 6: 42^\circ$
- Q14** Practice · Medium $x = 28: 5x + 40 = 180^\circ, \text{ so yes}$
- Q15** Practice · Medium $x = 23: \text{ largest angle } 115^\circ$

IDEA 03 · PERIMETER AND AREA (LINEAR)

- Q16** Dr Eslam · Perimeter $6x + 4 = 34, x = 5: 8 \times 9 = 72 \text{ cm}^2$
- Q17** Practice · Easy $4x + 8 = 45; x = 9.25$
- Q18** Practice · Medium $x = 6: 13 + 13 + 12 = 38 \text{ cm}$
- Q19** Practice · Hard $x = 3.5: 203 \text{ cm}^2$
- Q20** Practice · Hard $25(x + 6) - 52 = 258; x = 6.4$
- Q21** Edexcel 4MA1 · November 2021
· Paper 2H · Q2 $x = 2.5$
- Q22** Edexcel 4MA1 · November 2021
· Paper 1H · Q5 $5W = 4L, 9LW = 1620: L = 15, W = 12$

IDEA 04 · IT BECOMES A QUADRATIC

- Q23** Dr Eslam · Area to quadratic $x^2 + 3x - 40 = 0, (x+8)(x-5) = 0: x = 5$ ($x = -8$ rejected)
- Q24** Practice · Easy $(x+3)^2 = 10 \Rightarrow x^2 + 6x = 1$
- Q25** Practice · Hard $2x^2 - 5x + 2 > 0$ and $x > 1: x > 2$
- Q26** Practice · Very Hard $3x^2 = 36x, x = 12: AE = 36 \text{ cm}$
- Q27** Practice · Very Hard $h = \frac{32}{9}x: 27 : 32$
- Q28** Edexcel 4MA1 · November 2025
· Paper 2H · Q20 $a = 5, c = -7$

IDEA 05 · TWO UNKNOWNNS

- Q29** Dr Eslam · Pens and pencils $2p = 1.20: \text{pen } 60p, \text{pencil } 40p$
- Q30** Practice · Medium $L = 4, W = 3: 48 \text{ cm}^2$
- Q31** Practice · Medium £1.30
- Q32** Practice · Hard adult £8.50, child £4.50
- Q33** Practice · Hard tea £1.40, coffee £1.80
- Q34** Practice · Hard (a) $2(x+5) = y+8$ (b) $x = 7, y = 16$
- Q35** Practice · Very Hard 12 red, 9 green